Preprint

Light in Light Bubbles — Gravitational Light in the Phraqtal Field

Tension fronts as spectral carriers, matter formation and binding, and gravity from a common field action

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Abstract. This work develops an effective orientation and loading model of the Phraqtal Field within the Universal Quantum Foam Hypothesis. A common action determines the field equations, energy exchange, and gravitational source; general relativity describes its geometric response. Analytical comparisons of fronts relate encounter angle, phase, and energy to loading and twist. An arrival-time mapping distinguishes source cadence, geometric offset, and the selection of effective front portions. Static three-knot comparisons complement a stationary core solution. In the boundary-value case studied, a regime channel changes the central loading by a relative amount of 6.4×10^{-3}, the field energy by -2.1×10^{-6}, and the integrated active source by 2.6×10^{-6}. The outer envelope is a standing wave with vanishing net flux and requires incoming waves. Extending it unchanged to infinity would produce a linearly growing mass; it is therefore associated with a finite matching region. The regime channel is a quasistatically eliminated nearest-neighbor chain with potential U_{eff}. Its energy scale $\mathcal{E}_\Sigma = U(\zeta_c)/2$ is chosen at the reference point and subsequently held fixed. No quantitative mapping to baryons, SPARC, or a phenomenological Q-switch is made here. The chosen initial front data are topologically trivial. Smooth norm-preserving evolution with a fixed vacuum boundary preserves this class; forming a net Hopf charge requires an additional transition mechanism. The golden ratio is derived as a geometric and recursive comparison case.

Keywords: nonlinear field theory; regime coupling; tension fronts; arrival-time offset; triadic binding; gravity; golden ratio.

Contents

1 Introduction **6**

1.1 Research Questions and Calculated Contributions 6

1.2 Boundary Conditions and Scope of the Result 6

1.3 Related Field Models and Distinctions 7

1.4 Notation 7

2 Anun, Environment, and Support Across Nested Scales **8**

2.1 Field States Across Multiple Regimes 8

2.2 Response to the Anun Relative to a Moving Environment 9

2.3 Combined Gravity of the Anun and its Environment 10

3 The Effective Field Ansatz and its Interpretation **11**

3.1 Regime Information Already Included in the Model 12

4 Common Action and Derivation of the Field Equations **12**

5 Energy Exchange and the Gravitational Source **13**

5.1 Twist Direction, a Shared Axis, and Stress Moments 14

6 Weak Gravity: Motion, Range, and Strength **15**

6.1 How a Weak Gravitational Field Produces Motion 15

6.2 Environmental Contribution and Weak Coupling Strength 16

7 Source Difference and Additional Gravity **17**

7.1 The Source Difference in Terms of the Existing Field Variables 17

7.2 First Change and Locally Adjusted Loading 18

7.3 Numerical Comparison 19

7.4 From the Source Difference to the Additional Acceleration 20

7.5 Environmental Response and the Next Carrier Regime 20

7.6 From the Carrier Response to the Combined Source Response 21

8 A Regime Channel from the Same Field Action **22**

8.1 Channel Variables and Energy 22

8.2 Bounded Response and Unique Adjustment 23

8.3 Elimination and a Positive Adjusted Potential 23

8.4 Backreaction and the Stress-energy Tensor . . . 24
8.5 Energy Balance and a Moving Reference Environment . . . 24
8.6 Preserved Threshold and Free Fronts . . . 25
8.7 Local Comparison at the Reference Core . . . 25
8.8 Relations Among the Model Quantities . . . 26

9 Free Fronts, Spherical Propagation, and Lateral Transport 27
9.1 Geometric Dilution and Measurable Time Intervals . . . 28
9.2 Angular Structure and Mean Backreaction . . . 29

10 Arrival-Time Offsets and the Event Sequence of Spatially Offset Source Fronts 30
10.1 Arrival-Time Offset from Emission Position and Emission Time . . . 30
10.2 Bounds on the Geometric Offset . . . 30
10.3 The Effective Combined Sequence and its Regularity . . . 31
10.4 A Regular Sequence Constructed from Source Positions . . . 32
10.5 A Continuously Varying Mean Rate and Uniform Individual Intervals . . . 33
10.6 Connection to the Existing Field Description . . . 33

11 From Front Superposition to Bound Field Organization 34
11.1 Excitation of an Environment in Motion . . . 34
11.2 Angle and Profile Phase . . . 35
11.3 Spatial Extent and Amplification Duration . . . 35
11.4 Internal Orientation and Spatial Twist . . . 36
11.5 Energy Balance and Connection to the Core . . . 37
11.6 Scope and Targeted Overlap Comparison . . . 37
11.7 Norm-Preserving Initial Data from Finite Fronts . . . 39
11.8 Additive Initial Energy and Combined Energy Flux . . . 39
11.9 Loading Transfer and Remaining Directional Order . . . 40
11.10 Encounter Geometry and Control Variants . . . 41

12 Energy of Incoming Fronts and Generic Weighting 43
12.1 A Norm-Preserving Single-Front State . . . 43
12.2 Exact Energy of the Chosen Initial Configuration . . . 43
12.3 Analytical Normalization of a Localized Profile . . . 44
12.4 Generic Amplitude Weights and Energy Distribution . . . 44
12.5 Emergence of the Twist Contribution Under Superposition . . . 45

12.6 Scope and Connection to Matter Formation 46
12.7 Verification of the Derivation 46

13 Matter Formation: Front Coupling and Binding 46
13.1 Topological Condition for the Connection from Fronts 46
13.2 Organizational Roles of Three Fronts 47
13.3 Three Distinct Roles of Angles 47
13.4 Calculated Geometry and Phase Binding of Three Knots 47
13.5 The Effect of a Phase Change Depends on the Initial State 48
13.6 Local Energy Curvature and the Informative Value of the Ritz Sample 49
13.7 Connection to Persistent Matter Organization 49

14 Local Saturation and Spatial Binding 50
14.1 Size Balance and a Rotating Stationary Ansatz 51

15 A Regular Core and the Buildup of Loading 51
15.1 Central Expansion and a Finite Gravitational Source 51
15.2 Spatial Threshold of a Growing Loading Mode 52
15.3 Amplitude Limitation through Coupled Reorganization 53

16 A Specific Stationary Core Solution 54
16.1 Parameters and Results 54
16.2 Profile, Numerical Method, and Scope 55
16.3 Integrated Source Difference of the Core 56
16.4 Radial Profile and Weak Gravitational Response 57

17 Core Adjustment through the Regime Channel 58
17.1 Comparison at Identical Boundary Data 58
17.2 Energy, Stresses, and the Integrated Gravitational Source 59
17.3 Inner Enhancement and Outer Counteracting Change 60
17.4 Numerical Checks and Physical Connection 61

18 From the Nonlinear Core to the Outer Field Envelope 62
18.1 Exact Projection and Determination of the Exterior Amplitudes 62
18.2 Gravitational Monopole Contribution of the Weak Envelope 63
18.3 Unaveraged Angular Structure and Orbital Evaluation 64

19 Core, Transition, and Environment as a Single Boundary-Value Problem **65**
19.1 Orientation Matching and Energy Supply 66

20 Gravitational Backreaction on the Same Field Organization **67**
20.1 Stationary Geometry and Local Phase Rate 67
20.2 Explicit Gravitational Source and Time Measurement 67
20.3 First Geometrically Induced Field Adjustment 68

21 Scale Mapping, Size Balance, and Inertia **69**
21.1 Reciprocal Coupling with Different Regime Weights 69
21.2 Size Variation at Fixed Angular Momentum 70
21.3 Periodic Closure and Restoring Response of an Orbital Organization 70
21.4 Integrated Stress and a Shared Metric Response 71

22 The Golden Ratio as a Geometric and Recursive Comparison **72**
22.1 Equal Edges of an Icosahedral Arrangement 72
22.2 Golden-Ratio Decay in the Weak Regime Channel 72
22.3 Spiral Parameter and Distinction from the Orbital Period 73

23 Reference Values, Physical Scales, and Scope **74**
23.1 Dimensionless Specification and Units 74
23.2 Physical Context 74
23.3 Results and Physical Connection 75

24 Methods and Data Basis **76**
24.1 Data and Code Availability 77

A Spatial Hamiltonian Method **77**
A.1 Verification Equation and Discrete Energy 77
A.2 Positive Coupling Coefficients and the Nonlinear Operator 78
A.3 Flux Across a Cut and Coupled Perturbation Evolution 79

B Glossary of Terms **81**

1. Introduction

The UQSH describes matter as bound field motion and radiation as free reorganization of the same Phraqtal Field [1, 2]. This approach requires dynamics that calculate spatial binding, energy transfer, and gravitational effects from the same field variables. For this purpose, the present work studies a unit field $q \in S^2$ and a scalar loading ζ. Quadratic and quartic derivative terms describe orientation and twist; a loading potential describes their local reorganization. General relativity (GR) couples the full stress-energy tensor to the metric [12].

1.1. Research Questions and Calculated Contributions

The study addresses three questions: How does an encounter between free fronts excite loading and twist? What stationary bound configurations does the action support? How does their reorganization change the gravitational source relative to the same environment? The analysis combines analytical overlap and stability conditions with a stationary boundary-value problem. An additional scalar regime channel is eliminated from an energy with nearest-neighbor coupling. Its direct source contribution and the subsequent spatial adjustment of the profile can be calculated separately.

For the parameter case evaluated, the central loading increases by a relative amount of 6.36575×10^{-3}. The field energy decreases by 2.05250×10^{-6}, while the integrated active source increases by 2.62345×10^{-6}. The range of orders of magnitude, 10^{-3} to 10^{-6}, refers to these different relative model changes, not to a universal coupling constant. Their different signs arise because the active source includes spatial stresses as well as energy.

The sections on matter formation distinguish three results: norm-preserving initial front data with calculable energy (Section 12), static triadic phase binding (Section 13), and periodic closure of the stationary field ansatz (Section 21.3). Section 10 determines the geometric arrival-time offset and the conditions for a regularly thinned front sequence. Section 22 derives the geometric and recursive comparison cases involving the golden ratio.

1.2. Boundary Conditions and Scope of the Result

The stationary core is calculated on a finite domain with prescribed boundary values. The analytical weak outer envelope is a standing wave: the incoming and outgoing components have equal power, and the net flux vanishes. In an open exterior, the environment must supply the incoming component. With purely outgoing waves, the energy of the interior decreases unless another supply is present. The unchanged envelope has a source that asymptotically decreases, on average, as r^{-2}; its enclosed mass grows linearly with r. It therefore describes a finite matching interval rather than an isolated state of finite total mass in an infinite space.

Mathematically, the additional regime channel consists of a chain of scalar auxiliary variables with nearest-neighbor coupling. Their quasistatic elimination yields $U_{\mathrm{eff}}(\zeta)$. The example normalization $\mathcal{E}_\Sigma = U(\zeta_c)/2$ uses the central loading of the reference solution. Thereafter, $\mathcal{E}_\Sigma$ remains constant in all variations and comparison solutions; its value is not an independent

prediction. The channel contains no quantitative mapping to baryonic profiles, the SPARC data set, or the activation and shape axes of the phenomenological Q-switch in [1]. The orientation field q is not identical to these classification variables.

The chosen initial front data can be globally contracted to the vacuum. Under smooth norm-preserving evolution with a fixed vacuum boundary, their total Hopf class remains zero. A net knot charge requires boundary flux or a separate topological transition. The static knot comparisons are therefore independent binding diagnostics; they do not constitute a calculation of the formation of these knots from the fronts used here. Local twist and bound organization in the trivial sector remain accessible.

1.3. Related Field Models and Distinctions

Skyrme models and global monopoles. The combination of a unit field $q \in S^2$ with quadratic gradients and the square of $\partial_i q \times \partial_j q$ is closely related to the Faddeev–Skyrme model, which studies spatially localized, knotted field structures [3]. Gravitating skyrmions couple a nonlinear chiral field to the Einstein equations [4]. The present model additionally includes the loading ζ, its effect on the orientation coefficient, and an eliminable regime channel. The calculated family of cores is supported by finite boundary data; a topological charge is not assumed for this family. Global monopoles also illustrate how extended field gradients generate an exterior gravitational effect [5]. In that case, the asymptotic defect structure and a solid-angle deficit determine the geometry. The envelope considered here instead supports oscillating orientation dynamics with a positive leading contribution to the active source.

Born–Infeld-type nonlinearity. Born and Infeld introduce a nonpolynomial electromagnetic action with a characteristic field scale [6]. In the base model without the additional channel, a related square-root structure emerges only after local algebraic adjustment of ζ on the branch $X < 0$: $F(X) = b_*[1 - \sqrt{1 - AX/b_*}]$. The full action, by contrast, includes the derivatives of ζ and the quartic orientation term. A bounded coordinate or the decreasing factor $C(\zeta) = (1 + \zeta^2)^{-1}$ does not, by itself, bound either the total energy or all characteristics of the nonlinear equation.

Scalar-field dark matter and boson stars. Ultralight scalar fields use wave dynamics to describe galactic density profiles [7]. Boson stars provide self-gravitating, localized field configurations; their existence and dynamics serve as benchmarks for bound field organization [8, 9]. In the usual massive complex scalar-field models, a harmonic internal phase can support a stationary tensor. The rotating ansatz used here shares this property. Its weak orientation tail, however, is oscillatory and requires the environment described above. The present calculation determines source changes in an effective field model; fitting galactic dark matter is a separate calibration step.

1.4. Notation

Spacetime indices are Greek, and spatial indices are Latin. We use $x^0 = ct$ and the signature $(+ - - -)$. Primes on C, Z, U denote field derivatives; primes on radial profiles denote radial derivatives. The qualifiers below distinguish different uses of the same base letters. Physical terms and UQSH terminology are explained in the glossary (Appendix B).

Symbol	Meaning
q, ζ	Unit orientation and dimensionless loading.
$R_n, \Sigma_j^{A,U}$	Regime level and general saturation coordinates of two comparison states.
$\Sigma_n, \mathcal{E}_\Sigma, \eta_\Sigma$	Auxiliary response of the specific chain, constant energy-density scale, and participation.
Γ_m	Spatial matching surface between the core and its environment.
$\kappa, k_\zeta, \kappa_G$	Nearest-neighbor coupling; radial loading wavenumber; $4\pi G/c^2$.
$\mathcal{L}_K$	$\log K$ in the discrete interpolation scheme.
$\alpha, \beta_{\mathrm{pol}}, \phi_i$	Spatial encounter angle, internal direction angle, and profile phases of free fronts.
δ_2, δ_3	Internal phase shifts of the static three-knot configuration.
a_ζ	Amplitude of a normalized loading mode.
$\beta_4, \beta_R, \beta_z, \beta_c$	Quartic coupling coefficient and central curvature coefficients of the loading.
$\boldsymbol{\beta}_{\mathrm{sh}}$	Shift vector of the spatial 3+1 decomposition.
β_C, β_Z	Logarithmic derivatives C'/C and Z'/Z in the methods appendix.
Δ, Δ_g	Kinetic combination $S - k = -X$ in flat and curved geometry, respectively.
$\Delta Y, \delta Y, \nabla^2$	Finite state difference, first variation, and spatial Laplace operator.
G_f, H_g	$\lvert\nabla f\rvert^2$ and $h^{ab} f_a f_b$, respectively.
$\mathsf{H}_{ab}, \mathsf{H}_\Sigma, \mathsf{H}_f, \mathsf{H}_\zeta$	Hessian blocks and linear variational operators.
$H_h, \mathcal{H}(r), H_{\mathrm{W}}$	Discrete energy, oscillatory radial term, and complex envelope amplitude.
$H_0^1(D)$	Sobolev space of admissible test functions with vanishing boundary values.
$J_{\mu\nu}, J, Q_H$	Twist tensor, angular momentum magnitude, and Hopf diagnostic of the knot texture.
$\boldsymbol{Q}, \varepsilon, \Pi_{ij}$	Active gravitational source, energy density, and spatial stress tensor.
$E, \mathcal{F}_\Omega$	Field energy and stationary variational functional at fixed Ω.
$s_{\mathrm{dil}}, \mathcal{A}_{\mathrm{N}}, r_i^{\mathrm{amp}}$	Spatial dilation factor, front normalization, and generic amplitude weights.
$\lambda_X, \lambda_\zeta, \mu_{\mathrm{R}}, \epsilon_{\mathrm{R}}$	Eigenvalues of the front and loading operators; Ritz value and residual norm.
$\ell_{\mathrm{ang}}, \Lambda_{\mathrm{ang}}$	Spherical harmonic degree and angular eigenvalue $\ell_{\mathrm{ang}}(\ell_{\mathrm{ang}}+1)$.
$L_\parallel, L_{\parallel,i}, \ell_*, \ell_0$	Front length scales and physical units of length.
g_n, W_g	Reciprocal coupling coefficients and the associated weighted matrix.
$T_Q, N_Q, N_D, \delta_a^{\mathrm{arr}}, C_g$	Source period, number of source contributions and resolved arrival phases, arrival-time offset, and relative gap scatter.
$\varphi_{\mathrm{g}}, r_\kappa, b_{\mathrm{sp}}$	Golden ratio, decay ratio of the auxiliary chain, and logarithmic spiral parameter.

2. Anun, Environment, and Support Across Nested Scales

2.1. Field States Across Multiple Regimes

R_n denotes the selected observational regime, and $R_{n-1}, R_{n-2}, \ldots$ its finer carrier regimes. Coarser organizational structures are represented by $R_{n+1}, R_{n+2}, \ldots$. This sequence assumes neither a lowest nor a highest terminal regime. The indices count organizational levels; a

spatial distance within a regime is specified separately.

An Anun reaches the upper saturation configuration of its own regime. Its subregimes may largely retain their ordinary motion while carrying only a small additional change in tension. Further uptake can change the extent of the saturated region, its boundary structure, and its exchange processes. Local saturation, spatial size, and the exterior range of gravity are therefore distinct quantities [1].

Let Σ_j^U be the saturation coordinate of the moving reference environment and Σ_j^A that of the jointly reorganized state containing the Anun under consideration. With $j = n - k$, we have

$$\epsilon_j = \Sigma_j^A - \Sigma_j^U, \qquad \Sigma_n^A \simeq \Sigma_{A,n} < 1, \qquad |\epsilon_{n-k}| \ll 1 \quad \text{for weakly modified carrier channels.} \tag{1}$$

In the normalization used here, the upper saturation configuration is denoted by $\Sigma_{A,n}$. A small increment ϵ_j specifies how little this coordinate changes relative to the environment. It sets neither the background motion nor the total energy of the carrier organization to zero.

Description	Physical role
Anun in R_n	Saturated, dynamically maintained bound configuration with spatial extent.
Carrier regimes	Finer organization that supports both the Anun and its environment and responds in part to changes in them.
Free environment	Field organization that is unsaturated in the regime under consideration, with its own motion, radiation, and further underlying support.
Gravitational response	Geometry of the combined state, including interactions and environmental conditions.

At fixed depth k, the temporally or statistically averaged Anun contribution may decrease outward:

$$\overline{\epsilon}_{n-k}(r) \geq 0, \qquad \frac{d\overline{\epsilon}_{n-k}}{dr} \leq 0 \; (r > R_{A,n}), \qquad \overline{\epsilon}_{n-k}(r) \longrightarrow 0. \tag{2}$$

This condition describes the isolated comparison in its reference environment. In a galaxy, other knots and incoming fronts alter the exterior conditions. The condition alone implies neither a power law nor a sharp limit to the range.

2.2. Response to the Anun Relative to a Moving Environment

We use the following effective regime-dependent saturation equation:

$$\tau_j \partial_t \Sigma_j = -\Sigma_j + \tanh D_j, \qquad \tau_j > 0. \tag{3}$$

D_j combines local excitation with regime coupling mapped between levels; Σ_j, D_j are dimensionless. Time comparisons use the same selected observer time. With the same prescribed τ_j in both comparison states, subtraction gives exactly

$$\tau_j \partial_t \epsilon_j = -\epsilon_j + \tanh(D_j^U + \delta D_j) - \tanh D_j^U, \qquad \delta D_j = D_j^A - D_j^U. \tag{4}$$

The equation calculates the additional response about a background that is already moving. If τ_j also changes, the right-hand side acquires the term $-(\tau_j^A - \tau_j^U)\partial_t \Sigma_j^U$ when τ_j^A appears on the left-hand side.

For prescribed mapping operators satisfying $P_j[0] = Q_j[0] = 0$, the coupling relative to the reference can be written as

$$\delta D_j = \delta J_j + \kappa_j^- P_j[\epsilon_{j-1}] + \kappa_j^+ Q_j[\epsilon_{j+1}]. \tag{5}$$

P_j, Q_j map organization between neighboring regimes; δJ_j is the additional local excitation. For state-dependent mappings, their full state difference must instead be evaluated. For small δD_j, Taylor expansion gives

$$\tau_j \partial_t \epsilon_j = -\epsilon_j + \operatorname{sech}^2(D_j^U)\delta D_j + O\big((\delta D_j)^2\big). \tag{6}$$

The existing state thus determines the additional susceptibility. This expansion applies to weakly modified channels; the strongly saturated Anun organization is described by the full equation.

For a single linearized transfer path without further independent excitation, the decreasing participation can be expressed as

$$\epsilon_{n-k} = \mathsf{K}_k[\epsilon_{n-k+1}], \qquad \|\epsilon_{n-k}\| \le \left(\prod_{j=1}^{k} \eta_j\right)\|\epsilon_n\|, \qquad \|\mathsf{K}_j\| \le \eta_j < 1 \tag{7}$$

provided all participating steps can be linearized and the norms are defined comparably. Otherwise, the estimate begins only at a carrier channel that is already weakly modified. These factors concern state transfer. The gravitational contribution also depends on spatial scales, stresses, and geometric coupling. Even $\eta_j < 1$ alone does not force an infinite product to vanish.

Equation (3) is a phenomenological description of effective reorganization. Combining it with a conservative field action requires the receiving energy channels to be included. It is stated here as the existing regime description. Section 8 connects its quasistatic fixed point, for a specific input, to the common action. A finite relaxation time together with the receiving dynamical channels remains a distinct description.

2.3. Combined Gravity of the Anun and its Environment

In the static metric ansatz, we write

$$ds^2 = N^2 c^2 dt^2 - h_{ij} dx^i dx^j, \qquad \Xi = -\ln(N/N_{\mathrm{ref}}), \tag{8}$$

with a positive spatial metric h and a spatially constant clock normalization N_{ref}. For a test body initially at rest relative to a stationary observer and then released into free fall, that observer measures the initial acceleration $\mathbf{g}_{\mathrm{stat}} = -c^2 \operatorname{grad}_h \ln N = c^2 \operatorname{grad}_h \Xi$. Test bodies with general motion follow the full geodesic equation.

For the same clock normalization and corresponding locations in the comparison states, we

have

$$\Delta\Xi_A = -\ln(N_A/N_U), \qquad \Xi_A = \Xi_U + \Delta\Xi_A, \qquad \mathbf{g}_A = c^2 \operatorname{grad}_{h_A}(\Xi_U + \Delta\Xi_A). \tag{9}$$

The gravitational description includes the environment. The Anun changes its field organization; comparison with the reference does not remove the environment's existing gravity. In the weak-field region, with a shared approximately Euclidean geometry, $\delta\mathbf{g}_A \simeq c^2\nabla\Delta\Xi_A$. A positive $\Delta\Xi_A$ that decreases outward produces an inward additional acceleration.

Using the full stress-energy tensor of all independently accounted degrees of freedom and their interactions gives

$$Q_{\mathrm{tot}} = T^{\mathrm{tot}}_{\hat{0}\hat{0}} + \sum_i T^{\mathrm{tot}}_{\hat{i}\hat{i}}, \qquad \Delta_h\Xi - |\operatorname{grad}_h \Xi|^2 = -\frac{4\pi G}{c^4} Q_{\mathrm{tot}}. \tag{10}$$

The second relation follows by applying $N = N_{\mathrm{ref}} e^{-\Xi}$ to the static clock-rate equation $\Delta_h N = (4\pi G/c^4) N Q_{\mathrm{tot}}$ of the Einstein closure. Section 6 derives the weak-field limit from this relation. The total source is the target quantity of the full regime description; the explicit tensor introduced below calculates the selected effective part.

A decomposition by regime depth can be obtained without double counting through consistently refined solutions of the combined system. Let $\Xi_n^{[K]}$ be the response resolved down to depth K, with the same exterior environment and clock normalization; deeper carriers remain represented effectively in each case. Define

$$d_{n,k} = \Xi_n^{[k]} - \Xi_n^{[k-1]}, \qquad \Xi_n^{[K]} - \Xi_n^{[0]} = \sum_{k=1}^{K} d_{n,k}. \tag{11}$$

The stipulated decrease in the transferred gravitational contributions is specified, in the shared weak-field comparison region $\mathcal{D}$, by

$$a_{n,k} = \|c^2\nabla d_{n,k}\|_{\infty,\mathcal{D}}, \qquad a_{n,k+1} \le a_{n,k}, \qquad a_{n,k} \to 0, \qquad \sum_{k=1}^{\infty} a_{n,k} < \infty. \tag{12}$$

The last condition is a sufficient additional convergence requirement. These differences are resolution contributions to a combined response; the same nested energy is not added again as an independent subregime mass. The specific contributions and their directions must be determined from the coupled solutions.

3. The Effective Field Ansatz and its Interpretation

The Universal Quantum Foam Hypothesis (UQSH) describes matter, radiation, and the supporting environment as organizational forms of a continuous underlying field. Recursive support and regime-dependent coupling provide the conceptual background. The following calculation specifies a finite effective field ansatz with two variables: $q \in S^2$ represents an internal orientation, and $\zeta \in \mathbb{R}$ an additional loading and exchange channel. The spatial and temporal organization of these variables carries energy and stresses. A core denotes a locally

concentrated field state; its environment belongs to the same field.

We use the signature $(+---)$ and $x^0 = ct$. Greek indices range over spacetime, while Latin spatial indices range over three directions. A hat denotes components in a local orthonormal frame. q and ζ are dimensionless. An imagined matching surface subdivides the calculation; it does not define a regime transition.

Quantity	Meaning
q, ζ	Internal orientation and effective loading
$\mathcal{L}_f, T_{\mu\nu}$	Field action and full stress-energy tensor
$\mathcal{Q}$	Active source combination $\varepsilon + \sum_i p_i$
f, Ω	Profile and coordinate phase rate of the rotating ansatz
$\mathcal{F}_\Omega$	Stationary variational functional at fixed Ω
$N, \omega, h_{ab}, \varpi$	Clock rate, frame dragging, and spatial geometry

The mathematical results apply to the functions and state domains specified below. Their physical units are assigned in Section 23.

3.1. Regime Information Already Included in the Model

The model calculates orientation dynamics, loading, and their mutual energy exchange. Its environmental conditions capture the response of the exterior effective field. Here, ζ is a single effective state coordinate; it carries no regime index and has not yet been identified with all Σ_j or with their transfer fractions. Likewise, the core, transition region, and envelope describe spatial regions of this solution. Their radii do not automatically correspond to specified regime levels.

Connecting this model to the description across nested scales requires a mapping of the combined regime state onto q, ζ and their coefficients. At a consistent resolution, the gravitational source must reproduce the same physical response. The two variable names alone do not specify such a mapping. Section 8 defines one such mapping for exactly one scalar loading channel. The base-model core solution presented below provides the reference component with the new coupling switched off ($\eta_\Sigma = 0$). Section 17 adds the coupled spatial solution with an active regime channel. Its identification as an Anun additionally requires the regime-dependent saturation criterion and the combined carrier and environmental response from Section 2.

4. Common Action and Derivation of the Field Equations

The following action defines the base model and the existing core data. The common channel coupling in Section 8 extends only the potential term; its substitution rules apply to the correspondingly modified equations. The quartic coefficient is denoted by β_4. With $q \cdot q = 1$,

we choose

$$\mathcal{L}_f = \frac{AC(\zeta)}{2}\nabla_\mu q \cdot \nabla^\mu q - \frac{\beta_4}{4} J_{\mu\nu} \cdot J^{\mu\nu} + \frac{Z(\zeta)}{2}\nabla_\mu \zeta \nabla^\mu \zeta - V(q) - U(\zeta), \tag{13}$$

$$J_{\mu\nu} = \nabla_\mu q \times \nabla_\nu q, \qquad C = \frac{1}{1+\zeta^2}, \qquad Z = Z_0(1+\zeta^2), \tag{14}$$

$$U = \frac{b_* \zeta^4}{2(1+\zeta^2)}, \qquad V = C_s(1+q_3)^2, \qquad A, \beta_4, Z_0, b_* > 0, \quad C_s \geq 0. \tag{15}$$

The internal components of q are spacetime scalars. $J_{\mu\nu}$ measures the combined twist associated with two derivative directions. The field action is $S_f = c^{-1} \int \sqrt{-g}\, \mathcal{L}_f \, d^4x$.

An admissible variation satisfies $q \cdot \delta q = 0$. The antisymmetry of $J_{\mu\nu}$ and the product rule give

$$\frac{\partial \mathcal{L}_f}{\partial(\nabla_\mu q)} = AC\nabla^\mu q + \beta_4 J^{\mu\nu} \times \nabla_\nu q. \tag{16}$$

After integration by parts, the Euler equation is projected onto the tangent space of S^2 using $P_q = I - qq^{\mathsf{T}}$:

$$\boxed{P_q\left\{\nabla_\mu[AC\nabla^\mu q + \beta_4 J^{\mu\nu} \times \nabla_\nu q] + V_q\right\} = 0.} \tag{17}$$

Setting $X = \nabla_\mu q \cdot \nabla^\mu q$, the unconstrained variation of ζ gives

$$\boxed{Z\Box_g \zeta + \frac{Z'}{2}(\nabla\zeta)^2 - \frac{A}{2}C'X + U' = 0.} \tag{18}$$

A prime denotes differentiation with respect to ζ. In particular, $C' = -2\zeta/(1+\zeta^2)^2$ and $U' = b_*\zeta^3(\zeta^2+2)/(1+\zeta^2)^2$.

The metric variation of the same action gives

$$T_{\mu\nu} = AC\nabla_\mu q \cdot \nabla_\nu q - \beta_4 J_{\mu\alpha} \cdot J_\nu{}^\alpha + Z\nabla_\mu\zeta\nabla_\nu\zeta - g_{\mu\nu}\mathcal{L}_f. \tag{19}$$

For the metric closure at $\Lambda = 0$, we impose [12]:

$$\boxed{G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}.} \tag{20}$$

This Einstein form is retained as the geometric law of gravity. Variation of (13) determines the field equations and their source. With the extended channel, the full tensor from (84) takes its place. The geometric equation retains the same form. On solutions, $\nabla_\mu T^{\mu\nu} = 0$ holds, consistently with the Bianchi identity.

5. Energy Exchange and the Gravitational Source

In an orthonormal frame, let $v = \partial_{\hat{0}} q$, $a_i = \partial_{\hat{i}} q$, $w = \partial_{\hat{0}}\zeta$, and $z_i = \partial_{\hat{i}}\zeta$. Define

$$\mathcal{K}_2 = \frac{AC}{2}|v|^2 + \frac{Z}{2}w^2, \qquad \mathcal{G}_2 = \frac{AC}{2}\sum_i |a_i|^2 + \frac{Z}{2}\sum_i z_i^2, \tag{21}$$

$$\mathcal{E}_4 = \frac{\beta_4}{2}\left[\sum_i |v \times a_i|^2 + \sum_{i<j} |a_i \times a_j|^2\right], \qquad \mathcal{W} = U + V. \tag{22}$$

Equation (19) gives the energy density and spatial trace:

$$\varepsilon = \mathcal{K}_2 + \mathcal{G}_2 + \mathcal{E}_4 + \mathcal{W}, \tag{23}$$

$$\sum_i p_i = 3\mathcal{K}_2 - \mathcal{G}_2 + \mathcal{E}_4 - 3\mathcal{W}, \tag{24}$$

$$\boxed{\mathcal{Q} := \varepsilon + \sum_i p_i = 4\mathcal{K}_2 + 2\mathcal{E}_4 - 2\mathcal{W}.} \tag{25}$$

The gravitational source is thus expressed in terms of the field variables. The quadratic spatial contributions cancel in this combination; they remain relevant to the profiles and the other tensor components. Depending on the organization, $\mathcal{Q}$ may have either sign. A positive energy density alone does not determine its sign.

In flat space, we partition the energy as

$$\varepsilon_q = \frac{AC}{2}(|v|^2 + \sum_i |a_i|^2) + \mathcal{E}_4 + V, \qquad \varepsilon_\zeta = \frac{Z}{2}(w^2 + |\nabla\zeta|^2) + U. \tag{26}$$

Substitution of the field equations and application of the product rule give

$$\partial_0\varepsilon_q + \partial_i j_q^i = -\mathcal{P}_0, \qquad \partial_0\varepsilon_\zeta + \partial_i j_\zeta^i = \mathcal{P}_0, \tag{27}$$

$$\mathcal{P}_0 = \frac{A}{2}C'Xw, \qquad j_\zeta^i = -Zw\partial_i\zeta, \tag{28}$$

$$j_q^i = -ACv\cdot a_i - \beta_4\sum_j (v\times a_j)\cdot(a_i\times a_j). \tag{29}$$

In physical time, $\mathbf{S} = c\mathbf{j}$ and $\mathcal{P}_\zeta = c\mathcal{P}_0 = (A/2)C'X\dot{\zeta}$. The exchange cancels from the total balance and permits both transfer out and transfer back. Bound field energy is counted once in the full tensor.

For weak, approximately stationary gravity with $g_{00} = 1 + 2\Psi/c^2 + \cdots$, the temporal Einstein component gives

$$\nabla^2\Psi = \frac{4\pi G}{c^2}\mathcal{Q}, \qquad \mathbf{g}_{\text{slow}} = -\nabla\Psi. \tag{30}$$

Energy and stresses are evaluated before any averaging. Calculating the full deflection of light requires the entire metric from (20).

5.1. Twist Direction, a Shared Axis, and Stress Moments

The spatial twist can be directly related to the quartic energy and stress terms:

$$F_{ij} = q\cdot(\partial_{\hat{i}}q\times\partial_{\hat{j}}q), \qquad \mathbf{b} = (F_{23}, F_{31}, F_{12}). \tag{31}$$

Because $q \cdot q = 1$, the derivatives are tangential and $\partial_{\hat{i}} q \times \partial_{\hat{j}} q = F_{ij} q$. The purely spatial twist contribution therefore satisfies

$$\varepsilon_{\mathrm{V}} = \frac{\beta_4}{2}|\mathbf{b}|^2, \qquad \Pi^{\mathrm{V}}_{ij} = \beta_4 \left(\frac{1}{2}|\mathbf{b}|^2 \delta_{ij} - b_i b_j \right). \tag{32}$$

These are contributions already contained in the same tensor. Time-dependent organization additionally includes its J_{0i} contributions and momentum fluxes.

For a positive normalized averaging window and a common spatial reference frame, let

$$\mathbf{m} = \langle \mathbf{b} \rangle, \qquad \mathsf{W} = \langle \mathbf{b}\mathbf{b}^{\mathsf{T}} \rangle, \qquad \mathsf{W} - \mathbf{m}\mathbf{m}^{\mathsf{T}} \geq 0. \tag{33}$$

For constant β_4, one obtains

$$\overline{\varepsilon}_{\mathrm{V}} = \frac{\beta_4}{2} \operatorname{tr} \mathsf{W}, \tag{34}$$

$$\overline{\Pi}^{\mathrm{V}}_{ij} = \beta_4 \left[\frac{1}{2} \operatorname{tr} \mathsf{W}\, \delta_{ij} - W_{ij} \right], \tag{35}$$

$$\overline{\Pi}^{\mathrm{V,TF}}_{ij} = -\beta_4 \left[W_{ij} - \frac{1}{3} \operatorname{tr} \mathsf{W}\, \delta_{ij} \right]. \tag{36}$$

Oppositely directed contributions can retain a shared axis even when $\mathbf{m} = 0$. A separated eigenvalue of W identifies the axis; the directed residual $\mathbf{m}$ contains additional polarity information. Energy and stress therefore require the second moment.

For a prescribed linear transfer $\mathbf{b}_{\mathrm{an}} = \mathsf{L}\mathbf{b}_{\mathrm{aus}}$, $\mathbf{m}_{\mathrm{an}} = \mathsf{L}\mathbf{m}_{\mathrm{aus}}$ and $\mathsf{W}_{\mathrm{an}} = \mathsf{L}\mathsf{W}_{\mathrm{aus}}\mathsf{L}^{\mathsf{T}}$ hold. Multiple contributions transferred together also require their cross moments. These identities do not yet determine a spatial transfer operator. Relating the internal $\mathbf{b}$ to a magnetic field in teslas, charge, and induction remains part of the coupling to electromagnetic measurements. The scalar channel in Section 8 adds loading; directional information continues to be carried by q and the moments specified above.

6. Weak Gravity: Motion, Range, and Strength

6.1. How a Weak Gravitational Field Produces Motion

A weak gravitational field represents a small deviation of the measurement geometry from the chosen approximately flat reference. In the quasistatic region with negligible frame dragging,

$$N = 1 + \Psi/c^2 + O(\Psi^2/c^4), \qquad |\Psi|/c^2 \ll 1, \qquad h_{ij} \simeq \delta_{ij}. \tag{37}$$

The weak geometric response may occur far outside a strongly organized or saturated center. It requires neither the absence of subregimes nor an environment at rest or devoid of energy.

Using $\Xi = -\Psi/c^2 + \cdots$ in (10) and dropping the quadratic geometric terms gives

$$\nabla^2 \Psi = \frac{4\pi G}{c^2} \boldsymbol{Q}. \tag{38}$$

The full tensor at the chosen resolution is used here; in the explicit two-field model, $\boldsymbol{Q}$ is given by (25). The source contains energy and stresses. For ordinary, slowly moving matter with small pressures, approximately $\boldsymbol{Q} \simeq \rho c^2$, giving the Newtonian Poisson equation $\nabla^2\Psi = 4\pi G\rho$.

In the selected metric model, the equation of motion follows from the test-body action $S_{\mathrm{p}} = -mc^2 \int d\tau$. For $v \ll c$, the same metric yields

$$\frac{d\tau}{dt} = \sqrt{N^2 - v^2/c^2} \simeq 1 + \frac{\Psi}{c^2} - \frac{v^2}{2c^2}, \tag{39}$$

$$L_{\mathrm{p}} \simeq -mc^2 + \frac{1}{2}mv^2 - m\Psi, \qquad \frac{d^2\mathbf{x}}{dt^2} = -\nabla\Psi. \tag{40}$$

The mass m cancels from the equation of motion. This demonstrates universal free fall within the assumed metric coupling of the test body. It does not yet separately derive this coupling from internal field reorganization.

For a weak, spherically symmetric source, integration of (38) gives

$$M_{\mathrm{dyn}}(r) = \frac{4\pi}{c^2}\int_0^r \overline{\boldsymbol{Q}}(s)s^2\,ds, \qquad g_r = -\frac{GM_{\mathrm{dyn}}(r)}{r^2}. \tag{41}$$

Regularity at the center fixes the integration constant. For a strongly gravitating interior, the weak exterior solution is instead matched to its gravitational mass. For a positive enclosed source, g_r points inward. Outside a compact source, M_{dyn} is constant, and the $1/r^2$ law of Newtonian gravity emerges [13]. Here, it follows from the field equation and spherical symmetry; the dilution of radiation intensity over spherical surfaces would not, by itself, derive this law.

6.2. Environmental Contribution and Weak Coupling Strength

In the weak-field comparison on the same spatial reference, the two coupled sources $\boldsymbol{Q}_A, \boldsymbol{Q}_U$ are used:

$$\delta\boldsymbol{Q}_A = \boldsymbol{Q}_A - \boldsymbol{Q}_U, \qquad \nabla^2\delta\Psi_A = \frac{4\pi G}{c^2}\delta\boldsymbol{Q}_A, \qquad \mathbf{g}_A \simeq \mathbf{g}_U - \nabla\delta\Psi_A. \tag{42}$$

$\delta\boldsymbol{Q}_A$ also includes the reorganization of the environment caused by the Anun. The actual background response $\mathbf{g}_U$ remains included. The isolated limiting case with a decaying potential has the solution

$$\Psi(\mathbf{x}) = -\frac{G}{c^2}\int \frac{\boldsymbol{Q}(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|}\,d^3x'. \tag{43}$$

In an extended environment, its boundary conditions or a suitable background geometry must instead be included. The static expression describes a state that has already been established; it does not replace a propagation equation for temporal changes.

A distributed positive exterior source increases the enclosed active mass with r. For example,

$\overline{Q} = D_H/r^2$ in a finite exterior interval provides a contribution

$$M_{\mathrm{dyn}}(r) = M_0 + \frac{4\pi D_H}{c^2}(r - r_0), \qquad v_c^2 = \frac{GM_{\mathrm{dyn}}(r)}{r} \longrightarrow \frac{4\pi G D_H}{c^2} \quad (D_H > 0). \tag{44}$$

The arrow denotes the limiting behavior when the envelope contribution dominates within the interval of validity. Section 18 determines this source profile for the weak wave envelope chosen there. It is not a general law for every environment or every carrier regime.

Why can the gravitational response be small? The UQSH interpretation relates the small effective strength to the spatially extended contribution of the tension organization and its regime-dependent coupling. Only the change that acts jointly within this response channel determines the observed response. Large internal motion can coexist with small additional reorganization of deeper carriers and a small change in the exterior geometry. Contributions from more distant regime levels are transferred in part; the number and existence of the levels alone do not determine their strength.

What has been determined quantitatively? The model calculates the weak geometric response for a given G, a prescribed field action, and a prescribed source. Explaining why gravitational coupling has its observed small value relative to other interactions additionally requires a joint derivation of their couplings and measurement scales. Regime transfers and G have not yet been calculated in this way. The qualitative UQSH interpretation and the derived weak-field law thus have clearly identified, different scopes.

7. Source Difference and Additional Gravity

The weak additional gravitational effect of a reorganized field configuration is determined relative to its already moving environment. We therefore compare two corresponding states of the same field theory: the environmental state U and the reorganized state A. Both use the same model parameters, the same exterior time normalization, and mutually compatible environmental data. We denote their finite difference by

$$\Delta_A Y := Y^{(A)} - Y^{(U)} \tag{45}$$

The fields are evaluated together in each state; the nonlinear source also includes the modified environment.

7.1. The Source Difference in Terms of the Existing Field Variables

In the local orthonormal frame with signature $(+---)$, the relevant contributions to the action are

$$\mathcal{K}_2 = \frac{AC}{2}k + \frac{Z}{2}\mathcal{T}_\zeta, \qquad k = |\partial_{\hat{0}} q|^2, \qquad \mathcal{T}_\zeta = (\partial_{\hat{0}}\zeta)^2, \tag{46}$$

$$\mathcal{E}_4 = \frac{\beta_4}{2}\left[\sum_i |\partial_{\hat{0}} q \times \partial_{\hat{i}} q|^2 + \sum_{i<j} |\partial_{\hat{i}} q \times \partial_{\hat{j}} q|^2\right]. \tag{47}$$

Here, $q \cdot q = 1$. The base-model couplings remain

$$C = \frac{1}{1+\zeta^2}, \quad Z = Z_0(1+\zeta^2), \quad U = \frac{b_*\zeta^4}{2(1+\zeta^2)}, \quad V = C_s(1+q_3)^2. \tag{48}$$

The letter A without a state argument in parentheses denotes the coupling parameter. The active gravitational source is

$$\mathcal{Q} = \varepsilon + \sum_i p_i = 4\mathcal{K}_2 + 2\mathcal{E}_4 - 2(U+V). \tag{49}$$

The finite source difference is

$$\boxed{\Delta_A\mathcal{Q} = 2A\Delta_A(Ck) + 2\Delta_A(Z\mathcal{T}_\zeta) + 2\Delta_A\mathcal{E}_4 - 2\Delta_A V - 2\Delta_A U.} \tag{50}$$

It separates changes in internal motion, loading dynamics, quartic field organization, and potentials. The quadratic spatial-gradient contributions cancel in $\varepsilon + \sum_i p_i$. They remain effective through the field equations, the other tensor components, and the reorganization. Any averaging is performed after these nonlinear quantities have been constructed.

For loading that is constant in time in the chosen observer frame, $\mathcal{T}_\zeta = 0$. The orientation field may still rotate. Setting $u = 1 + \zeta^2 \geq 1$ gives

$$\mathcal{Q} = \frac{2Ak}{u} + 2\mathcal{E}_4 - 2V - b_*\left(u - 2 + \frac{1}{u}\right). \tag{51}$$

This formula does not yet require spatial loading gradients to be neglected. In particular,

$$\boxed{\begin{aligned}\Delta_A\mathcal{Q} = 2A\left(\frac{k^{(A)}}{u^{(A)}} - \frac{k^{(U)}}{u^{(U)}}\right) + 2\Delta_A\mathcal{E}_4 - 2\Delta_A V \\ - b_*\left[u^{(A)} - u^{(U)} + \frac{1}{u^{(A)}} - \frac{1}{u^{(U)}}\right].\end{aligned}} \tag{52}$$

The environmental state may already carry motion and stress in every contribution. The finite difference does not require a small change in loading.

7.2. First Change and Locally Adjusted Loading

For small changes about a selected reference state, δ now denotes the first variation. Equation (51) gives

$$\boxed{\delta\mathcal{Q} = \frac{2A}{u}\,\delta k + 2\delta\mathcal{E}_4 - 2\delta V - D_u\,\delta u, \qquad D_u = \frac{2Ak}{u^2} + b_*(1-u^{-2}) \geq 0.} \tag{53}$$

All coefficients are evaluated at the reference state. With $\delta u = 2\zeta\,\delta\zeta$, the change in loading is included directly. Its direct contribution can be outweighed by the simultaneous change in orientation.

On the locally adjusted loaded branch already used, with small loading derivatives, we

additionally have

$$u_* = \sqrt{1 + \frac{A(S-k)}{b_*}}, \qquad S = \sum_i |\partial_{\hat{i}} q|^2, \qquad S - k > 0. \tag{54}$$

Within this branch,

$$\delta u_* = \frac{A}{2 b_* u_*} (\delta S - \delta k), \tag{55}$$

$$\delta \mathcal{Q}_{\mathrm{loc}} = \left(\frac{2A}{u_*} + \frac{A D_u}{2 b_* u_*} \right) \delta k - \frac{A D_u}{2 b_* u_*} \delta S + 2\delta \mathcal{E}_4 - 2\delta V. \tag{56}$$

These quantities change together with the field. The formula provides a local readout of such a change and connects to the geometric backreaction on $S - k$ already formulated. In transition regions with significant loading gradients, the full loading equation is included.

7.3. Numerical Comparison

The completed core calculation contains an unloaded state and a coupled loaded state with the same boundary data. It uses $L = 1$, $f(L, \theta) = b_\partial \sin\theta$, $\zeta(L, \theta) = 0$, fixed Ω, and $b_\partial = b_{\partial,c} + 0.02$. The following values come from this comparison in model units:

Central quantity	Unloaded	Loaded
a in $f = aR + O(r^3)$	2.44848522	2.45517343
ζ_0	0	0.12499957
$\mathcal{Q}_0$	4.48759425	4.53658784

For $\beta_4 = 0.1248600866454894$ and $b_* = 1$,

$$\mathcal{Q}_0 = \beta_4 a^4 - \frac{b_* \zeta_0^4}{1 + \zeta_0^2}. \tag{57}$$

The difference therefore separates into

$$\Delta \mathcal{Q}_0 = \underbrace{0.0492339753}_{\text{orientation change}} - \underbrace{0.0002403813}_{\text{direct loading correction}} = 0.0489935940. \tag{58}$$

The central source increases by approximately 1.09 %. This comparison demonstrates why the effect of the reorganized configuration is determined from all contributions. It provides a completed core calculation; selecting a maximally saturated Anun configuration constitutes a separate physical application.

7.4. From the Source Difference to the Additional Acceleration

In the shared weak, approximately stationary metric comparison on the same leading-order spatial geometry,

$$\boxed{\nabla^2 \Delta_A \Psi = \frac{4\pi G}{c^2} \Delta_A Q, \qquad \Delta_A \mathbf{g} = -\nabla \Delta_A \Psi.} \tag{59}$$

The total acceleration remains $\mathbf{g}^{(A)} = \mathbf{g}^{(U)} + \Delta_A \mathbf{g}$. The weak-field metric expansion does not itself require linearization of the underlying field organization.

For a regular, spherically averaged difference without an additional central point source,

$$\boxed{\Delta_A g_r(r) = -\frac{4\pi G}{c^2 r^2} \int_0^r s^2 \Delta_A \overline{Q}(s)\, ds.} \tag{60}$$

A positive enclosed source difference produces an additional inward contribution. Angularly dependent sources are evaluated through their corresponding potential response. When only an exterior region is treated, the interior matching data must be included:

$$\Delta_A g_r(r) = -\frac{G \Delta_A M_m}{r^2} - \frac{4\pi G}{c^2 r^2} \int_{r_m}^r s^2 \Delta_A \overline{Q}(s)\, ds. \tag{61}$$

$\Delta_A M_m$ is then the dynamical source parameter determined by the interior matching.

7.5. Environmental Response and the Next Carrier Regime

The coupled variational problem is already available for spatial matching:

$$\delta(\mathcal{F}_{\Omega,\mathrm{in}} + \Gamma_{\mathrm{env}}[\gamma]) = 0, \qquad \gamma = (f, \zeta)|_{\Gamma_m}. \tag{62}$$

Γ_{env} is the exterior field component of the same theory, eliminated under stationary conditions. It includes the response of the spatial environment and, together with the interior, determines boundary values and normal fluxes. This spatial partition must be distinguished from a partition into regime scales.

For a further resolved carrier level, the coupled Hessian already used provides the appropriate procedure. For small quasistatic changes x in the observed organization and y in independent carrier variables, write

$$\begin{pmatrix} \mathsf{H}_{xx} & \mathsf{H}_{xy} \\ \mathsf{H}_{yx} & \mathsf{H}_{yy} \end{pmatrix} \begin{pmatrix} \delta x \\ \delta y \end{pmatrix} = \begin{pmatrix} j_x \\ j_y \end{pmatrix}, \tag{63}$$

If H_{yy} is invertible in the variational space under consideration, this gives

$$\left(\mathsf{H}_{xx} - \mathsf{H}_{xy} \mathsf{H}_{yy}^{-1} \mathsf{H}_{yx} \right) \delta x = j_x - \mathsf{H}_{xy} \mathsf{H}_{yy}^{-1} j_y. \tag{64}$$

This is the same elimination structure used to evaluate the orientation backreaction on the loading. For a carrier regime, the blocks are determined from the same action governing that regime and from a specified scale mapping. The formula gives the procedure; it does not automatically turn the existing variables q, ζ into two fully resolved regimes.

Scope and combined balance. Both field regions are evaluated using (50) and (52), respectively, together with (59). The comparison states and their boundary data determine the specific source profile. The simple Poisson difference applies at the specified weak-field metric order; a strongly curved interior is treated through the full geometric system and its exterior matching. General time-dependent effects require the corresponding dynamical metric response.

When a carrier level is added explicitly, the effects previously represented in C, Z, U or in the effective environmental data are identified. This ensures that the total energy is counted exactly once. The phenomenological saturation coordinate Σ_n and its tanh relation are not identified with ζ without an additional mapping. The metric closure and G remain the chosen basis of the coupling in this step.

The source is derived in Section 5, the local adjustment in Section 14, and the numerical comparison in Section 16. Spatial matching follows Section 19; geometric backreaction follows Section 20.

7.6. From the Carrier Response to the Combined Source Response

Eliminating the internal response also determines the first change in the gravitational source. The independent variables x, y and all coupling contributions are therefore included in a single stress-energy tensor. On the fixed leading-order comparison geometry, let $\boldsymbol{Q}_x, \boldsymbol{Q}_y$ be the linear derivative operators of the resulting source. They may contain spatial derivatives. Then

$$\delta\boldsymbol{Q} = \boldsymbol{Q}_x \delta x + \boldsymbol{Q}_y \delta y, \qquad \delta y = \mathsf{H}_{yy}^{-1}(j_y - \mathsf{H}_{yx}\delta x). \tag{65}$$

Substitution gives the desired source response:

$$\boxed{\delta\boldsymbol{Q} = (\boldsymbol{Q}_x - \boldsymbol{Q}_y \mathsf{H}_{yy}^{-1}\mathsf{H}_{yx})\delta x + \boldsymbol{Q}_y \mathsf{H}_{yy}^{-1} j_y.} \tag{66}$$

With $\mathsf{H}_{\mathrm{eff}} = \mathsf{H}_{xx} - \mathsf{H}_{xy}\mathsf{H}_{yy}^{-1}\mathsf{H}_{yx}$ and invertible $\mathsf{H}_{\mathrm{eff}}$, the full expression is

$$\delta\boldsymbol{Q} = (\boldsymbol{Q}_x - \boldsymbol{Q}_y \mathsf{H}_{yy}^{-1}\mathsf{H}_{yx})\mathsf{H}_{\mathrm{eff}}^{-1}(j_x - \mathsf{H}_{xy}\mathsf{H}_{yy}^{-1} j_y) + \boldsymbol{Q}_y \mathsf{H}_{yy}^{-1} j_y. \tag{67}$$

The order of the operators is essential. Boundary conditions, constraints, and the treatment of zero modes are part of the specified inverses. This quasistatic first variation extends the finite source difference to include the carrier response.

For $j_y = 0$ and positive H_{yy}, $\mathsf{H}_{xy}\mathsf{H}_{yy}^{-1}\mathsf{H}_{yx}$ is positive semidefinite when the coupling blocks are adjoints. The internal adjustment therefore reduces the corresponding effective stiffness. Whether the active gravitational source increases additionally depends on $\boldsymbol{Q}_x, \boldsymbol{Q}_y$ and the excitation. The source operators are determined from the full tensor; the energy Hessian alone does not replace this evaluation.

A condition for enhancement. For a single real response channel, let $\mathsf{H}_{xx} = a$, $\mathsf{H}_{yy} = d$, $\mathsf{H}_{xy} = \mathsf{H}_{yx} = b$, and $\delta\boldsymbol{Q} = a_Q \delta x + b_Q \delta y$. For $a, d > 0$, $ad > b^2$, and $j_y = 0$, one obtains

$$\delta\boldsymbol{Q}_{\mathrm{adjusted}} = \frac{a_Q d - b_Q b}{ad - b^2}\, j_x, \qquad \delta\boldsymbol{Q}_{\mathrm{fixed}} = \frac{a_Q}{a}\, j_x, \tag{68}$$

$$\delta Q_{\text{adjusted}} - \delta Q_{\text{fixed}} = \frac{b(a_Q b - a b_Q)}{a(ad - b^2)}\, j_x. \tag{69}$$

For $j_x > 0$, the source increases relative to the same channel with a fixed carrier if and only if $b(a_Q b - a b_Q) > 0$. This is a derived condition on the coupling and source coefficients that remain to be determined, not an imposed enhancement factor. The exterior gravitational response is then governed by the spatially enclosed source difference according to (60).

In simple terms, the carrier changes both the responsiveness of the observed organization and the total gravitational source. The two changes are calculated together. For a particular subregime, the following section supplies a concrete potential channel and its operators. The full scale mapping is specified during physical calibration.

8. A Regime Channel from the Same Field Action

8.1. Channel Variables and Energy

The quantities q and ζ remain the dynamical fields. An additional scalar organizational channel is now coupled to the loading potential $U(\zeta)$. Its response is distributed across neighboring regime levels. We use dimensionless auxiliary variables $\Sigma_n \in (-1, 1)$, $n \in \mathbb{Z}$, and exactly one directly excited input at $n = 0$. The zero index specifies the reference level of this channel; it denotes neither a spatial center nor a particular radius. After a common scale mapping, all quantities are evaluated at the same spacetime point.

The specific model choice is

$$h_n = h\,\delta_{n0}, \qquad h = \frac{\eta_\Sigma U(\zeta)}{\mathcal{E}_\Sigma}, \qquad 0 \le \eta_\Sigma < 1, \quad \mathcal{E}_\Sigma > 0, \quad 0 \le \kappa < \frac{1}{2}. \tag{70}$$

η_Σ is a dimensionless participation factor, $\mathcal{E}_\Sigma$ a constant energy density, and κ the symmetric nearest-neighbor coupling in the mapped coordinates. Choosing the potential as the input preserves the existing kinetic terms. It describes a quasistatically adjusted organizational contribution to the loading. The general regime hypothesis also allows other inputs; the present study examines this potential coupling.

Define the dimensionless function

$$\phi(s) = \frac{(1+s)\ln(1+s) + (1-s)\ln(1-s)}{2}, \qquad \phi'(s) = \operatorname{artanh} s, \tag{71}$$

$$\mathcal{V}_\kappa[\Sigma] = \sum_n \phi(\Sigma_n) - \kappa \sum_n \Sigma_n \Sigma_{n+1}. \tag{72}$$

The sums belong to the energy of a single effective channel. They do not recount masses already contained in q, ζ. For the localized input, we use a square-summable response that tends to zero along the regime sequence.

The field Lagrangian density $\mathcal{L}_f$ acquires the contribution

$$\boxed{\mathcal{L}_{\text{tot}} = \mathcal{L}_f - \mathcal{E}_\Sigma \mathcal{V}_\kappa[\Sigma] + \eta_\Sigma U(\zeta)\Sigma_0.} \tag{73}$$

The corresponding energy change at a fixed field state is

$$e_{\mathrm{K}} = \mathcal{E}_\Sigma \mathcal{V}_\kappa - \eta_\Sigma U \Sigma_0. \tag{74}$$

The first term is the organizational cost; the second describes its interaction with the loading. Both must be considered together. The original U is used as the potential before this channel is explicitly adjusted. The calculated adjustment then replaces part of this potential. This specification prevents the energy already adjusted from being counted again.

8.2. Bounded Response and Unique Adjustment

Variation with respect to Σ_n gives

$$\operatorname{artanh} \Sigma_n - \kappa(\Sigma_{n-1} + \Sigma_{n+1}) = h\delta_{n0}, \qquad \boxed{\Sigma_n = \tanh[h\delta_{n0} + \kappa(\Sigma_{n-1} + \Sigma_{n+1})].} \tag{75}$$

The regime law thus follows from the action. The right-hand side is a contraction in the supremum norm, with a factor no larger than $2\kappa < 1$. The Hessian of the channel cost also satisfies

$$(\mathsf{H}_\Sigma)_{nm} = \frac{\delta_{nm}}{1 - \Sigma_n^2} - \kappa(\delta_{n,m+1} + \delta_{n,m-1}), \qquad \mathsf{H}_\Sigma \geq (1 - 2\kappa) I. \tag{76}$$

The adjustment is therefore unique and minimizes the channel energy at fixed q, ζ. This statement concerns the auxiliary channel; the stability of the spatial fields is determined by their Hessian.

Linearization about the unloaded state gives

$$\Sigma_n = \frac{h}{\sqrt{1 - 4\kappa^2}}\, r_\kappa^{|n|} + O(h^3), \qquad r_\kappa = \frac{2\kappa}{1 + \sqrt{1 - 4\kappa^2}}. \tag{77}$$

For $\kappa = 0$, only the directly excited level remains. The decay concerns participation along the regime sequence. The spatial range continues to follow from q, ζ, the environment, and the metric.

The auxiliary variables are not assigned their own spacetime kinetic terms here. Equation (75) is a local quasistatic adjustment rule. It does not provide a relaxation time or a causal transport speed between regimes. Its use is restricted to processes in which the omitted adjustment is sufficiently fast relative to the field change being considered.

8.3. Elimination and a Positive Adjusted Potential

Define the uniquely determined function

$$W_\kappa(h) = \sup_\Sigma \{h\Sigma_0 - \mathcal{V}_\kappa[\Sigma]\}, \qquad W_\kappa'(h) = \Sigma_0(h), \qquad W_\kappa''(h) = (\mathsf{H}_\Sigma^{-1})_{00} > 0. \tag{78}$$

After adjustment, $e_{\mathrm{K}} = -\mathcal{E}_\Sigma W_\kappa(h)$. The total field action can equivalently be written with

$$\boxed{U_{\mathrm{eff}}(\zeta) = U(\zeta) - \mathcal{E}_\Sigma W_\kappa\left(\frac{\eta_\Sigma U(\zeta)}{\mathcal{E}_\Sigma}\right)} \tag{79}$$

in place of U. The explicit auxiliary channel and the eliminated potential are two forms of the same calculation.

From $\phi(s) \geq s^2/2$ and $|\sum_n \Sigma_n \Sigma_{n+1}| \leq \sum_n \Sigma_n^2$, it follows that $\mathcal{V}_\kappa \geq (1/2 - \kappa) \sum_n \Sigma_n^2 \geq 0$. For $h \geq 0$, we therefore have

$$0 \leq W_\kappa(h) \leq h, \qquad \boxed{(1 - \eta_\Sigma)U \leq U_{\text{eff}} \leq U}, \qquad \frac{dU_{\text{eff}}}{dU} = 1 - \eta_\Sigma \Sigma_0(h). \tag{80}$$

The adjusted potential remains nonnegative. As U grows, it continues to grow; the bounded coordinate Σ does not impose an absolute upper limit on energy. For $\kappa = 0$, the exactly solvable limiting case gives

$$\Sigma_0 = \tanh h, \qquad W_0(h) = \ln \cosh h. \tag{81}$$

8.4. Backreaction and the Stress-energy Tensor

The orientation equation retains its base-model form. The loading equation becomes directly

$$\boxed{Z \Box_g \zeta + \frac{Z'}{2}(\nabla\zeta)^2 - \frac{A}{2} C' X + (1 - \eta_\Sigma \Sigma_0) U' = 0.} \tag{82}$$

A change in ζ continues to act back on the coupled field organization through C, Z. The second potential derivative displays the additional adjustment response:

$$U''_{\text{eff}} = (1 - \eta_\Sigma \Sigma_0) U'' - \frac{\eta_\Sigma^2}{\mathcal{E}_\Sigma} (\mathsf{H}_\Sigma^{-1})_{00} (U')^2. \tag{83}$$

The second term is the specific Schur complement of this channel. It reduces the corresponding local stiffness; spatial gradients and the orientation response remain included in the full stability calculation.

Because the new coupling contains no metric derivatives, the same action gives the tensor

$$\boxed{T^{\text{tot}}_{\mu\nu} = T^f_{\mu\nu} + g_{\mu\nu} e_{\text{K}} = T^f_{\mu\nu} - g_{\mu\nu} \mathcal{E}_\Sigma W_\kappa(h).} \tag{84}$$

The last equality holds after the auxiliary channel has adjusted. GR uses this full tensor: $G_{\mu\nu} = (8\pi G/c^4) T^{\text{tot}}_{\mu\nu}$. For the same instantaneous fields, in a local orthonormal frame,

$$\delta_\Sigma \varepsilon = -\mathcal{E}_\Sigma W_\kappa, \qquad \delta_\Sigma p_i = +\mathcal{E}_\Sigma W_\kappa, \qquad \boxed{\mathcal{Q}_{\text{tot}} = \mathcal{Q}_f + 2\mathcal{E}_\Sigma W_\kappa(h).} \tag{85}$$

The active source combination grows here through the combined change in energy and pressure. It is not an additionally generated positive energy density.

8.5. Energy Balance and a Moving Reference Environment

In flat geometry and physical time, the adjusted regime response gives

$$\partial_t \varepsilon_f + \nabla \cdot \mathbf{S}_f = +\eta_\Sigma \Sigma_0 U' \dot{\zeta}, \tag{86}$$

$$\partial_t e_{\mathrm{K}} = -\eta_\Sigma \Sigma_0 U' \dot{\zeta}, \tag{87}$$

$$\partial_t(\varepsilon_f + e_{\mathrm{K}}) + \nabla\cdot \mathbf{S}_f = 0. \tag{88}$$

Here, ε_f is evaluated using the original U. In curved spacetime, $\nabla_\mu T^{\mu\nu}_{\mathrm{tot}} = 0$ likewise follows from the coupled field equations. The conservative exchange describes both transfer out and transfer back; an irreversible direction additionally requires a physical description of relaxation.

Now compare the already moving environment U with the reorganized Anun state A. The letter U as a state index remains distinct from the potential $U(\zeta)$. For identical parameters, exactly

$$\boxed{\Delta_A \boldsymbol{Q}_{\mathrm{tot}} = \Delta_A \boldsymbol{Q}_f + 2\mathcal{E}_\Sigma[W_\kappa(h_A) - W_\kappa(h_U)].} \tag{89}$$

The channel adjustment already present in the environment is included. On a common weak-field comparison geometry, this gives

$$\nabla^2 \delta\Psi_A = \frac{4\pi G}{c^2}\Delta_A \boldsymbol{Q}_{\mathrm{tot}}, \qquad \delta\mathbf{g}_A = -\nabla\delta\Psi_A. \tag{90}$$

The sign of the full change also depends on the adjusted fields in $\Delta_A \boldsymbol{Q}_f$. For the exterior effect, the spatial source difference is integrated and matched to the environment.

8.6. Preserved Threshold and Free Fronts

For small excitation,

$$W_\kappa(h) = \frac{h^2}{2\sqrt{1-4\kappa^2}} + O(h^4), \tag{91}$$

$$U_{\mathrm{eff}} = U - \frac{\eta_\Sigma^2 U^2}{2\mathcal{E}_\Sigma\sqrt{1-4\kappa^2}} + O(U^4). \tag{92}$$

With $U = b_*\zeta^4/[2(1+\zeta^2)]$, the additional correction starts at

$$U_{\mathrm{eff}} - U = -\frac{\eta_\Sigma^2 b_*^2}{8\mathcal{E}_\Sigma\sqrt{1-4\kappa^2}}\zeta^8 + O(\zeta^{10}). \tag{93}$$

At fixed parameters, the unloaded reference, the linear loading threshold, and the quartic amplitude coefficient are therefore preserved. The adjustment enters only at higher orders in loading. The quadratic free-vacuum limit and its speed c also remain unchanged. The free spherical solution continues to preserve the intervals between corresponding pulses. The new quasistatic channel alone does not specify any additional temporal thinning.

8.7. Local Comparison at the Reference Core

For a controlled constitutive comparison, the reference field state and its environmental data are held fixed. Its center is evaluated using $\zeta_c = 0.12499956848002458$, $b_* = 1$, and $\boldsymbol{Q}_{f,c} = 4.536587839739102$ in the existing dimensionless normalization. For this example

alone, we choose

$$\eta_\Sigma = \frac{1}{2}, \qquad \kappa = 0.2, \qquad \mathcal{E}_\Sigma = \frac{1}{2}U(\zeta_c) = 6.00953303877 \cdot 10^{-5}. \tag{94}$$

Once specified in this way, $\mathcal{E}_\Sigma$ is constant; it is not readjusted to ζ during variations. At the reference center, $h_c = 1$. The algebraic regime chain is solved on $-20 \le n \le 20$ with vanishing exterior neighbors. These are auxiliary indices, not spatial grid points.

Quantity	Value
Σ_0	0.787624239194
$\Sigma_{\pm1}$	0.162857017803
$\Sigma_{\pm2}$	0.0339765595633
$\mathcal{V}_\kappa$	0.328902952417
$W_\kappa(1)$	0.458721286777
$U(\zeta_c)$	$1.20190660775 \cdot 10^{-4}$
$U_{\mathrm{eff}}(\zeta_c)$	$9.26236534906 \cdot 10^{-5}$
$\delta_\Sigma \mathcal{Q}_c$	$5.51340145695 \cdot 10^{-5}$
$\mathcal{Q}_{\mathrm{tot},c}$ at fixed field	4.53664297375
Relative source change at fixed field	0.00121531901 %

The calculation determines the immediate energy, pressure, and source changes at the identical state. The subsequent spatial adjustment of q, ζ is solved as a coupled problem in Section 17. The initial profiles used here belong to the base model with $\eta_\Sigma = 0$; the local numerical value describes the immediate channel contribution at identical fields.

The interpretation of the existing results remains as follows: in the base model, the central increase of the loaded core relative to the unloaded reference is approximately 1.09176 %; the increase integrated over the calculated sphere is 0.0177447 %. These two comparisons are distinct from the new constitutive channel change.

8.8. Relations Among the Model Quantities

The explicit auxiliary channel now combines the input, organizational cost, backreaction, and gravitational source within a single action. The following substitutions incorporate it into the base-model equations:

Base model	Adjusted channel
Potential U	U_{eff} from (79)
Restoring term U'	$(1 - \eta_\Sigma \Sigma_0)U'$
Potential curvature U''	Expression (83)
Tensor $T^f_{\mu\nu}$	$T^f_{\mu\nu} - g_{\mu\nu}\mathcal{E}_\Sigma W_\kappa$
Source $\mathcal{Q}_f$	$\mathcal{Q}_f + 2\mathcal{E}_\Sigma W_\kappa$
Stationary functional	Replace U with U_{eff}

In this table, the second potential derivative denotes the local curvature of the potential. The kinetic terms, orientation coupling, and GR are retained. When the channel is active, the local algebraic saturation formulas of the base model must be obtained by a new variation using U_{eff}; their original closed-form expressions are not carried over unchanged.

The coupled spatial solution with the same boundary data is presented in Section 17. The connection to physical quantities requires scale mapping and calibration of $\eta_\Sigma, \mathcal{E}_\Sigma, \kappa$, together with the common exterior environment. Finite relaxation times and irreversible dedensification are studied within a corresponding dynamical description. If U already represents the same fully adjusted channel, its coefficients must be recalibrated accordingly before such a comparison. The comparison chosen here interprets U as the initial potential before this adjustment.

Directed twist organization remains contained in the full q tensor and its directional quantities. Σ_n describes this scalar loading channel; magnetic polarity, a photon interpretation, or a universal additional redshift belongs to the measurement and dynamical relations that must be determined in each case. The present formulation provides a constitutive extension with consistent balances, an algebraic carrier response, and a completed coupled stationary core comparison.

9. Free Fronts, Spherical Propagation, and Lateral Transport

Around $q_U = (0, 0, -1)$ and $\zeta_U = 0$, we set $q = (\pi_1, \pi_2, -\sqrt{1 - \pi_1^2 - \pi_2^2})$ and $\zeta = z$. The potentials and the twist term begin at fourth order. The quadratic action is therefore

$$\mathcal{L}_f^{(2)} = \frac{A}{2} \sum_{a=1}^{2} \partial_\mu \pi_a \partial^\mu \pi_a + \frac{Z_0}{2} \partial_\mu z \partial^\mu z, \qquad \Box \pi_a = \Box z = 0. \tag{95}$$

This free limit has the dispersion relation $\omega_{\mathrm{wave}}^2 = c^2 |\mathbf{k}|^2$. Exactly vanishing loading data remain in the sector $\zeta = 0$ for compatible boundary data. A small perturbation around a general q_U, by contrast, satisfies $Z_0 \Box_g z + A X_U z = 0$.

For the meridional sector $q = (\sin \vartheta, 0, -\cos \vartheta)$, all derivatives of q are parallel; hence $J_{\mu\nu}$ vanishes even in the presence of angular dependence. We write $\tau = x^0$, use ∇_{S^2} on the unit sphere, and set

$$X_a = (\partial_\tau a)^2 - (\partial_r a)^2 - r^{-2} |\nabla_{S^2} a|^2, \qquad a \in \{\vartheta, \zeta\}. \tag{96}$$

After integration over the spatial volume, the reduced action, up to the constant overall factor $1/c$, is

$$S_{\mathrm{mer}} = \int d\tau \, dr \, d\Omega_s \, r^2 \left[\frac{AC}{2} X_\vartheta + \frac{Z}{2} X_\zeta - C_s (1 - \cos \vartheta)^2 - U \right]. \tag{97}$$

With $p_\vartheta = AC \vartheta_\tau$, $p_\zeta = Z \zeta_\tau$, and $\mathcal{D}_r b = r^{-2} \partial_r (r^2 b)$, this directly gives

$$\partial_\tau p_\vartheta = \mathcal{D}_r (AC \vartheta_r) + r^{-2} \nabla_{S^2} \cdot (AC \nabla_{S^2} \vartheta) - V_\vartheta, \tag{98}$$

$$\partial_\tau p_\zeta = \mathcal{D}_r (Z \zeta_r) + r^{-2} \nabla_{S^2} \cdot (Z \nabla_{S^2} \zeta) + \frac{A}{2} C' X_\vartheta + \frac{Z'}{2} X_\zeta - U'. \tag{99}$$

The radial and tangential currents are

$$j_{\vartheta,r} = -AC\vartheta_\tau\vartheta_r, \qquad \mathbf{j}_{\vartheta,S^2} = -\frac{AC}{r}\vartheta_\tau\nabla_{S^2}\vartheta, \tag{100}$$

with the corresponding expressions for ζ. The balance equation contains $\mathcal{D}_r j_r + r^{-1}\nabla_{S^2}\cdot\mathbf{j}_{S^2}$. The angular integral of the second divergence vanishes. Lateral transport thus changes the local distribution and can thereby affect subsequent channel exchange.

9.1. Geometric Dilution and Measurable Time Intervals

In the radial free limit, $\vartheta_{\tau\tau} = r^{-2}\partial_r(r^2\vartheta_r)$. With $\psi = r\vartheta$, this becomes $\psi_{\tau\tau} = \psi_{rr}$, so that, outside the source,

$$\vartheta(\tau, r) = \frac{F(\tau - r)}{r}. \tag{101}$$

The amplitude decreases as r^{-1}. Matched features of the profile F retain their time intervals at a fixed detector. The energy flux gives, with differentiation with respect to τ,

$$L_\tau := 4\pi r^2 j_{\vartheta,r} = 4\pi A\left[(F')^2 + \frac{FF'}{r}\right], \tag{102}$$

$$X_\vartheta = -\frac{2FF'}{r^3} - \frac{F^2}{r^4}. \tag{103}$$

The integral of the second flux term over a complete compact pulse is a vanishing boundary term. The physical power flux is cL_τ. Geometric dilution alone therefore preserves the integrated pulse energy available and the matched time intervals. Active loading can additionally exchange energy through $(A/2)C'X_\vartheta\zeta_\tau$; the direction of this exchange follows from the state.

The model probe $q_1 = \sin\vartheta$ defines a specific readout channel. For two matched events, let

$$T_D = \tau^{\text{proper}}_{D,m+1} - \tau^{\text{proper}}_{D,m}, \qquad z_T = \frac{T_D}{T_{\text{em}}} - 1. \tag{104}$$

The cadence parameter z_T is evaluated separately from spectral modulation. Section 10 addresses the arrival-time offsets of different front portions that remain effective. Pulse height, pulse width, peak separation, and the centroid of the energy flux are distinct observables. For a specified pulse window I_m, the last of these is, for example,

$$\bar{t}_m = \frac{\int_{I_m} tL(t,r)\,dt}{\int_{I_m} L(t,r)\,dt}, \qquad \bar{T}_m = \bar{t}_{m+1} - \bar{t}_m, \tag{105}$$

provided that the denominator is nonzero. The signed flux is used in full.

In a static geometry, for static emitters and receivers and an undisturbed propagating periodic wave,

$$1 + z_{\text{grav}} = \frac{N_{\text{obs}}}{N_{\text{em}}}. \tag{106}$$

This follows from the shared coordinate period and $d\tau^{\text{proper}} = N dt$. The metric contribution

is included once in an overall balance. Identifying the internal field modes with electromagnetic photon polarizations and relating their energy to $h\nu$ are additional tasks of physical identification.

9.2. Angular Structure and Mean Backreaction

A small quadrupole component uses $Y = P_2(\cos\theta)$, $\Delta_{S^2} Y = -6Y$. For $y = (\vartheta, \zeta, p_\vartheta, p_\zeta)^{\mathsf{T}}$, we set

$$y_\epsilon = y_0 + \epsilon s Y + \epsilon^2 (m + y_2^\perp) + O(\epsilon^3), \qquad \langle y_2^\perp \rangle_{S^2} = 0, \quad s = (\eta, \xi, \pi_\vartheta, \pi_\zeta)^{\mathsf{T}}. \tag{107}$$

With $v = p_\vartheta/(AC)$ and $u = p_\zeta/Z$, the radial operator is fully specified by

$$\mathcal{F}_0[y] = \begin{pmatrix} v \\ u \\ \mathcal{D}_r(AC\vartheta_r) - V_\vartheta \\ \mathcal{D}_r(Z\zeta_r) + \frac{A}{2}C'(v^2 - \vartheta_r^2) + \frac{Z'}{2}(u^2 - \zeta_r^2) - U' \end{pmatrix}. \tag{108}$$

Taylor expansion of (98)–(99) and $\langle Y^2 \rangle = 1/5$, $\langle |\nabla_{S^2} Y|^2 \rangle = 6/5$, together with $\Lambda_{\mathrm{ang}} = 6$, gives

$$\partial_\tau y_0 = \mathcal{F}_0[y_0], \tag{109}$$

$$\partial_\tau s = D\mathcal{F}_0[y_0]s - \frac{\Lambda_{\mathrm{ang}}}{r^2}(0, 0, AC\eta, Z\xi)^{\mathsf{T}}, \tag{110}$$

$$\partial_\tau m = D\mathcal{F}_0[y_0]m + \frac{1}{10}D^2\mathcal{F}_0[y_0][s, s] - \frac{\Lambda_{\mathrm{ang}}}{10r^2}(0, 0, 0, AC'\eta^2 + Z'\xi^2)^{\mathsf{T}}. \tag{111}$$

$D\mathcal{F}_0[y]s := \partial_\epsilon \mathcal{F}_0[y + \epsilon s]|_0$ and the second derivative are thus derivatives of the explicitly written operator. The factor $1/10$ is the Taylor factor $1/2$ times $1/5$. The angular divergence integrates to zero; the quadratic angular gradients yield the final source term.

For $a \in \{\vartheta, \zeta\}$, let $K_\vartheta = AC$, $K_\zeta = Z$, $s_\vartheta = \eta$, $s_\zeta = \xi$, and $e_{a,r} = p_a^2/(2K_a) + K_a a_r^2/2 + V_a$, with $V_\vartheta = V$, $V_\zeta = U$. The energy coefficient is

$$E_a^{[2]} = 4\pi \int r^2 \left[De_{a,r}[y_0]m + \frac{1}{10}D^2 e_{a,r}[y_0][s, s] + \frac{\Lambda_{\mathrm{ang}} K_a s_a^2}{10r^2} \right] dr. \tag{112}$$

For $\mathcal{P}_r = (A/2)C'(v^2 - \vartheta_r^2)u$,

$$\mathcal{P}^{[2]} = D\mathcal{P}_r[y_0]m + \frac{1}{10}D^2\mathcal{P}_r[y_0][s, s] - \frac{\Lambda_{\mathrm{ang}} AC' u\eta^2}{10r^2}. \tag{113}$$

With vanishing boundary flux, $\Delta E_\vartheta^{[2]} = -Q^{[2]}$ and $\Delta E_\zeta^{[2]} = Q^{[2]}$ follow, with $Q^{[2]} = 4\pi \int d\tau \int r^2 \mathcal{P}^{[2]} dr$. The initial quadratic energy is also subtracted for $m(0) = 0$. The mean is closed to this perturbative order; a complete finite angular organization additionally contains $y_2^\perp$ and higher orders.

10. Arrival-Time Offsets and the Event Sequence of Spatially Offset Source Fronts

The dedensification of the tension skin and its exchange with the subregimes are adopted as an existing model component. The subject here is the temporal arrangement of the front portions that remain effective in the observational regime. The source cadence, the geometric arrival-time offsets, and the intervals in the detected combined sequence are calculated separately.

The geometric reference calculation uses straight propagation paths at speed c and a shared time coordinate in a flat background at rest. Each event is a matched front feature. Its subsequent identification with a measured signal requires these features to be resolved separately.

10.1. Arrival-Time Offset from Emission Position and Emission Time

A front feature originates at time t_i^{em} at position $\mathbf{x}_i$. A detector at rest is located at $\mathbf{X} = R\widehat{\mathbf{n}}$, where $|\widehat{\mathbf{n}}| = 1$. Its arrival time is

$$t_i^{\mathrm{arr}} = t_i^{\mathrm{em}} + \frac{d_i}{c}, \qquad d_i = |R\widehat{\mathbf{n}} - \mathbf{x}_i|. \tag{114}$$

For two matched features, this gives exactly

$$\boxed{\Delta t_{ij}^{\mathrm{arr}} = \Delta t_{ij}^{\mathrm{em}} + \frac{d_j - d_i}{c}.} \tag{115}$$

The second term is the geometric arrival-time offset. It can change the ordering of different subsequences. Within a subsequence with a fixed point of origin, however, a constant emission interval is preserved.

For periodic subsequences with a shared source interval T_Q, we write $t_{m,a}^{\mathrm{em}} = mT_Q + \eta_a$. The index a denotes a region of origin, m its successive emissions, and η_a the emission offset. Then

$$t_{m,a}^{\mathrm{arr}} = mT_Q + \frac{R}{c} + \delta_a^{\mathrm{arr}}(R, \widehat{\mathbf{n}}), \tag{116}$$

$$\delta_a^{\mathrm{arr}} := \eta_a + \frac{|R\widehat{\mathbf{n}} - \mathbf{x}_a| - R}{c}. \tag{117}$$

A dense combined sequence can thus be formed from several offset subsequences, even though each subsequence retains its own interval T_Q. The shared T_Q is an assumption of this periodic comparison case; Equation (114) does not require it.

10.2. Bounds on the Geometric Offset

The reverse triangle inequality gives

$$|d_j - d_i| \le |\mathbf{x}_j - \mathbf{x}_i|. \tag{118}$$

For points of origin within a region of diameter D_Q, therefore,

$$\left|\Delta t_{ij}^{\mathrm{arr}} - \Delta t_{ij}^{\mathrm{em}}\right| \le \frac{D_Q}{c}. \tag{119}$$

This bound also applies to different points of origin for successive emissions, provided that all positions considered lie within this region. The purely spatial offset of a bounded source thus produces a bounded spread in travel times.

For $|\mathbf{x}_a| \le a_Q \ll R$,

$$\delta_a^{\mathrm{arr}} = \eta_a - \frac{\widehat{\mathbf{n}} \cdot \mathbf{x}_a}{c} + \frac{|\mathbf{x}_a|^2 - (\widehat{\mathbf{n}} \cdot \mathbf{x}_a)^2}{2cR} + O\left(\frac{a_Q^3}{cR^2}\right). \tag{120}$$

The leading arrival-time offset is determined by the source depth along the line of sight. The correction due to front curvature decreases as R^{-1}. A greater travel distance therefore does not amplify a fixed source offset without bound. Visibility changes along the path enter through the existing dedensification.

10.3. The Effective Combined Sequence and its Regularity

Let $w_a(R, \widehat{\mathbf{n}})$ be the effectiveness of a front contribution adopted from the existing field and coupling description. A specified readout channel assigns it a detection indicator, for example,

$$\chi_a(R, \widehat{\mathbf{n}}) = \begin{cases} 1, & w_a(R, \widehat{\mathbf{n}}) \ge w_{\min}, \\ 0, & w_a(R, \widehat{\mathbf{n}}) < w_{\min}. \end{cases} \tag{121}$$

This describes the selection process; it is not a new dedensification law. Let the weights and the source geometry be constant in time over the interval considered.

For the effective contributions, the arrival-time offsets are ordered modulo T_Q. This produces N_D distinct, resolved arrival phases

$$0 \le \theta_1 < \theta_2 < \cdots < \theta_{N_D} < T_Q. \tag{122}$$

Simultaneous or unresolved contributions are not counted as separate events. The cyclic intervals are

$$g_j = \theta_{j+1} - \theta_j \quad (1 \le j < N_D), \tag{123}$$

$$g_{N_D} = T_Q + \theta_1 - \theta_{N_D}. \tag{124}$$

Regardless of their uniformity, it follows that

$$\sum_{j=1}^{N_D} g_j = T_Q, \qquad \overline{g} = \frac{T_Q}{N_D}, \qquad \Gamma_D = \frac{N_D}{T_Q}. \tag{125}$$

Γ_D is the event rate of the combined sequence. Fewer effective arrival phases increase their mean interval.

The sequence is exactly regular if and only if

$$\boxed{\theta_j \equiv \theta_* + (j-1)\frac{T_Q}{N_D} \pmod{T_Q}.} \tag{126}$$

The remaining arrival phases must therefore divide the entire period T_Q uniformly. Merely reducing their number does not determine this subdivision.

For approximate regularity, we use

$$C_g := \frac{\sqrt{{N_D}^{-1}\sum_{j=1}^{N_D}(g_j - \overline{g})^2}}{\overline{g}}. \tag{127}$$

When $C_g = 0$, all intervals are equal. If, after cyclic matching, the phases lie at most ϵ_t from a regular arrangement, then

$$|g_j - T_Q/N_D| \le 2\epsilon_t, \qquad C_g \le \frac{2N_D\epsilon_t}{T_Q}. \tag{128}$$

This estimate directly translates inaccuracies in the source ordering into a tolerance for the measured intervals.

10.4. A Regular Sequence Constructed from Source Positions

A specific geometry consists of N_Q regions of origin along the line connecting them to the detector:

$$\mathbf{x}_a = \frac{acT_Q}{N_Q}\widehat{\mathbf{n}}, \qquad a = 0, \dots, N_Q - 1, \qquad \eta_a = 0. \tag{129}$$

The detector lies beyond all regions of origin. Within each source period, these regions emit simultaneously. Their different distances give exactly

$$t^{\mathrm{arr}}_{m,a} = mT_Q + \frac{R}{c} - \frac{aT_Q}{N_Q}. \tag{130}$$

The combined arrival sequence has the interval T_Q/N_Q. Here, this interval has been derived from the source positions.

If precisely the contributions $a = a_0 + kd$ modulo N_Q remain effective, where d is a divisor of N_Q and $k = 0, \dots, N_Q/d - 1$, then

$$N_D = \frac{N_Q}{d}, \qquad g_j = \frac{dT_Q}{N_Q} = \frac{T_Q}{N_D}. \tag{131}$$

The source geometry and a correspondingly ordered selection can thus produce a slower combined sequence that remains regular. For $N_Q = 8$, the table shows several selections. Gaps are given in units of $T_Q/8$, up to cyclic ordering.

Effective contributions a	Gaps	$\overline{g}$	C_g
$0, 1, \ldots, 7$	$1, 1, 1, 1, 1, 1, 1, 1$	$T_Q/8$	0
$0, 2, 4, 6$	$2, 2, 2, 2$	$T_Q/4$	0
$0, 4$	$4, 4$	$T_Q/2$	0
0	8	T_Q	0
$0, 1, 2, 3$	$5, 1, 1, 1$	$T_Q/4$	$\sqrt{3}/2$
$0, 1, 4, 6$	$2, 2, 3, 1$	$T_Q/4$	$1/(2\sqrt{2})$

In particular, removing a contiguous source region in this example produces a longer pause between the remaining groups. A uniform increase in all intervals, by contrast, requires a selection distributed across the arrival phases.

The construction refers to the chosen line of sight. A different viewing direction changes the projections of the source positions and hence the arrival phases. This directional dependence is part of the assessment of a specific source organization.

10.5. A Continuously Varying Mean Rate and Uniform Individual Intervals

For N_Q initially uniformly arranged arrival phases, $g_0 = T_Q/N_Q$ is the smallest time increment. If contributions are only removed and no arrival times are shifted, each remaining gap is an integer multiple of g_0. Exact regularity with N_D remaining events therefore requires N_Q/N_D to be an integer.

For an arbitrary N_D, we set

$$u := \frac{N_Q}{N_D}, \qquad h := \lfloor u \rfloor, \qquad \alpha := u - h. \tag{132}$$

The most uniform possible selection has gaps hg_0 and $(h+1)g_0$. The fraction of longer gaps is α. This gives the sharp lower bound

$$\boxed{C_{g,\min} = \frac{\sqrt{\alpha(1-\alpha)}}{u}.} \tag{133}$$

It is attained by selections distributed as uniformly as possible. The proof follows from the fact that, at a fixed sum, the variance of integer gaps is minimal when they differ by at most one.

Many subsequences allow fine changes in the mean rate N_D/T_Q. At a fixed noninteger ratio u, however, a larger number N_Q does not eliminate the relative scatter in Equation (133). Continuous stretching with very uniform individual intervals would also require the arrival phases to shift appropriately or a different temporal source ordering to underlie the sequence.

10.6. Connection to the Existing Field Description

The cadence assessment requires the point of origin, emission time, and adopted effectiveness of each matched contribution together:

$$\left(\mathbf{x}_i, t_i^{\mathrm{em}}, w_i(R, \widehat{\mathbf{n}})\right). \tag{134}$$

The arrival times are calculated from these quantities, and the events that remain effective are ordered. The mean and scatter of their intervals are evaluated separately. A larger mean with a small C_g describes regular temporal thinning; a larger mean with a large C_g describes a more interrupted sequence.

The specific connection consists of linking the weights obtained from the existing dedensification ansatz to the geometric arrival phases. The question to be tested is whether strong and weak contributions are distributed across the phases such that Equation (126) remains approximately satisfied. The weights are not subsequently tailored to a uniform sequence for this test.

Scope. The derivation describes arrival times and event intervals in the stated reference geometry. The periodic example assumes a temporally ordered source and resolved front features. The geometric construction does not specify a source structure selected by the field equation. The existing dedensification and spectral modulation remain separate input components; their laws are not changed here. Gravitational travel-time effects, moving sources, and a propagation environment that varies in time can additionally be incorporated through their respective arrival-time mappings.

11. From Front Superposition to Bound Field Organization

The guiding idea of UQSH connects matter formation with the densification, selective coupling, and spatial binding of free field motion [2]. The existing action allows the first part of this process to be investigated: fronts change the combined orientation organization, which affects the loading, and both carry energy and stress. The following analytical connection assigns angles, profile phase, internal orientation, and environmental loading to their respective terms. The stationary cores serve as previously calculated comparison states for bound organization.

The global front ansätze belong to the topologically trivial sector. Its preservation and the conditions for a transition to a net knot charge are addressed in Section 13.1.

11.1. Excitation of an Environment in Motion

After adjustment of the regime channel, the loading equation (18) applies with U_{eff} in place of U. We consider a background solution (q_U, ζ_U) on a fixed geometry and set $z = \delta\zeta$, $q_U \cdot \delta q = 0$, and $\delta X = 2\nabla_\mu q_U \cdot \nabla^\mu \delta q$. The first variation gives

$$\mathcal{D}_{\zeta,U} z := \nabla_\mu (Z_U \nabla^\mu z) + \mathcal{M}_{\zeta,U} z = \frac{A}{2} C'_U \, \delta X, \tag{135}$$

$$\mathcal{M}_{\zeta,U} = Z'_U \Box_g \zeta_U + \frac{Z''_U}{2} (\nabla\zeta_U)^2 - \frac{A}{2} C''_U X_U + U''_{\text{eff},U}. \tag{136}$$

The subscript U denotes evaluation at the environmental state; primes here denote derivatives with respect to ζ. Existing loading thus couples directly, through $C'_U \delta X$, to changes in the front organization. The change δq is part of the coupled field response; the equation identifies its effect on the loading component. The perturbative order at which the first contribution enters also depends on the existing orientation gradients.

For the special background $\zeta_U = 0$, C'_U and $U''_{\mathrm{eff},U}$ vanish. With $C''(0) = -2$,

$$Z_0 \Box_g z + A X_{\mathrm{bg}} z = 0. \tag{137}$$

Here, X_{bg} denotes the chosen orientation solution on the zero branch, including any existing fronts. This background is determined accordingly for each of the four comparison cases. Exactly vanishing loading data remain on the zero branch for compatible boundary data. An existing small loading perturbation, by contrast, can be amplified or restored by X_{bg}. $\zeta_U = 0$ denotes a state of this loading channel; the orientation and its environment may be in motion. According to (93), the channel correction begins at ζ^8 and leaves this linear initial response unchanged.

11.2. Angle and Profile Phase

For a locally planar, meridional comparison, we write

$$q = (\sin\vartheta, 0, -\cos\vartheta), \qquad \vartheta = \vartheta_U + f_1(\xi_1) + f_2(\xi_2), \qquad \xi_i = x^0 - \mathbf{n}_i \cdot \mathbf{x}, \quad |\mathbf{n}_i| = 1. \tag{138}$$

In this kinematic ansatz, the profiles propagate at c. With $k_{i\mu} = \partial_\mu \xi_i$, $k_i \cdot k_i = 0$ and $k_1 \cdot k_2 = 1 - \cos\alpha$, where $\cos\alpha = \mathbf{n}_1 \cdot \mathbf{n}_2$. For $\vartheta_U = 0$, it follows directly that

$$\boxed{X = 2(1 - \cos\alpha) f_1'(\xi_1) f_2'(\xi_2).} \tag{139}$$

Here, f_i' denotes the derivative with respect to ξ_i. Each individual planar profile has $X_i = 0$. In the combined state, the product of the intersecting profile slopes determines the sign: slopes in the same sense give $X > 0$, and slopes in opposite senses give $X < 0$, provided that $\alpha \neq 0$. The relative phase determines which slopes encounter one another.

The environment in motion can also be included. Let $X^{(0)}, X^{(1)}, X^{(2)}, X^{(12)}$ be the expressions for ϑ_U, $\vartheta_U + f_1$, $\vartheta_U + f_2$, and $\vartheta_U + f_1 + f_2$. Then, for this profile comparison,

$$\Delta_{12} X := X^{(12)} - X^{(1)} - X^{(2)} + X^{(0)} = 2(1 - \cos\alpha) f_1' f_2'. \tag{140}$$

The individual interaction terms with ϑ_U cancel in this difference. Planar fronts propagating in the same direction have no mutual X cross term here; their individual coupling to an environment in motion remains included in the combined state. After the four states undergo independent nonlinear evolution, their actual fields are evaluated.

11.3. Spatial Extent and Amplification Duration

In addition to the local response, Equation (137) includes spatial stiffness. In a locally flat comparison region, it reads, in physical time,

$$\partial_t^2 z + c^2 \mathcal{H}_X z = 0, \qquad \mathcal{H}_X = -\nabla^2 + \frac{A}{Z_0} X_{\mathrm{bg}}. \tag{141}$$

For a bounded test domain D, perturbations that vanish at the boundary, and a background held fixed during the comparison, the Rayleigh quotient determines the smallest eigenvalue:

$$\lambda_{X,\min} = \inf_{0 \neq v \in H_0^1(D)} \frac{\displaystyle\int_D \left[|\nabla v|^2 + \frac{A}{Z_0} X_{\mathrm{bg}} v^2 \right] d^3x}{\displaystyle\int_D v^2 \, d^3x}. \tag{142}$$

Here, $H_0^1(D)$ comprises perturbations with a square-integrable first derivative and vanishing boundary value. A negative value requires enough spatially effective negative contributions from X_{bg} to outweigh the positive gradient contribution. The sign at a single point does not determine this.

For a time-independent negative eigenvalue λ_X and the corresponding modal component $a(t)$,

$$\gamma = c\sqrt{-\lambda_X}, \qquad a(t_0 + T) = a_0 \cosh(\gamma T) + \frac{\dot{a}_0}{\gamma} \sinh(\gamma T). \tag{143}$$

The initial data determine the contribution of the growing solution. For appreciable amplification, the appropriate overlap state must persist long enough; γT is the dimensionless comparison quantity for this purpose. For spatially constant $X_{\mathrm{bg}} = X_c < 0$ and an eigenmode of $-\nabla^2$ with eigenvalue k_D^2, the condition becomes particularly transparent:

$$\frac{A}{Z_0}|X_c| > k_D^2, \qquad \gamma = c\sqrt{\frac{A}{Z_0}|X_c| - k_D^2}. \tag{144}$$

This is an analytical criterion for the fixed background. For time-varying X_{bg}, the time-dependent equation is evaluated; parametric amplification may then also play a role. A negative frozen eigenvalue is therefore not a general necessary criterion for every dynamical excitation.

11.4. Internal Orientation and Spatial Twist

The meridional comparison aligns all derivatives with the same internal direction and has $J_{\mu\nu} = 0$. To connect to twist, we therefore allow both tangent directions around $q_U = (0, 0, -1)$:

$$q = (\pi_1, \pi_2, -\sqrt{1 - |\boldsymbol{\pi}|^2}), \qquad \boldsymbol{\pi} = \epsilon \left[\mathbf{e}_1 f_1(\xi_1) + \mathbf{e}_2 f_2(\xi_2) \right], \quad \mathbf{e}_i \in \mathbb{R}^2, \quad |\mathbf{e}_i| = 1. \tag{145}$$

Set $\rho_e = \mathbf{e}_1 \cdot \mathbf{e}_2$ and $\sigma_e = \det(\mathbf{e}_1, \mathbf{e}_2)$, so that $\rho_e^2 + \sigma_e^2 = 1$. The internal directions are distinct from the spatial propagation directions $\mathbf{n}_i$. To second order in amplitude,

$$X = 2\epsilon^2 (1 - \cos\alpha) \rho_e f_1' f_2' + O(\epsilon^4). \tag{146}$$

The oriented area density of the spherical chart used is $q \cdot (\partial_{\pi_1} q \times \partial_{\pi_2} q) = -(1 - |\boldsymbol{\pi}|^2)^{-1/2}$. For $\mathbf{b} = (F_{23}, F_{31}, F_{12})$, the following holds exactly within this chart:

$$\boxed{\mathbf{b} = -\frac{\epsilon^2 \sigma_e f_1' f_2'}{\sqrt{1 - |\boldsymbol{\pi}|^2}} (\mathbf{n}_1 \times \mathbf{n}_2).} \tag{147}$$

The domain is $|\boldsymbol{\pi}| < 1$. The two expressions distinguish two organizational roles. Parallel internal directions have $|\rho_e| = 1$ but $\sigma_e = 0$: the scalar cross term can be pronounced while this spatial twist vanishes. Orthogonal internal directions have $\rho_e = 0$ and $|\sigma_e| = 1$: spatial twist can be present while the scalar cross term vanishes at second order. Higher orders in amplitude remain part of the full action.

For exactly counterpropagating planar profiles, $1 - \cos\alpha = 2$, but $\mathbf{n}_1 \times \mathbf{n}_2 = 0$. In this comparison, the largest geometric factor of the scalar cross term therefore does not coincide with spatial twist. Temporal-spatial components J_{0i} may still be present for different internal directions. Spatial twist and scalar loading are therefore evaluated together but separately.

11.5. Energy Balance and Connection to the Core

In the adjusted regime channel, U_{eff} is assigned to the loading energy. In flat geometry,

$$\varepsilon_{\zeta,\text{eff}} = \frac{Z}{2}\left[c^{-2}\dot{\zeta}^2 + |\nabla\zeta|^2\right] + U_{\text{eff}}, \tag{148}$$

$$\mathbf{S}_\zeta = -Z\dot{\zeta}\nabla\zeta, \qquad \mathcal{P}_\zeta = \frac{A}{2}C' X\dot{\zeta}. \tag{149}$$

For $E_{\zeta,D} = \int_D \varepsilon_{\zeta,\text{eff}}\, d^3x$, the balance of the two fields gives

$$\frac{dE_{\zeta,D}}{dt} = -\int_{\partial D} \mathbf{S}_\zeta \cdot d\mathbf{A} + \int_D \mathcal{P}_\zeta\, d^3x. \tag{150}$$

For small loading, $\mathcal{P}_\zeta = -AX\zeta\dot{\zeta} + O(\zeta^3\dot{\zeta})$. When $X < 0$ and the magnitude of the loading grows, $\zeta\dot{\zeta} > 0$, this exchange is positive. The orientation energy carries the same term with the opposite sign. The buildup of loading is thus described as a redistribution of existing field energy.

The nonlinear connection leads to the local restoring response in Section 14 and to spatial profiles such as those in Section 17. With an active channel, the adjustment is determined by U_{eff}. The transition from initial amplification to bound organization is assessed through the remaining structure, its energy flux, and its response to perturbations. The tensor source in (84) simultaneously includes its energy and stresses in the gravitational description.

11.6. Scope and Targeted Overlap Comparison

The analytical relations specify the decisive comparison quantities. Their application to a front passage is made concrete by the following conditions:

1. **Shared environment and finite fronts.** The profile, spatial embedding, angle, relative phase, internal orientations, and initial loading are prescribed. The planar formulas describe the local overlap region; curvature, lateral boundaries, and backreaction are included in the full evolution.

2. **Regime response and timescales.** The auxiliary channel has no time-kinetic term of its own. For its quasistatic use, its physical adjustment must be sufficiently fast relative to

the relevant change in the fronts. Such an adjustment time is not determined by the algebraic channel equation alone.

3. **A four-state comparison.** The same environment is investigated without an additional front, with front one, with front two, and with both fronts. For an observable O, $\Delta_{12}O = O^{(12)} - O^{(1)} - O^{(2)} + O^{(0)}$ is evaluated. The energy balance includes all incoming and outgoing fluxes. For initially spatially separated compact additional profiles, the supplied energy can be specified without initial mutual overlap.

4. **Post-passage evolution and physical identification.** Loading, twist moments, spatially remaining energy, and outflow are tracked after the front passage as well. An energy cross term initially captures the superposition; lasting binding additionally requires persistent post-passage organization and an appropriate perturbation response. Frequency comparisons use a prescribed supplied energy. Model profiles are identified with measured gamma radiation through shared scales and an electromagnetic readout channel.

To establish the connection, loading amplification and the required twist order are first assessed for the same background; the resulting state is then connected to the binding and gravitational description already formulated. The following subsections specify the coupled initial data, close the overlap balance, and define a reproducible encounter geometry and its post-passage evolution.

11.7. Norm-Preserving Initial Data from Finite Fronts

The overlap comparison can be closed with defined initial data. We first consider local field evolution on the same flat geometry. On the initial surface, the environment in motion has the data (q_U, v_U, ζ_U, w_U), with $v_U = \partial_0 q_U$, $w_U = \partial_0 \zeta_U$, $|q_U| = 1$, and $q_U \cdot v_U = 0$. It evolves in every case. A small existing loading is prescribed in common; an additional comparison with $\zeta_U = w_U = 0$ checks the exact zero branch.

For each additional front, we define an internal rotation $R_i = \exp(\psi_i [\mathbf{u}_i]_\times)$ about a fixed unit axis $\mathbf{u}_i$, where $[\mathbf{u}_i]_\times \mathbf{a} = \mathbf{u}_i \times \mathbf{a}$. The four initial states are denoted by $a, b \in \{0, 1\}$:

$$q^{(ab)} = R_1^a R_2^b q_U, \tag{151}$$

$$v^{(ab)} = R_1^a R_2^b v_U + (\partial_0 R_1^a) R_2^b q_U + R_1^a (\partial_0 R_2^b) q_U, \tag{152}$$

$$\zeta^{(ab)} = \zeta_U, \qquad w^{(ab)} = w_U. \tag{153}$$

Here and below, $00, 10, 01, 11$ correspond to the cases $0, 1, 2, 12$. The orthogonality of R_i and antisymmetry of $R_i^\mathsf{T} \partial_0 R_i$ ensure exactly $|q^{(ab)}| = 1$ and $q^{(ab)} \cdot v^{(ab)} = 0$. The rotations specify the initial data. Each case then evolves according to the coupled field equations.

Finite fronts are assigned a smooth longitudinal profile F_i and a smooth lateral cutoff G_i:

$$\psi_i = \varepsilon_i F_i(\xi_i / L_{\parallel,i}) G_i(\rho_i / R_i^{\mathrm{tr}}) \cos(k_i \xi_i + \phi_i), \tag{154}$$

$$\xi_i = c(t - t_0) - \mathbf{n}_i \cdot (\mathbf{x} - \mathbf{x}_i), \qquad \rho_i = |(I - \mathbf{n}_i \mathbf{n}_i^\mathsf{T})(\mathbf{x} - \mathbf{x}_i)|. \tag{155}$$

Here, $F_i(s) = 0$ for $|s| \geq 1$; $G_i(r) = 1$ for $0 \leq r \leq 1/2$ and $G_i(r) = 0$ for $r \geq 1$. All transitions are smooth. R_i^{tr} denotes the transverse radius and is distinct from the rotation matrix R_i. The velocity data follow by differentiating this profile, giving $\partial_0 \psi_i = \partial_{\xi_i} \psi_i$. The lateral edges are included in the field evolution; the planar relations apply directly in the inner plateau of the prescribed profile.

The momentum data with respect to x^0 also follow from the same action. With $d_i = \partial_i q$, the derivative with respect to $v = \partial_0 q$ gives

$$\pi_q = \mathsf{M}_q v, \qquad \mathsf{M}_q = AC\, I + \beta_4 \sum_i (|d_i|^2 I - d_i d_i^\mathsf{T}), \qquad \pi_\zeta = Z w. \tag{156}$$

For every vector h, $h^\mathsf{T} \mathsf{M}_q h = AC|h|^2 + \beta_4 \sum_i |h \times d_i|^2 > 0$ for $h \neq 0$. This positive map relates initial velocity and canonical momentum without an additional inertia ansatz.

11.8. Additive Initial Energy and Combined Energy Flux

The two fronts are placed on the initial surface with disjoint compact supports. At each position, there is then either no additional front or exactly one. Owing to the smooth cutoffs, this also holds for the required derivatives. For any local state quantity $\mathcal{O}$, in particular energy density, energy flux, and active source, it therefore follows that

$$\Delta_{12} \mathcal{O}(t_0, \mathbf{x}) := \mathcal{O}^{(11)} - \mathcal{O}^{(10)} - \mathcal{O}^{(01)} + \mathcal{O}^{(00)} = 0. \tag{157}$$

The existing coupling of each individual front to the environment remains included. For a domain D containing the initial profiles, the individual energy changes $E_{\mathrm{in},1} = E_D^{(10)}(t_0) - E_D^{(00)}(t_0)$ and $E_{\mathrm{in},2} = E_D^{(01)}(t_0) - E_D^{(00)}(t_0)$ are additive:

$$E_D^{(11)}(t_0) - E_D^{(00)}(t_0) = E_{\mathrm{in},1} + E_{\mathrm{in},2}. \tag{158}$$

These differences retain their signs. Positive values denote an energy input relative to the environment. In frequency or orientation comparisons, the energy inputs actually calculated are matched or reported alongside the results; equal profile amplitude alone does not guarantee equal energy in an environment in motion.

Let $K \subset D$ be a fixed encounter domain shared by all cases. We define the accumulated flux, positive outward,

$$E_K^{(ab)}(t) = \int_K \varepsilon^{(ab)}\, d^3x, \qquad \mathcal{F}_K^{(ab)}(t) = \int_{t_0}^{t} ds \int_{\partial K} \mathbf{S}^{(ab)} \cdot \mathbf{n}\, dA. \tag{159}$$

Local energy conservation and (157) give the closed overlap balance

$$\boxed{\Delta_{12} E_K(t) + \Delta_{12} \mathcal{F}_K(t) = 0.} \tag{160}$$

A positive additional energy contribution in the encounter domain is thus tied to a corresponding change in flux. The difference flux can be negative even when individual outgoing fluxes are positive. For the entire domain, $\Delta_{12} E_D = 0$ remains valid as long as its outer difference flux vanishes. Reorganization is therefore captured through its spatial distribution and energy exchange; the global energy difference alone is unsuitable for this purpose.

11.9. Loading Transfer and Remaining Directional Order

Because the loading data are initially identical in all cases, their cross difference also has no initial energy contribution. With the decomposition in (149),

$$\boxed{\Delta_{12} E_{\zeta,K}(t) + \Delta_{12} \mathcal{F}_{\zeta,K}(t) = \int_{t_0}^{t} ds \int_K \Delta_{12} \mathcal{P}_\zeta\, d^3x.} \tag{161}$$

The right-hand side measures the additional net transfer into the loading component. Together with (160), it separates storage, transport, and exchange without introducing a second energy reservoir. The same calculation includes the source combination Q from (25), with the adjusted potential, as a gravitational diagnostic.

Directional order is evaluated using the previously defined first and second twist moments. For a uniform window in the domain K,

$$\mathbf{m}_K^{(ab)} = \frac{1}{|K|} \int_K \mathbf{b}^{(ab)}\, d^3x, \qquad \mathsf{W}_K^{(ab)} = \frac{1}{|K|} \int_K \mathbf{b}^{(ab)} \mathbf{b}^{(ab)\mathsf{T}}\, d^3x. \tag{162}$$

The trace of W_K determines the spatial twist-energy contribution; eigenvalues and eigendirections describe the axial order. Oppositely directed contributions can cancel $\mathbf{m}_K$ while W_K remains. Each individual matrix $\mathsf{W}_K^{(ab)}$ is positive semidefinite; its cross difference need not

have this property and is treated as a signed response.

For the post-passage evolution, a shared window $[t_*, t_* + T]$ is chosen after the expected passage of the free reference fronts. A time-averaged quantity is

$$\overline{\Delta_{12}\mathcal{O}}_K = \frac{1}{T}\int_{t_*}^{t_*+T} \Delta_{12}\mathcal{O}_K(t)\,dt. \tag{163}$$

The individual time histories and their changes relative to case 00 are also tracked. Energy and loading in the domain, fluxes, twist moments, and the spatial position of the organization are compared together. This distinguishes a departing front from a longer-lasting concentration. Such post-passage behavior provides the initial state for the subsequent assessment of its dynamical persistence.

11.10. Encounter Geometry and Control Variants

A specific geometric starting point uses equal length scales $L_{\parallel,1} = L_{\parallel,2} = L_\parallel$ and $R_1^{\mathrm{tr}} = R_2^{\mathrm{tr}} = 4L_\parallel$. We choose

$$\mathbf{n}_1 = (\sqrt{3}/2, 0, 1/2), \qquad \mathbf{n}_2 = (-\sqrt{3}/2, 0, 1/2), \qquad \mathbf{x}_i = -8L_\parallel\,\mathbf{n}_i. \tag{164}$$

The encounter angle is 120°; the kinematic profile centers reach the origin after $8L_\parallel/c$. The initial separation $8\sqrt{3}L_\parallel$ exceeds $2\sqrt{17}L_\parallel$, the sum of the radii of enclosing spheres. The supports are therefore disjoint. A shared encounter domain is, for example, $K = B_{2L_\parallel}(0)$. The freely translated reference profiles have completely left this domain after $11L_\parallel/c$. A specific post-passage window is therefore $c(t - t_0)/L_\parallel \in [12, 16]$. This time choice describes the geometric reference passage; in the coupled case, the remaining organization is measured. These length ratios specify a reproducible test geometry; they do not physically calibrate $L_\parallel$.

In the local reference state $q_U = (0, 0, -1)$, we choose the internal tangent directions $\mathbf{e}_1 = (1, 0, 0)$ and $\mathbf{e}_2 = (1, 1, 0)/\sqrt{2}$, together with $\mathbf{u}_i = q_U \times \mathbf{e}_i$. The internal angle is $\beta_{\mathrm{pol}} = 45^\circ$. For even longitudinal envelopes with $F_i(0) = 1$ and $\phi_1 = \pi/2$, $\phi_2 = -\pi/2$, oppositely directed profile slopes meet in the kinematic center comparison. The leading scalar cross term is negative, and both geometric twist factors are nonzero. The subsequent field evolution determines the actual response.

Variation	Analytical comparison in the planar, weak-profile region
$\alpha = 120^\circ$, $\beta_{\mathrm{pol}} = 45^\circ$	The scalar cross term and spatial twist are jointly accessible.
$\beta_{\mathrm{pol}} = 0$	The spatial twist of the meridional profile ansatz vanishes.
$\beta_{\mathrm{pol}} = 90^\circ$	The second-order scalar cross term vanishes; spatial twist remains possible.
$\alpha = 180^\circ$	The spatial cross-product factor vanishes; temporal-spatial twist may remain.
$\phi_2 \mapsto \phi_2 + \pi$	The leading scalar cross term and leading directed twist change sign; the leading twist-energy contribution formed from them remains the same.

Each variant again includes all four cases. The relations in the table refer to the prescribed local

profile comparison. Lateral edges, an environment in motion, and nonlinear backreaction are captured through the full individual cases and their combined differences.

An implementation additionally specifies the model coefficients, environmental data, amplitudes, wavenumbers, outer boundary, post-passage duration, and resolution. The regular hyperbolic state domain is also checked; the positive momentum map alone does not replace this check. Outer reflections are excluded or included in the balance. The fixed geometry defines the local field comparison here; evolving gravity as well additionally requires the metric initial conditions for each of the four cases. The initial comparison, energy check, and readout define the initial-value problem for numerical front evolution.

12. Energy of Incoming Fronts and Generic Weighting

12.1. A Norm-Preserving Single-Front State

The preparation for triadic front coupling begins with the energy of a single localized excitation. The reference states are $q_{\mathrm{U}} = (0, 0, -1)$ and $\zeta_{\mathrm{U}} = 0$. A constant internal unit vector $\mathbf{e}$ satisfies $\mathbf{e} \cdot q_{\mathrm{U}} = 0$. For a dimensionless profile u, we set

$$q = \cos(\vartheta u) q_{\mathrm{U}} + \sin(\vartheta u)\mathbf{e}, \qquad \zeta = au. \tag{165}$$

The parameters ϑ and a weight orientation and loading. The norm constraint $q \cdot q = 1$ is satisfied exactly. The internal direction $\mathbf{e}$ must be distinguished from the spatial propagation direction $\widehat{\mathbf{n}}$.

In this ansatz, all derivatives of q are proportional to the same internal tangent vector. Therefore,

$$\partial_\mu q \cdot \partial_\nu q = \vartheta^2 \partial_\mu u \, \partial_\nu u, \qquad J_{\mu\nu} = \partial_\mu q \times \partial_\nu q = 0. \tag{166}$$

The single-front ansatz contains an orientation excitation but does not yet contain a two-dimensional internal twist structure. This property follows from the chosen polarization; it does not hold for arbitrary individual tension fronts with more complex internal organization.

12.2. Exact Energy of the Chosen Initial Configuration

We use the signature $(+, -, -, -)$ and $x^0 = ct$. The field action with

$$C(\zeta) = \frac{1}{1+\zeta^2}, \qquad Z(\zeta) = Z_0(1+\zeta^2), \tag{167}$$

$$V(q) = C_s(1+q_3)^2, \qquad U(\zeta) = \frac{b_* \zeta^4}{2(1+\zeta^2)} \tag{168}$$

gives, on (165),

$$K(u) = \frac{A\vartheta^2}{1+a^2u^2} + Z_0 a^2(1+a^2u^2), \tag{169}$$

$$W(u) = C_s[1-\cos(\vartheta u)]^2 + \frac{b_* a^4 u^4}{2(1+a^2u^2)}, \tag{170}$$

$$\mathcal{L}\big|_{\mathrm{front}} = \frac{K(u)}{2}\left(\frac{\dot{u}^2}{c^2} - |\nabla u|^2\right) - W(u). \tag{171}$$

The corresponding energy density and energy flux are

$$\boxed{\varepsilon = \frac{K(u)}{2}\left(\frac{\dot{u}^2}{c^2} + |\nabla u|^2\right) + W(u), \qquad \mathbf{S} = -K(u)\dot{u}\,\nabla u.} \tag{172}$$

This evaluates the entire field ansatz on the specified initial data. It does not assume that evolution in u alone is a closed reduction of the coupled equations.

When the previously introduced quasistatically adjusted regime channel is included, $U_{\mathrm{eff}}(au)$

replaces $U(au)$ in the potential contribution. The kinetic contributions remain unchanged for this potential coupling. A dynamically independent carrier channel would require its own energy and flux balance.

12.3. Analytical Normalization of a Localized Profile

With a fixed front axis, we define

$$s = \widehat{\mathbf{n}} \cdot (\mathbf{x} - \mathbf{b}), \qquad r_\perp = \left| (I - \widehat{\mathbf{n}}\widehat{\mathbf{n}}^{\mathsf{T}})(\mathbf{x} - \mathbf{b}) \right|. \tag{173}$$

At the initial time, let

$$u = F(s)G(r_\perp), \qquad \dot{u} = -c\,\partial_s u, \tag{174}$$

$$F(s) = \exp\left(-\frac{s^2}{2{L_\parallel}^2}\right)\cos(ks + \varphi), \qquad G(r_\perp) = \exp\left(-\frac{r_\perp^2}{2w^2}\right). \tag{175}$$

Here, $L_\parallel$ is the length scale of the longitudinal envelope, and w is the lateral width. The initial velocity describes a directed incoming excitation. This does not assume a packet that permanently retains its shape while propagating at c with finite width and nonlinear coupling.

In the limit of small $|au|$ and $|\vartheta u|$, $K_0 = A\vartheta^2 + Z_0 a^2$. The potential energy begins at fourth order in the excitation amplitudes. The quadratic energy contribution is

$$\boxed{E^{(2)} = \pi K_0 \left(w^2 I_1 + \frac{I_0}{2}\right),} \tag{176}$$

with the exactly evaluable integrals

$$I_0 = \int_{-\infty}^{\infty} F^2\, ds = \frac{\sqrt{\pi} L_\parallel}{2}\left[1 + e^{-(kL_\parallel)^2}\cos(2\varphi)\right], \tag{177}$$

$$I_1 = \int_{-\infty}^{\infty} (F')^2\, ds = \frac{\sqrt{\pi}}{4L_\parallel}\left[1 + 2(kL_\parallel)^2 + e^{-(kL_\parallel)^2}\cos(2\varphi)\right]. \tag{178}$$

Here, $\int G^2\, d^2x_\perp = \pi w^2$ and $\int |\nabla_\perp G|^2\, d^2x_\perp = \pi$ have been used. The formula separates amplitude mixing, carrier wavenumber, front length, and lateral width.

12.4. Generic Amplitude Weights and Energy Distribution

As an asymmetric comparison case, we choose the dimensionless amplitude weights

$$(r_1^{\mathrm{amp}}, r_2^{\mathrm{amp}}, r_3^{\mathrm{amp}}) = (1, 0.98, 0.44). \tag{179}$$

The weights are freely chosen initial parameters. They determine neither frequencies nor incidence angles nor orbital periods. For the same mixture of orientation and loading, we set

$$a_i = \mathcal{A}_{\mathrm{N}} r_i^{\mathrm{amp}}\, \bar{a}, \qquad \vartheta_i = \mathcal{A}_{\mathrm{N}} r_i^{\mathrm{amp}}\, \bar{\vartheta}. \tag{180}$$

With the profile factors and the shared coefficient

$$\mathcal{P}_i = \pi \left(w_i^2 I_{1,i} + \frac{I_{0,i}}{2} \right), \qquad \overline{K}_{\mathrm{F}} = A\bar{\vartheta}^2 + Z_0 \bar{a}^2 \tag{181}$$

, the quadratic limit gives

$$\boxed{E_i^{(2)} = \mathcal{A}_{\mathrm{N}}^2 (r_i^{\mathrm{amp}})^2 \overline{K}_{\mathrm{F}} \mathcal{P}_i.} \tag{182}$$

For equal profile factors, therefore, exactly

$$E_1^{(2)} : E_2^{(2)} : E_3^{(2)} = 1 : 0.9604 : 0.1936. \tag{183}$$

Different wavenumbers change the energy distribution. Equal widths $w_i = L_{\parallel,i} = L_{\parallel}$, phases $\varphi_i = 0$, and $(k_1 L_{\parallel}, k_2 L_{\parallel}, k_3 L_{\parallel}) = (100, 100, 10)$ give, for example,

$$E_1^{(2)} : E_2^{(2)} : E_3^{(2)} \simeq 1 : 0.9604 : 0.00195516. \tag{184}$$

This dimensionless example compares front profiles. It does not establish an electromagnetic frequency calibration.

For sufficiently separated weak fronts with a prescribed quadratic total energy $E_{\mathrm{in}}^{(2)}$,

$$\mathcal{A}_{\mathrm{N}} = \left[\frac{E_{\mathrm{in}}^{(2)}}{\overline{K}_{\mathrm{F}} \sum_i (r_i^{\mathrm{amp}})^2 \mathcal{P}_i} \right]^{1/2}. \tag{185}$$

For overlapping or strong fronts, the full energy, including its interaction terms, is used. The nonlinear energy formulation applies to the saturation region.

12.5. Emergence of the Twist Contribution Under Superposition

For multiple fronts, different internal tangent directions $\mathbf{e}_i$ are allowed. With

$$\mathbf{v} = \sum_i \vartheta_i u_i \mathbf{e}_i, \qquad q = \cos|\mathbf{v}|\, q_{\mathrm{U}} + \frac{\sin|\mathbf{v}|}{|\mathbf{v}|}\mathbf{v} \tag{186}$$

, the norm is preserved. To leading order in small orientation amplitudes,

$$J_{\mu\nu} \simeq \sum_{i<j} \vartheta_i \vartheta_j (\mathbf{e}_i \times \mathbf{e}_j) \left(\partial_\mu u_i \, \partial_\nu u_j - \partial_\nu u_i \, \partial_\mu u_j \right). \tag{187}$$

Such a contribution requires both nonparallel internal orientations and correspondingly independent spacetime profile gradients. For parallel internal directions, the entire orientation remains on a great circle, and $J_{\mu\nu} = 0$. The incidence geometry and internal orientation thus have separate, explicitly calculable roles.

Two suitable fronts can already activate this term. The special role of the third front must therefore be demonstrated through its effect on localization, circulation, and perturbation response. The existence of a twist term alone does not select a triad and does not guarantee binding.

12.6. Scope and Connection to Matter Formation

The derivation provides norm-preserving initial data and an analytical energy normalization for the weak limit. It makes the geometric weighting a defined, variable initial condition. For highly stressed states, the nonlinear energy is used, including the chosen regime channel. The frequency, length, and energy scales, as well as the mixture of orientation and loading, must subsequently be assigned a physical interpretation.

The tangent representation is globally homotopic to the vacuum. The resulting topological condition for the connection to matter is derived in Section 13.1.

12.7. Verification of the Derivation

The norm identity, the vanishing of $J_{\mu\nu}$ for the single-front ansatz, and the energy density were checked symbolically. The two longitudinal integrals were checked in 36 combinations of width, wavenumber, and phase by independent numerical quadrature. The scaled difference $|I_{\mathrm{quad}} - I_{\mathrm{analytic}}|/\max(1, |I_{\mathrm{analytic}}|)$ remained below 10^{-12}. These checks concern the analytical formulas; no time evolution of the field was performed.

13. Matter Formation: Front Coupling and Binding

Within UQSH, matter formation denotes the transition from freely propagating field excitation to a spatially bound organization in continuous motion. The front comparison describes its excitation; the static three-knot calculation investigates phase binding between already localized textures. The transition between these two tasks also depends on the topological class of the initial data.

13.1. Topological Condition for the Connection from Fronts

For the globally defined tangent representation $q = \exp_{q_{\mathrm{U}}}(\mathbf{v})$, where $\mathbf{v}$ vanishes at the vacuum boundary,

$$q_s = \exp_{q_{\mathrm{U}}}(s\mathbf{v}), \qquad 0 \le s \le 1, \tag{188}$$

forms a continuous connection to the vacuum. The initial front data constructed in Section 12 are therefore topologically trivial. In the smooth continuum, their total Hopf class is preserved under norm-preserving evolution with an unchanged vacuum boundary. An organization with nonzero net Hopf charge requires boundary flux or a transition in which at least one of these conditions ceases to hold.

Local twist, binding with vanishing total charge, and oppositely oriented substructures remain possible within the trivial sector. The static knot comparisons address a separate texture family. Their numerical Hopf integrals are discretization diagnostics; for example, $\widehat{Q}_1 \simeq 0.8665$ is not a certified integer topological charge. A dynamical path from the free fronts chosen here to the knot texture is therefore treated as a separate formation problem.

13.2. Organizational Roles of Three Fronts

The scenario considered assigns three tasks to fronts with different effects: a strongly loading, high-frequency excitation produces a preferred local tension direction. A second front, with a suitable orientation, contributes to deflection into circulating field motion. A third, low-frequency excitation is intended to support the restoring response of the bound organization. The gamma-, high-, and low-frequency assignments describe these model roles; incidence direction, phase, field strength, overlap, and regime coupling determine their specific effects.

Saturation is understood here as dynamic redistribution through transfer, scattering, and regime exchange. A persistent organization requires periodic closure, a balanced power budget, and a restoring response to perturbations. These conditions are formulated for the stationary ansatz in Section 21.3.

13.3. Three Distinct Roles of Angles

The spatial encounter angle α measures the angle between the propagation directions $\mathbf{n}_1, \mathbf{n}_2$. The angle β_{pol}, by contrast, measures the orientation of the internal tangent directions $\mathbf{e}_1, \mathbf{e}_2$. In the planar weak-front approximation, the geometric factors from (146) and (147) are

$$X = 2\epsilon^2(1-\cos\alpha)\cos\beta_{\mathrm{pol}}\, f_1' f_2' + O(\epsilon^4), \tag{189}$$

$$|\mathbf{b}| = \frac{\epsilon^2 |f_1' f_2'|\,|\sin\beta_{\mathrm{pol}}\sin\alpha|}{\sqrt{1-|\boldsymbol{\pi}|^2}}. \tag{190}$$

The expression for $\mathbf{b}$ holds exactly within the spherical chart used. For the chosen encounter, $\alpha = 120^\circ$, $\beta_{\mathrm{pol}} = 45^\circ$,

$$2(1-\cos\alpha)\cos\beta_{\mathrm{pol}} = \frac{3}{\sqrt{2}}, \qquad |\sin\beta_{\mathrm{pol}}\sin\alpha| = \sqrt{\frac{3}{8}}. \tag{191}$$

Oppositely directed profile slopes give $X < 0$ and a nonvanishing spatial twist contribution. These angles define an analytically controlled test case. The internal phase shifts δ_2, δ_3 of the following three-knot configuration, by contrast, are compared energetically; they are not spatial incidence angles.

13.4. Calculated Geometry and Phase Binding of Three Knots

Relative to a shared reference point, the knot centers are located at

$$\mathbf{x}_1 = (-35, 0, 0)\,\Delta x, \qquad \mathbf{x}_2 = (35, 0, 0)\,\Delta x, \tag{192}$$

$$\mathbf{x}_3 = (0, 24, -4)\,\Delta x. \tag{193}$$

The base has length $70\Delta x$, and both legs have length $\sqrt{1817}\Delta x$. The spatial apex angle is 110.38804°, and each of the two base angles is 34.80598°. The internal reference phases are $0^\circ, 120^\circ, 240^\circ$; δ_2 and δ_3 denote the additional changes in the second and third phases. Spatial triangle angles and internal phases are distinct quantities.

A preliminary static investigation combines three identically scaled, compactified unit-field knots at fixed spatial geometry and $q_{\mathrm{vac}} = (0, 0, -1)$. The cubic grid contains 208^3 points with $\Delta x = 0.04490924551148$. The discrete normalization uses $C_2 = 0.5408967213314$, $C_4 = 0.05401712081422$, and $C_Q = 0.9735759389529$. These prototype coefficients are not equated here with the coefficients of the later core calculation. The isolated reference knot has the dimensionless energy $E_1 = 94.587602122427$ and the numerical Hopf diagnostic $\widehat{Q}_1 = 0.866496376711$.

The binding difference and its nonadditive remainder are defined by

$$\Delta E_{123} = E_{123} - 3E_1, \tag{194}$$

$$\Delta E_{\mathrm{pair}} = \Delta E_{12} + \Delta E_{13} + \Delta E_{23}, \tag{195}$$

$$\Delta E_{\mathrm{nonadd}} = \Delta E_{123} - \Delta E_{\mathrm{pair}}. \tag{196}$$

For the reference configuration $(\delta_2, \delta_3) = (7.5^\circ, 0^\circ)$, the results are

Quantity	Dimensionless value
ΔE_{123}	-7.599487700546
ΔE_{pair}	-7.314722118453
$\Delta E_{\mathrm{nonadd}}$	-0.284765582093
$(\widehat{Q}_{123} - 3\widehat{Q}_1)/(3\widehat{Q}_1)$	$-8.1130866 \times 10^{-3}$
Minimum neighboring alignment	0.14273124

The combined field organization thus has a lower energy than the three isolated reference knots. The sum of the separately determined pair contributions does not fully capture this reduction. Such nonadditivity can arise from the nonlinear field geometry; it does not require an additionally postulated fundamental three-body term.

Successive refinement of a local phase window, with 25 valid grid points at each stage, yields the best discrete point

$$\delta_2 = 11.7861328125^\circ, \qquad \delta_3 = 1.4580078125^\circ, \tag{197}$$

$$\Delta E_{123} = -7.603577132901, \qquad \Delta E_{\mathrm{improvement}} = -0.004089432355. \tag{198}$$

The many decimal places identify the coordinates of the grid examined. They do not imply a corresponding physical angular precision. The numerical selection criteria are a relative texture error below 0.03, positive neighboring alignment, and a boundary error below 10^{-5}. The energy reduction persists in the grid and interpolation comparisons considered; the ranking of very close phase points can change. The reported phases therefore identify the selected discrete comparison point.

13.5. The Effect of a Phase Change Depends on the Initial State

The two paths from the reference configuration to the refined point give the following energy changes:

Order	First change	Second change
First δ_2, then δ_3	-0.0025695750	-0.0015198573
First δ_3, then δ_2	$+0.0025302779$	-0.0066197103

The same change in δ_3 raises or lowers the energy depending on the phase already assigned to δ_2. The sums agree to within rounding. This finding describes a coupled energy landscape; irreversible hysteresis is not inferred from this static comparison.

13.6. Local Energy Curvature and the Informative Value of the Ritz Sample

A separate discrete knot calculation uses the initial state

$$E = E_2 + E_4 + E_{\text{sat}} = 104.3467647, \qquad \|\nabla_T E\|_{\text{RMS}} = 5.62884 \times 10^{-4}. \tag{199}$$

The remaining tangent gradient is nonzero. This value does not establish sufficiently converged stationarity of the state. The following curvature diagnostic must be distinguished from the stationary core calculation.

Three Lanczos samples of 20 steps each yield the smallest Ritz values μ_{R} and corresponding residual norms ϵ_{R}; the interval bounds are rounded:

Sample	μ_{R}	ϵ_{R}	$[\mu_{\text{R}} - \epsilon_{\text{R}}, \mu_{\text{R}} + \epsilon_{\text{R}}]$
1	0.00944936	0.02003408	$[-0.010585,\ 0.029483]$
2	0.00945256	0.02055771	$[-0.011105,\ 0.030010]$
3	0.00949997	0.02028866	$[-0.010789,\ 0.029789]$

For a self-adjoint discrete Hessian operator H_T, a normalized Ritz vector v, and $r = \mathsf{H}_T v - \mu_{\text{R}} v$,

$$\operatorname{dist}\big(\mu_{\text{R}}, \operatorname{spec} \mathsf{H}_T\big) \le \|r\|_2 = \epsilon_{\text{R}}. \tag{200}$$

The interval therefore contains at least one eigenvalue; it is not a lower bound on the smallest eigenvalue of the entire operator [11]. The residuals exceed all reported Ritz values, and all three intervals include negative values. **The sample is not sufficiently informative to exclude negative curvature directions.** Together with the residual gradient, it does not determine the stability of this knot. The positive curvatures of the later stationary profile basis concern a different state and a different variational space.

13.7. Connection to Persistent Matter Organization

The calculations quantitatively organize two components of matter formation: front geometry and profile phase determine the excitation term, while internal phases change the energy of a localized configuration. A persistent orbit additionally requires periodic closure, a balanced power budget, and a restoring response to admissible perturbations. Section 21.3 formulates these conditions for the rotating field ansatz. The specific identification of such an organization with the electron subsequently requires energy, charge, angular momentum, and electromagnetic coupling to be determined together. The phase values in (198) define a specific static comparison for this purpose, not an already calculated electron orbit.

14. Local Saturation and Spatial Binding

For a fixed spatial orientation field, we set $s_q^2 = \sum_i |\partial_i q|^2$ and $\chi = \zeta^2 \geq 0$. In the local static comparison without loading derivatives, the energy contribution dependent on χ is

$$e_{\mathrm{loc}}(\chi, s_q) = \frac{A s_q^2 + b_* \chi^2}{2(1+\chi)}, \quad \partial_\chi e_{\mathrm{loc}} = \frac{b_* \chi(\chi+2) - A s_q^2}{2(1+\chi)^2}. \tag{201}$$

The derivative changes from negative to positive exactly once. The local minimum and the resulting restoring response are

$$\chi_* = \sqrt{1 + A s_q^2 / b_*} - 1, \tag{202}$$

$$e_{\mathrm{adjusted}} = b_* \left(\sqrt{1 + A s_q^2 / b_*} - 1 \right), \tag{203}$$

$$\mathcal{R}(s_q) = \frac{A s_q}{\sqrt{1 + A s_q^2 / b_*}} \longrightarrow \sqrt{A b_*}. \tag{204}$$

It is this restoring response that is bounded. At high loading, the energy continues to grow, approximately linearly in s_q. $\mathcal{R}$ has the dimensions of energy per area; it is neither a radial pressure nor an upper bound on all source contributions.

For q in motion, $X = k - S$, with $k = |\partial_{\hat{0}} q|^2$, $S = \sum_i |\partial_{\hat{i}} q|^2$, and $\Delta = S - k$. The algebraic limit of (18) gives

$$\frac{\zeta}{(1+\chi)^2} [b_* \chi(\chi+2) - A\Delta] = 0. \tag{205}$$

On the adjusted branch with $\Delta > 0$,

$$\chi_* = \sqrt{1 + A\Delta / b_*} - 1, \qquad u_* := 1 + \chi_*. \tag{206}$$

For $\Delta \leq 0$, the local minimum is at $\zeta = 0$. The zero branch remains a solution otherwise as well. The selection and accessibility of a branch follow from the spatial and temporal dynamics.

Algebraic substitution into the action gives

$$\mathcal{L}_{\mathrm{eff}} = F(X) - \frac{\beta_4}{4} J^2 - V, \qquad F(X) = \begin{cases} b_* [1 - \sqrt{1 - AX/b_*}], & X < 0, \\ AX/2, & X \geq 0. \end{cases} \tag{207}$$

With $u_* = 1$ on the zero branch, $2F_X = A/u_*$. This factor remains inside the derivative in the orientation equation. The corresponding active source is

$$Q_{\mathrm{loc}} = \frac{2Ak}{u_*} + 2\mathcal{E}_4 - 2V - b_*(u_* - 2 + u_*^{-1}). \tag{208}$$

This reduction applies when the omitted loading derivatives are small. Spatially structured cores are treated below using the full loading equation.

14.1. Size Balance and a Rotating Stationary Ansatz

In the static, isolated, unloaded sector, let $E = E_2 + E_4 + E_V$, with

$$E_2 = \frac{A}{2}\int |\nabla q|^2 dV, \quad E_4 = \frac{\beta_4}{2}\int \sum_{i<j} |\partial_i q \times \partial_j q|^2 dV, \quad E_V = \int V dV. \tag{209}$$

Under $q_{s_{\rm dil}}(\mathbf{x}) = q(\mathbf{x}/s_{\rm dil})$,

$$E(s_{\rm dil}) = s_{\rm dil}E_2 + s_{\rm dil}^{-1}E_4 + s_{\rm dil}^3 E_V, \quad E_4 = E_2 + 3E_V, \quad E''(1) = 2E_4 + 6E_V > 0 \tag{210}$$

for a nontrivial stationary state. The relation tests the balance with respect to a change in size. Existence and stability against other perturbations require the full field problem.

For the core in motion, we use cylindrical coordinates (R, z, φ):

$$q = (\sin f \cos\psi, \sin f \sin\psi, -\cos f), \qquad \psi = \varphi + \Omega x^0, \quad f = f(R,z), \quad \zeta = \zeta(R,z). \tag{211}$$

The invariants and the source can be stationary even though q carries internal motion. $\Omega > 0$ is the chosen coordinate phase rate of this ansatz. Define

$$P = R^{-2} - \Omega^2, \quad G_f = |\nabla f|^2, \quad \Delta = G_f + P\sin^2 f, \quad K = AC + \beta_4 P \sin^2 f. \tag{212}$$

The relations $X = -\Delta$ and $J^2 = 2P\sin^2 f\, G_f$ hold. With $dV = R\, dR\, dz\, d\varphi$, $-\int \mathcal{L}_f dV$ becomes

$$\mathcal{F}_\Omega = \int \left[\frac{AC}{2}\Delta + \frac{\beta_4}{2}P\sin^2 f\, G_f + \frac{Z}{2}|\nabla\zeta|^2 + V + U\right] dV, \quad V = C_s(1-\cos f)^2. \tag{213}$$

This is the stationary variational functional at fixed Ω. Its difference between solutions is not released physical energy. Its variation gives the two spatial equations

$$0 = \nabla\cdot(K\nabla f) - P\sin f\cos f(AC + \beta_4 G_f) - V_f, \tag{214}$$

$$0 = Z\nabla^2\zeta + \frac{Z'}{2}|\nabla\zeta|^2 - \frac{A}{2}C'\Delta - U'. \tag{215}$$

Here, $\nabla\cdot\mathbf{b} = R^{-1}\partial_R(Rb_R) + \partial_z b_z$. The corresponding active source is

$$Q = 2AC\Omega^2\sin^2 f + \beta_4\sin^2 f(\Omega^2 + R^{-2})G_f - 2V - 2U. \tag{216}$$

15. A Regular Core and the Buildup of Loading

15.1. Central Expansion and a Finite Gravitational Source

A smooth, axisymmetric, and equatorially symmetric core has the expansion near $r^2 = R^2 + z^2 = 0$

$$f = aR + h_c R^3 + j_c R z^2 + O(r^5), \qquad \zeta = \zeta_c + \beta_R R^2 + \beta_z z^2 + O(r^4). \tag{217}$$

sin f/R is finite; the apparent factors R^{-2} therefore do not produce singular field components. With $u_c = 1 + \zeta_c^2$, the constant term of (215) gives

$$Z_0 u_c(4\beta_R + 2\beta_z) = \frac{\zeta_c}{u_c^2}[b_*\zeta_c^2(\zeta_c^2 + 2) - 2Aa^2]. \tag{218}$$

For the special isotropic choice $\beta_R = \beta_z = \beta_c$,

$$\beta_c = \frac{\zeta_c[b_*\zeta_c^2(\zeta_c^2 + 2) - 2Aa^2]}{6Z_0 u_c^3}. \tag{219}$$

Spatial curvature thus contributes to the loading balance. Algebraic adjustment alone would set the bracketed expression to zero. The leading orientation equation additionally determines

$$\left(\frac{A}{u_c} + \beta_4 a^2\right)(8h_c + 2j_c) - \frac{4Aa\zeta_c\beta_R}{u_c^2} + \frac{2Aa^3}{3u_c} - \frac{\beta_4 a^5}{3} + \Omega^2 a\left(\frac{A}{u_c} - 2\beta_4 a^2\right) = 0. \tag{220}$$

These relations follow by substituting the series into the full equations, not by prescribing a source density.

For $Q = Q_c + Q_R R^2 + Q_z z^2 + O(r^4)$,

$$Q_c = \beta_4 a^4 - \frac{b_*\zeta_c^4}{u_c}, \tag{221}$$

$$Q_R = \frac{2A\Omega^2 a^2}{u_c} + \beta_4(8a^3 h_c - a^6/3 + \Omega^2 a^4) - 2U'(\zeta_c)\beta_R, \tag{222}$$

$$Q_z = 4\beta_4 a^3 j_c - 2U'(\zeta_c)\beta_z, \qquad \overline{Q}_2 = \frac{2Q_R + Q_z}{3}. \tag{223}$$

The angular mean is $\overline{Q} = Q_c + \overline{Q}_2 r^2 + O(r^4)$. Integrating (30) gives the inward-directed monopole contribution

$$g_0(r) = \frac{4\pi G}{c^2}\left(\frac{Q_c}{3}r + \frac{\overline{Q}_2}{5}r^3 + O(r^5)\right). \tag{224}$$

A finite positive central source therefore produces a gravitational field that begins regularly and linearly. The extent of the core is determined by the full profile and its matching.

15.2. Spatial Threshold of a Growing Loading Mode

Let f_0 be a stationary unloaded reference solution with the same environmental data. For $\zeta = \epsilon\eta$, the linear order of (18) gives

$$Z_0\partial_0^2\eta + \mathsf{H}_\zeta\eta = 0, \quad \mathsf{H}_\zeta = -Z_0\nabla^2 - A\Delta_0, \quad \Delta_0 = |\nabla f_0|^2 + P\sin^2 f_0. \tag{225}$$

Because $C'(0) = 0$, backreaction on the orientation begins at second order in the loading. An eigenmode $\mathsf{H}_\zeta u = \lambda_\zeta u$ with $\lambda_\zeta < 0$ has a growing component $\exp[c\sqrt{-\lambda_\zeta/Z_0}\,t]$. For

Dirichlet variations in the finite domain $\mathcal{D}$,

$$\lambda_{\zeta,\min} = \inf_{u\neq 0} \frac{Z_0 \int |\nabla u|^2 dV - A\int \Delta_0 u^2 dV}{\int u^2 dV}. \tag{226}$$

The weighted spatial orientation effect must exceed the cost of distributing the loading. High loading at individual points alone does not determine this threshold.

In the comparison with constant $\Delta_0 = \Delta_c > 0$ inside a sphere $r < L$, at $\lambda_\zeta = 0$,

$$u(r) = u_c \frac{\sin(k_\zeta r)}{k_\zeta r}, \quad k_\zeta^2 = \frac{A\Delta_c}{Z_0}, \quad u'(L) + \sigma_U u(L) = 0 \implies k_\zeta L \cot(k_\zeta L) = 1 - \sigma_U L. \tag{227}$$

Environment	Boundary response	Threshold $A\Delta_c L^2/Z_0$
Vanishing gradient	$u'(L) = 0$	0
Weak exterior continuation	$u'(L) = -u(L)/L$	$\pi^2/4$
Fixed value	$u(L) = 0$	π^2

The middle row follows from $u_{\text{out}} \propto 1/r$ when the exterior Δ_0 is negligible. For a growing mode $\lambda_\zeta = -Z_0 s^2$, outside the sphere $u \propto e^{-sr}/r$, and

$$k_\zeta \cot(k_\zeta L) = -s, \qquad k_\zeta^2 + s^2 = A\Delta_c/Z_0. \tag{228}$$

The $1/r$ limit is a threshold solution; this exterior form is square-integrable only for $s > 0$.

Multiplying (215) by ζ and integrating by parts also gives, nonlinearly,

$$\int_{\mathcal{D}} \left[Z_0(1+2\zeta^2)|\nabla\zeta|^2 + \frac{b_*\zeta^4(\zeta^2+2)}{(1+\zeta^2)^2} \right] dV = A\int_{\mathcal{D}} \frac{\zeta^2\Delta}{(1+\zeta^2)^2} dV + \int_{\partial\mathcal{D}} Z\zeta\partial_n\zeta\, dS. \tag{229}$$

The boundary term is a spatial variational quantity. At a positive interior maximum, furthermore, $b_*\zeta_m^2(\zeta_m^2+2) \le A\Delta(\mathbf{x}_m)$.

15.3. Amplitude Limitation through Coupled Reorganization

At fixed f_0, the exact loading contribution is

$$\mathcal{F}_\Omega[f_0,\zeta] - \mathcal{F}_\Omega[f_0,0] = \int \left[\frac{Z_0}{2}(1+\zeta^2)|\nabla\zeta|^2 + \frac{-A\Delta_0\zeta^2 + b_*\zeta^4}{2(1+\zeta^2)} \right] dV. \tag{230}$$

For a simple mode with $\int u^2 dV = 1$ and $\zeta = a_\zeta u$, near the threshold,

$$\Delta\mathcal{F}_\Omega = \frac{\lambda_\zeta}{2}a_\zeta^2 + \frac{g_4}{2}a_\zeta^4 + O(a_\zeta^6), \tag{231}$$

$$g_4 = Z_0\int u^2|\nabla u|^2 dV + \int (A\Delta_0 + b_*)u^4 dV. \tag{232}$$

Multiplying $\mathsf{H}_\zeta u = \lambda_\zeta u$ by u^3 and integrating by parts gives, at $u|_{\partial\mathcal{D}} = 0$,

$$g_4 = 4Z_0 \int u^2 |\nabla u|^2 dV + (b_* - \lambda_\zeta) \int u^4 dV > 0 \quad (\lambda_\zeta \le 0). \tag{233}$$

The orientation responds with $f = f_0 + a_\zeta^2 \xi_f + O(a_\zeta^4)$. Its second variation is

$$\mathsf{H}_f \xi_f = -\nabla\cdot(K_0 \nabla \xi_f) + W_f \xi_f, \quad K_0 = A + \beta_4 P \sin^2 f_0, \tag{234}$$

$$W_f = P\cos(2f_0)(A + \beta_4 |\nabla f_0|^2) + V_{ff}(f_0) - \nabla\cdot(2\beta_4 P \sin f_0 \cos f_0 \nabla f_0), \tag{235}$$

$$b_f = A\nabla\cdot(u^2 \nabla f_0) - A u^2 P \sin f_0 \cos f_0. \tag{236}$$

Here, $V_{ff} = 2C_s[\sin^2 f + (1 - \cos f)\cos f]$. Through fourth order, the combined change is

$$\Delta\mathcal{F}_\Omega = \frac{\lambda_\zeta}{2} a_\zeta^2 + a_\zeta^4 \left[\frac{g_4}{2} + \langle b_f, \xi_f \rangle + \frac{1}{2} \langle \xi_f, \mathsf{H}_f \xi_f \rangle \right] + \cdots . \tag{237}$$

For invertible H_f with the same boundary data, variation with respect to ξ_f gives

$$\xi_f = -\mathsf{H}_f^{-1} b_f, \qquad g_{\text{coupled}} = g_4 - \langle b_f, \mathsf{H}_f^{-1} b_f \rangle. \tag{238}$$

If H_f is positive, the adjustment reduces the quartic coefficient. For $g_{\text{coupled}} > 0$,

$$a_\zeta^2 = -\frac{\lambda_\zeta}{2 g_{\text{coupled}}} + O({\lambda_\zeta}^2). \tag{239}$$

At a simple threshold crossing, a small stationary loaded continuation emerges. For $g_{\text{coupled}} \le 0$ or an additional zero mode, higher orders or further modes, respectively, must be included. This statement concerns the stationary bifurcation at fixed Ω; its general temporal stability is a separate perturbation problem.

16. A Specific Stationary Core Solution

16.1. Parameters and Results

The following calculation uses the dimensionless reference values from Section 23, $L = 1$, $\Omega^2 = 0.5767110308943739$, and the boundary data

$$f(L, \theta) = b_\partial \sin\theta, \qquad \zeta(L, \theta) = 0. \tag{240}$$

b_∂ is a boundary parameter, and b_* is a potential parameter. The axis and center are regular; both profiles are axisymmetric and equatorially symmetric. (214) and (215) are solved together. The metric is flat in this comparison, corresponding to leading field order in weak gravity.

The unloaded reference reaches the loading threshold crossing at

$$b_{\partial,c} \simeq 2.089890081467. \tag{241}$$

At this threshold, the results are

Quantity	Value in model units
Smallest eigenvalue of H_f within the ansatz	24.2437533
g_4 at fixed orientation profile	12.5618159
$\langle b_f, \mathsf{H}_f^{-1} b_f \rangle$	1.2417016
g_{coupled}	11.3201143

The spatial orientation response reduces the quartic coefficient by approximately 9.9 %; it remains positive. At $b_\partial = b_{\partial,c} + 0.02$, the coupled nonlinear solution gives:

Quantity at identical boundary data	Unloaded	Loaded
$\zeta(0)$	0	0.12499957
Central slope a	2.44848522	2.45517343
Central active source Q_c	4.48759425	4.53658784
Quadratic mean coefficient $\overline{Q}_2$	−3.51992806	−3.87945361

The central loading is determined by the solution and is not an additional input. At this point, the unloaded reference has $\lambda_\zeta = -0.18032212$ and $g_{\text{coupled}} = 11.53290722$. From (239), $a_\zeta \simeq 0.08841788$; the projection of the nonlinear solution onto the normalized mode is 0.08930812. The deviation from the leading approximation is approximately 1.01 %.

The loaded state has $\Delta\mathcal{F}_\Omega = -3.5718472 \cdot 10^{-4}$ and the smallest stationary Hessian eigenvalue 0.35304236 in the resolved ansatz at fixed Ω and fixed boundary values. Its source begins with $\overline{Q} = 4.53658784 - 3.87945361 r^2 + O(r^4)$. The positive Hessian refers to this variational space. Time-dependent, nonaxisymmetric, and additional phase perturbations are not fully included in it.

16.2. Profile, Numerical Method, and Scope

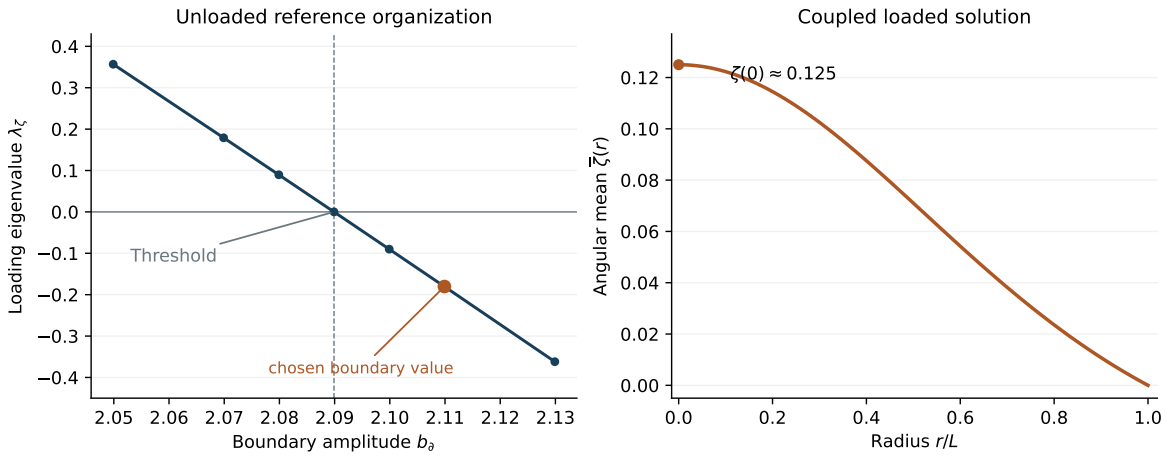


Figure 1. Left: the lowest loading eigenvalue along the stationary unloaded reference. Right: the angular mean of the loading in the coupled state. The lines on the left connect the

calculated points. On the right, the loading reaches approximately 0.125 at the center and satisfies the fixed zero value at $r = L$.

The Galerkin representation used satisfies regularity and boundary values through its basis. With $s = r/L$ and $\mu = \cos\theta$, the radial factor is

$$s^{\ell_{\mathrm{ang}}}(1 - s^2)P_n^{(2,\ell_{\mathrm{ang}}+1/2)}(2s^2 - 1). \tag{242}$$

For $f - b_\partial s\sqrt{1-\mu^2}$, it is multiplied by $\sqrt{1-\mu^2}P'_{\ell_{\mathrm{ang}}}(\mu)$ for odd ℓ_{ang}, and for ζ, by $P_{\ell_{\mathrm{ang}}}(\mu)$ for even ℓ_{ang}. $P_n^{(a,b)}$ are Jacobi polynomials, and $P_{\ell_{\mathrm{ang}}}$ are Legendre polynomials. Normalization uses the measure $2\pi r^2 dr\, d\mu$.

Two previously calculated resolutions use $40 = 8 \times 5$ and $84 = 12 \times 7$ coefficients per field, respectively. The finer quadrature with 96×96 points was checked against 128×128 points. For the strong-form equation residuals E_f, E_ζ, the normalized quantities are

$$\epsilon_f = \frac{L^2\|E_f\|_{L^2}}{A\|f\|_{L^2}}, \qquad \epsilon_\zeta = \frac{L^2\|E_\zeta\|_{L^2}}{Z_0\|\zeta\|_{L^2}}. \tag{243}$$

Diagnostic	40 + 40 coefficients	84 + 84 coefficients
$\zeta(0)$	0.12499979	0.12499957
ϵ_f	$7.08 \cdot 10^{-5}$	$1.10 \cdot 10^{-6}$
ϵ_ζ	$3.80 \cdot 10^{-4}$	$1.22 \cdot 10^{-6}$

These quantities measure equation residuals. The numerical checks also include the variation of the functional, its Hessian, the projection of the strong-form equations, and the spatial balance (229). Both sides of this balance give 0.08046686698 in the finer quadrature.

The comparison yields a stationary loaded profile and the linear growth mode of its unloaded reference. No temporal formation path was integrated. Because $f(L, \pi/2) \simeq 2.11$, the boundary still lies in the nonlinear region. The outer envelope amplitude follows only from a subsequent coupled transition solution.

16.3. Integrated Source Difference of the Core

The central increase alone does not determine the gravitational effect of the entire sphere. The loaded and unloaded profiles are therefore integrated at identical boundary data over the domain $0 \le r \le L = 1$ considered. For the stationary ansatz,

$$Q = \frac{2A\Omega^2 \sin^2 f}{1+\zeta^2} + \beta_4 \sin^2 f \left(\Omega^2 + R^{-2}\right)|\nabla f|^2 - 2C_s(1-\cos f)^2 - \frac{b_*\zeta^4}{1+\zeta^2}. \tag{244}$$

With $\overline{Q} = \frac{1}{2}\int_{-1}^{1} Q\, d\mu$ and the dimensionless quantities

$$I_X(r) = 4\pi \int_0^r s^2 \overline{Q}_X(s)\, ds, \qquad \Delta I = I_A - I_U \tag{245}$$

, the finer solution gives:

Quantity	Value at $r = 1$
Unloaded integrated source I_U	11.43802199
Loaded integrated source I_A	11.44005163
Difference ΔI	0.00202964
Relative increase $100\Delta I/I_U$	0.0177447 %

The four source terms contribute to the integrated difference, in the order given above,

$$-0.00392770 + 0.00611545 - 0.00010894 - 0.00004917 = 0.00202964. \tag{246}$$

The quartic orientation coupling thus outweighs the three negative changes. In the angular mean, the local difference changes sign at $r \simeq 0.587752$: the source increases inside and decreases outside. Consequently, the central increase of approximately 1.09 % amounts to an increase of only approximately 0.0177 % for the entire sphere examined.

The evaluation uses only the determined 84 + 84 coefficients. Gaussian quadratures with 96 and 128 points per direction differ by at most $5.3 \cdot 10^{-14}$ in the integrated individual contributions. The reconstructed angular means reproduce the reference values to within $1.8 \cdot 10^{-15}$. This agreement checks the quadrature of the fixed profiles. Estimating the spatial discretization error additionally requires refinement of the profile basis.

16.4. Radial Profile and Weak Gravitational Response

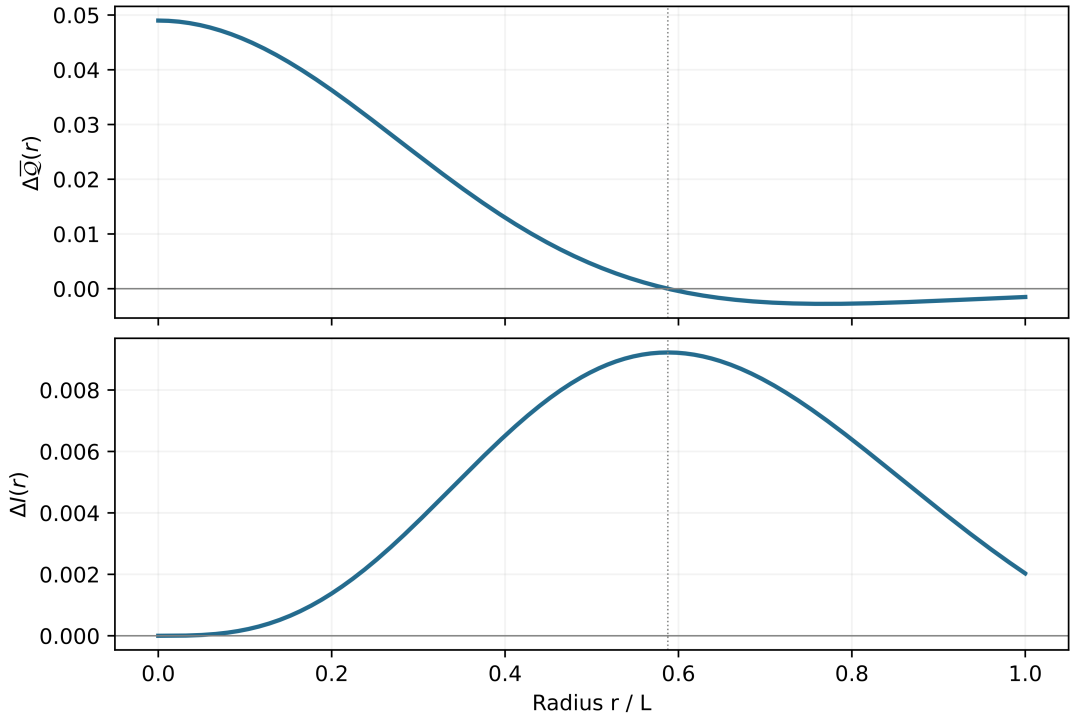


Figure 2. Top: the local source difference. Bottom: its cumulative integral up to the given radius. The outer negative contribution reduces the inner increase but does not cancel it within the domain considered. The curves are based on the same profiles.

For a physical length scale ℓ_0 and source scale Q_0 with $r_{\mathrm{phys}} = \ell_0 r$, the weak, quasistatic monopole branch gives

$$\Delta M_{\mathrm{dyn}}(< r) = \frac{Q_0 \ell_0^3}{c^2} \Delta I(r), \qquad \Delta g_r(r) = -\frac{G Q_0 \ell_0}{c^2} \frac{\Delta I(r)}{r^2}. \tag{247}$$

The additional radial monopole contribution at the sphere boundary is directed inward. Its magnitude is 0.0177447 % of the unloaded monopole contribution there generated by this sphere. At the center, the acceleration vanishes regularly because $\Delta I = O(r^3)$. The normalized acceleration change is $-\Delta I(r)/r^2$.

This statement concerns the weak metric response to the calculated core. It does not assume small field amplitudes, but it does assume a weak gravitational potential. The axisymmetric solution may additionally have higher multipoles; the radial monopole contribution was calculated here. The sphere does not yet have complete exterior matching. Contributions from a responding environment and conversion to kilograms or acceleration units require this matching and the physical scales, respectively.

17. Core Adjustment through the Regime Channel

The spatial solution with an active regime channel connects the local carrier response to the reorganization of the entire core considered. In this comparison, the central loading increases by approximately 0.637 %. The field energy decreases slightly, while the integrated active gravitational source increases. Both changes arise from the same stationary field organization and its stress balance.

17.1. Comparison at Identical Boundary Data

The calculation uses $L = 1$, $\Omega^2 = 0.5767110308943739$, and

$$f(L, \theta) = 2.1098900814674373 \sin\theta, \qquad \zeta(L, \theta) = 0. \tag{248}$$

Regularity, axial symmetry, and equatorial symmetry match the existing core representation. The channel uses $\eta_\Sigma = 0.5$, $\kappa = 0.2$, and the fixed energy scale $\mathcal{E}_\Sigma = 6.009533038767751 \times 10^{-5}$. The auxiliary indices range from −20 to 20; outer neighbors vanish. All numbers are given in dimensionless model units.

Three clearly identified states are compared:

State	Meaning
B	Already loaded stationary reference without an additional channel.
F	The same spatial fields with the channel switched on.
A	Jointly adjusted stationary fields with an active channel.

F is the fixed intermediate state used to evaluate the immediate channel contribution. For a

target quantity Y, exactly

$$\Delta Y = Y_A - Y_B = \underbrace{Y_F - Y_B}_{\Delta Y_{\mathrm{direct}}} + \underbrace{Y_A - Y_F}_{\Delta Y_{\mathrm{adjustment}}} . \tag{249}$$

The reference B is already loaded. The comparison between unloaded and loaded cores in Section 16 addresses a different change of state.

The two profiles f, ζ are determined from the functional

$$\mathcal{F}_\Omega^\Sigma[f,\zeta] = \mathcal{F}_\Omega^0[f,\zeta] - \mathcal{E}_\Sigma \int_{B_L} W_\kappa\!\left(\frac{\eta_\Sigma U(\zeta)}{\mathcal{E}_\Sigma}\right) \mathrm{d}V \tag{250}$$

The channel initially changes the loading variation; the existing mutual coupling simultaneously adjusts the orientation. The central loading and active source give

Quantity	Reference B	Adjusted A	Relative change
$\zeta(0)$	0.1249995685	0.1257952840	+0.636575 %
$\lVert\zeta\rVert_{L^2}$	0.0893091203	0.0898702369	+0.628286 %
$Q(0)$	4.5365878397	4.5372626934	+0.0148758 %

17.2. Energy, Stresses, and the Integrated Gravitational Source

The field energy E and the functional $\mathcal{F}_\Omega$ varied at fixed Ω are evaluated separately. The integrated source is

$$I_X(r) = 4\pi \int_0^r s^2 \overline{Q}_X(s)\,\mathrm{d}s, \qquad \overline{Q}_X = \frac{1}{2}\int_{-1}^{1} Q_X\,\mathrm{d}\mu. \tag{251}$$

The entire sphere gives

Quantity	Reference B	Adjusted A
E	20.91454452193	20.91450159478
$\mathcal{F}_\Omega$	17.89951067439	17.89950829458
$I(L)$	11.44005162965	11.44008164208

The decomposition in (249) makes the mutual adjustment quantitatively explicit:

Quantity	Immediate channel effect	Field adjustment	Total
ΔE	-2.324694×10^{-6}	-4.060246×10^{-5}	-4.292715×10^{-5}
$\Delta\mathcal{F}_\Omega$	-2.324694×10^{-6}	-5.512126×10^{-8}	-2.379815×10^{-6}
$\Delta I(L)$	$+4.649387\times10^{-6}$	$+2.536304\times10^{-5}$	$+3.001243\times10^{-5}$

Of the integrated source increase, 15.49 % comes from the immediate channel contribution and 84.51 % from the field adjustment. Relative to B, E changes by -0.000205250 % and $I(L)$ by $+0.000262345$ %.

These signs follow from the distinction between field energy and active source $Q = \varepsilon + \sum_i p_i$. For identical fields, $\Delta U = -\mathcal{E}_\Sigma W_\kappa$ produces the changes

$$\Delta\varepsilon_{\text{direct}} = -\mathcal{E}_\Sigma W_\kappa, \quad \Delta p_{i,\text{direct}} = +\mathcal{E}_\Sigma W_\kappa, \quad \Delta Q_{\text{direct}} = +2\mathcal{E}_\Sigma W_\kappa. \tag{252}$$

After profile adjustment, the integrated sum of stresses satisfies $\Pi_{\text{int}} = \int \sum_i p_i \, \mathrm{d}V$

$$\Delta\Pi_{\text{int}} = \Delta I - \Delta E = 7.29395817 \times 10^{-5}. \tag{253}$$

A stronger active source is thus compatible with lower field energy: the stress response outweighs the decrease in energy.

The four source contributions from temporal orientation, quartic field organization, and the two potentials contribute:

$$\Delta I = \left(-4.93378618 + 7.70882178 - 0.13664959 + 0.36285719\right) \times 10^{-5}. \tag{254}$$

The quartic contribution outweighs the changes in the opposite direction. In addition, $E - \mathcal{F}_\Omega = 2T_{\text{rot}}$ is reproduced in all three states. With $\Delta T_{\text{rot}} = -2.02736674 \times 10^{-5}$, correspondingly, $\Delta E = \Delta\mathcal{F}_\Omega + 2\Delta T_{\text{rot}}$. The comparison concerns stationary states at fixed Ω; it does not determine radiative energy released during a temporal transition.

17.3. Inner Enhancement and Outer Counteracting Change

The local source change is distributed nonuniformly in space. In the angular mean, the source increases in the inner region; farther out, it decreases relative to the reference. The radial sample points constrain the sign change to $0.5791 < r/L < 0.5912$. The immediate channel contribution is nonnegative; the outer negative contribution arises from profile adjustment.

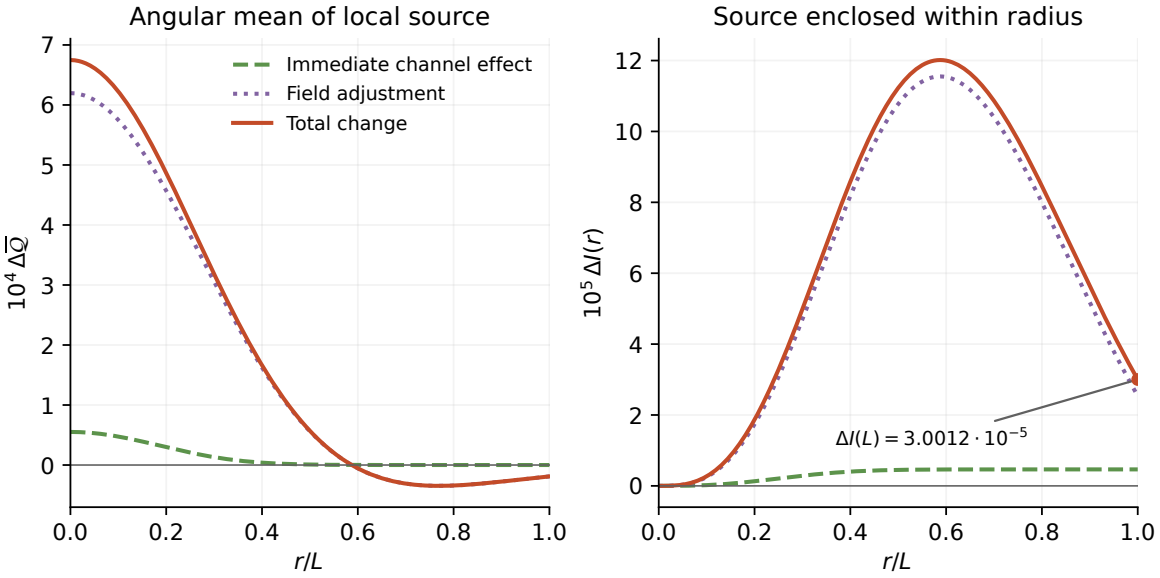


Figure 3. Left: the change in the angular mean of the active source. Right: its cumulative integral up to the given radius. The lines connect the radial sample points of the solution. The axes show scaled dimensionless quantities.

Among the radial points evaluated, ΔI reaches its largest value, $1.20142591 \times 10^{-4}$, at $r/L = 0.5875$. By the boundary, this inner increase has fallen to $3.00124320 \times 10^{-5}$. The outer region thus compensates for approximately 75.02 % of the excess enclosed at this sample point. The additional enclosed source remains positive at all evaluated radii $r > 0$. The central relative increase of 0.0148758 % and the integrated increase of 0.000262345 % therefore describe different aspects of the same reorganization.

With $r_{\mathrm{phys}} = \ell_* \widetilde{r}$ and $Q_{\mathrm{phys}} = (E_*/\ell_*^3)\widetilde{Q}$, the additional weak radial monopole contribution is

$$\Delta M_{\mathrm{dyn}}(< \widetilde{r}) = \frac{E_*}{c^2} \Delta I(\widetilde{r}), \qquad \Delta g_{0,r}(\widetilde{r}) = -\frac{G E_*}{c^2 \ell_*^2} \frac{\Delta I(\widetilde{r})}{\widetilde{r}^2}. \tag{255}$$

At the sphere boundary, this contribution is directed inward. Its magnitude equals 0.000262345 % of the monopole contribution generated by the reference sphere. The orientation of the acceleration vector follows from the minus sign; $\Delta I > 0$ increases the magnitude of the attraction. This statement concerns the source in the calculated domain and the weak metric connection.

17.4. Numerical Checks and Physical Connection

The stationary state with a regime channel is determined from the reference without an additional channel. The Galerkin basis contains $12 \times 7 = 84$ coefficients each for orientation and loading. No field time steps were performed. The stationary solution uses a quadrature of 96×96 points; the same coefficients are subsequently evaluated with 128×128 points.

Diagnostic	Result
Gradient norm in the stationary solve	1.48×10^{-14}
Gradient norm with the check quadrature	6.35×10^{-11}
Normalized strong-form orientation residual	1.10×10^{-6}
Normalized strong-form loading residual	8.74×10^{-6}
Smallest Hessian eigenvalue in the fixed ansatz	0.346256
Largest quadrature difference in the energies	8.89×10^{-14}
Largest quadrature difference in the integrated sources	2.67×10^{-14}
Largest error in $E - \mathcal{F}_\Omega - 2T_{\mathrm{rot}}$	4.89×10^{-15}

The positive Hessian eigenvalue describes the local stationary restoring response in the chosen variational space at fixed boundary values and fixed Ω. The agreement of the integrals checks the quadrature of the same spatial representation.

The functional derivatives were checked for the base model and the active channel in three directions with step sizes 3×10^{-5}, 1.5×10^{-5}, and 7.5×10^{-6}. Across all cases tested, two Richardson estimates yield gradient errors below 3.35×10^{-10} and Hessian-action errors below 1.38×10^{-13} in the chosen normalization. The previously dominant raw finite-difference error decreases by approximately a factor of four when the step size is halved; for this check, its cause is thus identified as the leading quadratic step-size contribution. The field equation was not changed for this purpose.

Arithmetic consistency checks of the source decomposition, energy identities, and derivative values give a maximum deviation below 1.25×10^{-14}.

Scope and next development. The comparison gives an explicit stationary response to the specified regime channel: direct potential adjustment, reorganization, stress change, and gravitational source increase are quantitatively connected. The percentages belong to this single parameter case. For the physical connection, the core, transition, environment, and metric are solved together, and the length and energy scales are calibrated. Refinement of the spatial basis of the adjusted state and a general dynamical stability analysis remain separate verification steps. Finite relaxation times, irreversible dedensification, and detector cadence require corresponding time evolution. The metric closure uses GR with invariant c.

18. From the Nonlinear Core to the Outer Field Envelope

18.1. Exact Projection and Determination of the Exterior Amplitudes

The lowest angular component of a smooth axisymmetric profile is

$$F(r) = \frac{3}{4} \int_0^\pi f(r, \theta) \sin^2 \theta \, d\theta. \tag{256}$$

For $f = F \sin \theta$, the projection gives exactly F. The full orientation equation can be written as $\mathcal{L}_\Omega f = \mathcal{N}$, where

$$\mathcal{L}_\Omega = \partial_R^2 + R^{-1} \partial_R + \partial_z^2 - P, \tag{257}$$

$$A\mathcal{N} = P\{\sin f \cos f (AC + \beta_4 G_f) - Af\} + V_f - \nabla \cdot [(K - A) \nabla f]. \tag{258}$$

Angular projection gives the identity

$$F'' + \frac{2}{r} F' + \left(\Omega^2 - \frac{2}{r^2} \right) F = S_1(r), \quad S_1 = \frac{3}{4} \int_0^\pi \mathcal{N} \sin^2 \theta \, d\theta. \tag{259}$$

S_1 contains the loading, nonlinearity, and other angular components of the same profile.

In the weak unloaded exterior interval, S_1 is small. The homogeneous basis functions and their Wronskian are

$$y_c = \frac{\cos x - \sin x / x}{r}, \quad y_s = \frac{\sin x + \cos x / x}{r}, \quad x = \Omega r, \qquad y_c y_s' - y_c' y_s = \frac{\Omega}{r^2}. \tag{260}$$

From $F = d_c y_c + d_s y_s$, at every matching point,

$$d_c = \frac{r^2}{\Omega} (y_s' F - y_s F'), \qquad d_s = \frac{r^2}{\Omega} (y_c F' - y_c' F). \tag{261}$$

For the full inhomogeneous profile, differentiation gives $d_c' = -r^2 y_s S_1 / \Omega$ and $d_s' = r^2 y_c S_1 / \Omega$. With $F = ar + (4h_c + j_c) r^3 / 5 + O(r^5)$ at the center,

$$d_c(r_m) = -\frac{3a}{\Omega^2} - \frac{1}{\Omega} \int_0^{r_m} s^2 y_s(s) S_1(s) \, ds, \tag{262}$$

$$d_s(r_m) = \frac{1}{\Omega}\int_0^{r_m} s^2 y_c(s) S_1(s)\, ds. \tag{263}$$

The integrals are regular at the center. They determine how the core and nonlinear transition supply the exterior amplitude. No central series is extrapolated into the far field for this purpose. The envelope coefficient is

$$D_H = \frac{2A\Omega^2}{3}(d_c^2 + d_s^2). \tag{264}$$

In the homogeneous exterior interval, D_H is constant; changing the evaluation surface within that interval introduces no new parameter.

18.2. Gravitational Monopole Contribution of the Weak Envelope

For the lowest linear exterior mode, angular averaging of (216) gives

$$\overline{Q}_{\mathrm{lin}} = \frac{4A\Omega^2}{3}F^2. \tag{265}$$

Set $\mathcal{A}_H = d_c^2 + d_s^2$. An exact antiderivative is

$$\mathcal{J}(r) = \frac{\mathcal{A}_H}{2}r + \mathcal{H}(r), \qquad \mathcal{J}'(r) = r^2F^2, \tag{266}$$

$$\mathcal{H}(r) = \frac{d_c^2 - d_s^2}{4\Omega}\sin(2\Omega r) - \frac{d_c d_s}{2\Omega}\cos(2\Omega r) - \frac{[d_s\cos(\Omega r) - d_c\sin(\Omega r)]^2}{\Omega^2 r}. \tag{267}$$

The derivative of the last line, together with $\mathcal{A}_H/2$, reproduces the square of the combination of the two basis functions. Let the already enclosed active source be

$$M_m = \frac{4\pi}{c^2}\int_0^{r_m} s^2\overline{Q}_{\mathrm{full}}(s)\, ds. \tag{268}$$

Radial Poisson integration in the exterior interval then gives

$$M_{\mathrm{dyn}}(r) = M_m + \frac{16\pi A\Omega^2}{3c^2}[\mathcal{J}(r) - \mathcal{J}(r_m)]. \tag{269}$$

The core is included once, with its full source. For $v_0^2(r) := r\Psi_0'(r) = GM_{\mathrm{dyn}}(r)/r$,

$$v_0^2(r) = v_H^2 + \frac{C_H}{r} + \frac{\Gamma_H\mathcal{H}(r)}{r}, \tag{270}$$

$$\Gamma_H = \frac{16\pi GA\Omega^2}{3c^2}, \quad v_H^2 = \frac{\Gamma_H\mathcal{A}_H}{2} = \frac{4\pi GD_H}{c^2}, \quad C_H = GM_m - \Gamma_H\mathcal{J}(r_m). \tag{271}$$

For $\mathcal{A}_H > 0$, the bound is

$$\frac{|\Gamma_H\mathcal{H}(r)/r|}{v_H^2} \le \frac{1}{2\Omega r} + \frac{2}{(\Omega r)^2}. \tag{272}$$

If these terms and C_H/r are small, the constant contribution dominates. This integrated monopole mechanism requires no additional radial phase averaging. A pointwise smooth source D_H/r^2, by contrast, requires corresponding averaging.

The result applies to the weak exterior interval and the specified angular component. Continuing it unchanged to $r = \infty$ would produce a linearly growing enclosed source; a physical application requires the transition to the actual environment. A specific galactic rotation curve, including higher angular components, follows from the coupled solution and its physical calibration.

18.3. Unaveraged Angular Structure and Orbital Evaluation

The direction-dependent potential response is also given before radial averaging. Set $Q_0(r) = 4A\Omega^2 F(r)^2/3$ and $\kappa_G = 4\pi G/c^2$. For the same lowest exterior mode,

$$Q_{\mathrm{grav}}(r,\theta) = Q_0(r)[1 - P_2(\cos\theta)], \qquad P_2(\mu) = \frac{3\mu^2 - 1}{2}. \tag{273}$$

In the monopole–quadrupole sector, we write $\Psi = \Psi_0(r) + \Psi_2(r)P_2(\cos\theta)$. Then

$$(r^2\Psi_0')' = \kappa_G r^2 Q_0, \qquad (r^2\Psi_2')' - 6\Psi_2 = -\kappa_G r^2 Q_0. \tag{274}$$

A complete quadrupole solution in the interval $r_m < r < r_e$ is

$$\boxed{\Psi_2(r) = \frac{\kappa_G}{5}\left[\frac{1}{r^3}\int_{r_m}^{r} s^4 Q_0(s)\,ds + r^2\int_{r}^{r_e}\frac{Q_0(s)}{s}\,ds\right] + a_2 r^2 + \frac{b_2}{r^3}.} \tag{275}$$

The constants a_2, b_2 contain the matching to the core and environment. They are determined together with the monopole boundary data. The integral form contains the radial wave structure without phase averaging.

For slow gravitational test motion in the equatorial plane, at $r = R$, the exact relations within this potential approximation are

$$\boxed{\begin{aligned} \omega_{\mathrm{orb}}^2(R) &= \frac{\Psi_0'(R) - \frac{1}{2}\Psi_2'(R)}{R}, \\ \nu_z^2(R) &= \omega_{\mathrm{orb}}^2(R) + \frac{3\Psi_2(R)}{R^2}, \qquad v_{\mathrm{circ}}^2 = R^2\omega_{\mathrm{orb}}^2. \end{aligned}} \tag{276}$$

The rates refer to physical time. A circular orbit requires $\omega_{\mathrm{orb}}^2 > 0$; positive ν_z^2 characterizes the vertical restoring response in this approximation. Circular motion and the vertical restoring response can thus be calculated from the same unaveraged source. Higher gravitational angular components are added for a correspondingly extended field organization.

Setting $Q_0 = D_H/r^2$ in the leading averaged exterior region and considering the particular solution with $v_H^2 = \kappa_G D_H$ gives

$$\Psi_0 = v_H^2 \ln(r/r_0), \qquad \Psi_2 = \frac{v_H^2}{6}, \qquad \frac{\nu_z^2}{\omega_{\mathrm{orb}}^2} = \frac{3}{2}. \tag{277}$$

The averaged ratio is thus identified as a limiting case. For the unaveraged full solution, the integrals and boundary contributions from (275) are used; the ratio is not prescribed there.

19. Core, Transition, and Environment as a Single Boundary-Value Problem

At an imaginary surface Γ_m, we choose the normal to point from the interior to the exterior. The boundary term in the variation of (213) is

$$\delta\mathcal{F}_\Omega\big|_{\partial\mathcal{D}} = \int_{\partial\mathcal{D}} (K\partial_n f\ \delta f + Z\partial_n\zeta\ \delta\zeta) dS. \tag{278}$$

In the absence of a separate surface source, therefore,

$$[f] = [\zeta] = 0, \qquad [K\partial_n f] = [Z\partial_n\zeta] = 0. \tag{279}$$

The jumps use the same normal direction. On a regular exterior solution branch with matching value $\gamma = (f_{\Gamma_m}, \zeta_{\Gamma_m})$ and prescribed outer data $\mathcal{B}_{\text{env}}$, let

$$\Gamma_{\text{env}}[\gamma] = \mathcal{F}_{\Omega,\text{out}}[f_\gamma, \zeta_\gamma; \mathcal{B}_{\text{env}}]. \tag{280}$$

The outward domain normal of the exterior subregion points inward at Γ_m. Therefore, $\delta\Gamma_{\text{env}}/\delta\gamma = -(K\partial_n f, Z\partial_n\zeta)_{\text{out}}$, and the combined problem is

$$\delta(\mathcal{F}_{\Omega,\text{in}} + \Gamma_{\text{env}}) = 0. \tag{281}$$

Γ_{env} is the eliminated exterior field contribution from the same action. Its form is determined by the exterior solution.

In the weak comparison with negligible exterior Δ_0 and small nonlinear loading terms, $\nabla^2\zeta_{\text{out}} = 0$. This gives

$$\zeta_{\text{out}} = \sum_{\ell_{\text{ang}}} z_{\ell_{\text{ang}}} (r_m/r)^{\ell_{\text{ang}}+1} P_{\ell_{\text{ang}}}(\cos\theta), \tag{282}$$

$$\partial_r\zeta_{\ell_{\text{ang}}}(r_m) = -\frac{\ell_{\text{ang}}+1}{r_m} z_{\ell_{\text{ang}}}, \tag{283}$$

$$\Gamma^{(2)}_{\zeta,\text{out}} = -\frac{Z_0}{2}\int_{r=r_m} \zeta\partial_r\zeta\, dS = 2\pi Z_0 r_m \sum_{\ell_{\text{ang}}} \frac{\ell_{\text{ang}}+1}{2\ell_{\text{ang}}+1} z^2_{\ell_{\text{ang}}}. \tag{284}$$

The last line follows by integrating the harmonic exterior energy by parts. With equatorial symmetry, only even ℓ_{ang} are present. A fixed $\zeta(r_m) = 0$ alone does not provide smooth matching to $\zeta_{\text{out}} \equiv 0$; the normal derivative would also have to vanish.

If an exterior zero value occurs only at $r_e > r_m$, the general harmonic form $a_{\ell_{\text{ang}}} r^{\ell_{\text{ang}}} + b_{\ell_{\text{ang}}} r^{-\ell_{\text{ang}}-1}$, with $\rho_m = r_m/r_e$, gives

$$\partial_r\zeta_{\ell_{\text{ang}}}(r_m) = -\frac{(\ell_{\text{ang}}+1) + \ell_{\text{ang}}\rho_m^{2\ell_{\text{ang}}+1}}{r_m(1-\rho_m^{2\ell_{\text{ang}}+1})}\, z_{\ell_{\text{ang}}}. \tag{285}$$

The exterior environment helps determine the inner boundary response.

19.1. Orientation Matching and Energy Supply

In the linear exterior orientation region, let $F(r_e) = F_e$ be prescribed. Using the basis functions from Section 18, define

$$\mathcal{U}(r) = y_c(r)y_s(r_e) - y_s(r)y_c(r_e). \tag{286}$$

Solving the two linear boundary equations gives, for $\mathcal{U}(r_m) \neq 0$,

$$F'(r_m) = \frac{\mathcal{U}'(r_m)}{\mathcal{U}(r_m)}F(r_m) + \frac{\Omega}{r_m^2\mathcal{U}(r_m)}F_e. \tag{287}$$

When the denominator vanishes, the original two boundary equations remain decisive; they determine resonance compatibility. The nonlinear transition must be completed before this linear approximation is applied.

The effect of the remaining inhomogeneity on the amplitude can be bounded using (259). Set

$$\mathcal{B}_{12} = \frac{1}{\Omega}\int_{r_1}^{r_2} r\sqrt{1 + \frac{1}{\Omega^2 r^2}}\,|S_1(r)|\,dr. \tag{288}$$

Because $y_c^2 + y_s^2 = r^{-2}[1 + (\Omega r)^{-2}]$,

$$\|\mathbf{d}(r_2) - \mathbf{d}(r_1)\| \le \mathcal{B}_{12}, \qquad \frac{|D_H(r_2) - D_H(r_1)|}{D_H(r_1)} \le 2\eta_H + \eta_H^2, \quad \eta_H = \frac{\mathcal{B}_{12}}{\|\mathbf{d}(r_1)\|}, \tag{289}$$

where $\mathbf{d} = (d_c, d_s)$ and $\|\mathbf{d}(r_1)\| > 0$. A small field amplitude, small loading, and the remaining angular structure remain additional conditions for the full linear exterior description.

For a complex weak exterior amplitude

$$H_{\mathrm{W}} = c_{\mathrm{out}}(y_c - iy_s) + c_{\mathrm{in}}(y_c + iy_s), \quad q_1 + iq_2 \simeq H_{\mathrm{W}}\sin\theta\, e^{i(\varphi+\Omega x^0)} \tag{290}$$

, flux integration gives

$$L_r = \frac{8\pi A c\Omega^2}{3}(|c_{\mathrm{out}}|^2 - |c_{\mathrm{in}}|^2). \tag{291}$$

The real stationary profile has $c_{\mathrm{out}} = (d_c + id_s)/2$ and $c_{\mathrm{in}} = (d_c - id_s)/2$, and hence vanishing net flux. A configuration with an outgoing energy flux requires a corresponding energy balance for the interior.

For a background held unchanged in the interior, a negative Dirichlet test direction from (226) persists under mere enlargement of the domain: extending the test function by zero gives

$$\inf \mathcal{R}_{\mathrm{tot}} \le \inf \mathcal{R}_{\mathrm{in}}^{\mathrm{Dirichlet}}. \tag{292}$$

With coupled adjustment, however, f_0 and hence Δ_0 can change. The threshold of the overall system is therefore determined with its own environment.

20. Gravitational Backreaction on the Same Field Organization

20.1. Stationary Geometry and Local Phase Rate

For a stationary, axisymmetric, circular geometry in the domain $N > 0$, we set

$$ds^2 = N^2(dx^0)^2 - h_{ab}dx^a dx^b - \varpi^2(d\varphi - \omega dx^0)^2, \quad (x^1, x^2) = (R, z). \tag{293}$$

h_{ab} is positive definite, and ϖ is the circumferential radius. The functions are independent of x^0 and φ. For observers normal to the spatial slices,

$$n = N^{-1}(\partial_0 + \omega\partial_\varphi), \qquad d\tau^{\text{proper}} = N dt, \quad \widehat{\Omega} := n(\psi) = \frac{\Omega + \omega}{N}. \tag{294}$$

The plus sign follows from $\psi = \varphi + \Omega x^0$; the pattern rotation has coordinate angular velocity $-c\Omega$. The local phase rate of this ansatz is $c\widehat{\Omega}$. The normalization of coordinate time is part of the environmental specification.

Direct contraction gives

$$H_g = h^{ab} f_a f_b, \quad P_g = \varpi^{-2} - \widehat{\Omega}^2, \quad \Delta_g = H_g + P_g \sin^2 f, \quad X = -\Delta_g, \quad J^2 = 2P_g \sin^2 f \, H_g. \tag{295}$$

With $w_g = N\varpi\sqrt{h_2}$ and $h_2 = \det h_{ab}$, the stationary functional at fixed geometry is

$$\mathcal{F}_{\Omega,g} = \int w_g \left[\frac{AC}{2}\Delta_g + \frac{\beta_4}{2} P_g \sin^2 f \, H_g + \frac{Z}{2} h^{ab}\zeta_a\zeta_b + V + U \right] dR\, dz\, d\varphi. \tag{296}$$

With $K_g = AC + \beta_4 P_g \sin^2 f$, its field variation gives

$$0 = w_g^{-1}\partial_a(w_g K_g h^{ab} f_b) - P_g \sin f \cos f (AC + \beta_4 H_g) - V_f, \tag{297}$$

$$0 = Z w_g^{-1}\partial_a(w_g h^{ab}\zeta_b) + \frac{Z'}{2} h^{ab}\zeta_a\zeta_b - \frac{A}{2}C'\Delta_g - U'. \tag{298}$$

The metric additionally satisfies (20). For $N = 1$, $\omega = 0$, $\varpi = R$, $h_{ab} = \delta_{ab}$, the flat-space stationarity equations are recovered. Time measurement, spatial distances, and frame dragging thus jointly change the field profiles.

20.2. Explicit Gravitational Source and Time Measurement

Relative to n, $n(\zeta) = 0$. Substituting the rotating ansatz into (19) gives

$$Q_g = 2AC\widehat{\Omega}^2 \sin^2 f + \beta_4 \sin^2 f(\widehat{\Omega}^2 + \varpi^{-2})H_g - 2V - 2U, \tag{299}$$

$$T_{\hat{0}\hat{\varphi}} = \frac{\widehat{\Omega}}{\varpi} \sin^2 f(AC + \beta_4 H_g). \tag{300}$$

The azimuthal momentum component is generally nonzero. The corresponding Einstein components therefore determine the frame dragging ω together with the field.

The positive spatial metric is $h_{ij}dx^i dx^j = h_{ab}dx^a dx^b + \varpi^2 d\varphi^2$. With its covariant derivative

D_i and the shift $\boldsymbol{\beta}_{\rm sh} = -\omega\partial_\varphi$, we define

$$\mathcal{K}_{ij} = \frac{D_i\beta_{{\rm sh},j} + D_j\beta_{{\rm sh},i} - \partial_0 h_{ij}}{2N}. \tag{301}$$

Stationarity and axial symmetry give

$$\mathcal{K}^i{}_i = 0, \quad \mathcal{K}_{a\varphi} = -\frac{\omega^2}{2N}\partial_a\omega, \quad \mathcal{K}_{ij}\mathcal{K}^{ij} = \frac{\omega^2}{2N^2}h^{ab}\omega_a\omega_b. \tag{302}$$

The trace evolution equation of the Einstein system is

$$(\partial_0 - \mathcal{L}_{\boldsymbol{\beta}_{\rm sh}})\mathcal{K} = -D_iD^iN + N\left[\mathcal{K}_{ij}\mathcal{K}^{ij} + \frac{4\pi G}{c^4}(\varepsilon + \sum_i p_i)\right]. \tag{303}$$

This geometric identity follows from the trace of the spatial Einstein evolution equations using the Hamiltonian constraint; a standard presentation is given in [10]. With $\mathcal{K} = 0$, one obtains here

$$\boxed{D_iD^iN = \frac{4\pi G}{c^4}N\mathcal{Q}_g + \frac{\omega^2}{2N}h^{ab}\omega_a\omega_b,} \quad D_iD^iN = \frac{1}{\omega\sqrt{h_2}}\partial_a(\omega\sqrt{h_2}h^{ab}N_b). \tag{304}$$

The remaining Einstein components determine the spatial metric and the shift. (304) is an explicit component of the full coupled system.

In the weak limit $N = 1 + \Psi/c^2 + \cdots$, $\mathcal{K}_{ij}\mathcal{K}^{ij}$ and products of geometric perturbations are of higher order. This gives $\nabla^2\Psi = 4\pi G\mathcal{Q}_0/c^2$ and hence (30). The regular core source and the exterior monopole response are limiting cases of this metric closure.

20.3. First Geometrically Induced Field Adjustment

Around the flat reference, write

$$N = 1 + \nu, \quad h_{ab} = \delta_{ab} + \sigma_{ab}, \quad \varpi = R(1+\rho), \qquad \nu, \sigma_{ab}, \rho, \omega = O(\epsilon). \tag{305}$$

At fixed environment-referenced Ω, the first variation of (294) and the contraction give

$$\delta P_g = -2\rho/R^2 + 2\Omega^2\nu - 2\Omega\omega, \tag{306}$$

$$\delta\Delta_g = -\sigma^{ab}f_{0,a}f_{0,b} + \delta P_g\sin^2 f_0 + 2\nabla f_0\cdot\nabla\delta f + 2P_0\sin f_0\cos f_0\,\delta f, \quad P_0 = R^{-2} - \Omega^2. \tag{307}$$

On the locally adjusted branch with $\Delta_0 > 0$, (206) gives

$$\delta\chi_* = \frac{A}{2b_*u_{*,0}}\delta\Delta_g, \qquad u_{*,0} = \sqrt{1 + A\Delta_0/b_*}. \tag{308}$$

At fixed $\Omega + \omega$, a smaller N increases the local phase rate and reduces its immediate contribution to Δ_g. Spatial geometry, frame dragging, and profile adjustment have additional effects. The total loading change follows from their sum.

The full spatial response follows from the functional. With $dV_0 = R\,dR\,dz\,d\varphi$, $\vartheta_g = \nu + \rho + \frac{1}{2}\sigma^a{}_a$,

and $\mathcal{E}_{\Omega,0}$ as the integrand of (213), the first change at fixed fields is

$$\mathcal{J}_g[f,\zeta] = \int dV_0 \Big[\vartheta_g \mathcal{E}_{\Omega,0} - \frac{K_0}{2}\sigma^{ab} f_a f_b - \frac{Z}{2}\sigma^{ab}\zeta_a\zeta_b + \frac{1}{2}(AC + \beta_4 |\nabla f|^2)\sin^2 f\, \delta P_g \Big], \quad K_0 = AC + \beta_4 P_0 \sin^2 f. \tag{309}$$

On a regular reference branch with a coupled Hessian H_0,

$$\mathsf{H}_0 \begin{pmatrix} \delta f \\ \delta\zeta \end{pmatrix} = - \begin{pmatrix} \delta\mathcal{J}_g/\delta f \\ \delta\mathcal{J}_g/\delta\zeta \end{pmatrix}\Big|_{f_0,\zeta_0}. \tag{310}$$

The correspondingly linearized environmental conditions are part of this system. When the exterior field is eliminated, its variations enter the Hessian and the driving term; a shared smooth geometry is simultaneously continued across the imaginary surface.

In an expansion in weak gravitational coupling, $T^{(0)}$ first gives the initial metric change. This is followed by the field response (310). The change δT feeds back together with quadratic geometric terms at the next coupling order. The formulas determine this backreaction mechanism analytically.

21. Scale Mapping, Size Balance, and Inertia

21.1. Reciprocal Coupling with Different Regime Weights

The symmetric normalization in Section 8 is a specific choice. An explicit mapping between regimes with different weights includes the measure and the definition of the adjoint. For a quadratic mismatch term

$$-\frac{1}{2}\sum_n w_n g_n |\chi_{n+1} - C_n\chi_n|^2 \tag{311}$$

, variation with shared spacetime measures and constant coefficients gives, in channel n,

$$0 = Z_n \Box\chi_n + V_n'(\chi_n) + g_n C_n^\dagger (C_n\chi_n - \chi_{n+1}) \tag{312}$$

$$+ \frac{w_{n-1}}{w_n} g_{n-1}(\chi_n - C_{n-1}\chi_{n-1}). \tag{313}$$

The weight relation is part of the reciprocal coupling. Different measures also change the adjoint. This form illustrates the required mapping; it is not added to (73) as a second channel energy.

With $\mathsf{W}_g = \mathrm{diag}(w_n g_n)$ and the Euclidean adjoint operator $D^\dagger = D^\mathsf{T}$ in a shared real coordinate representation, for the mismatches $D\chi$ and a shared direction v with $Dv = 0$, $v^\mathsf{T} D^\dagger \mathsf{W}_g D\chi = 0$. A large response in this direction therefore requires a corresponding excitation in the rest of the field problem. A low mismatch cost alone does not generate this excitation.

21.2. Size Variation at Fixed Angular Momentum

For a fully localized, stationarily rotating organization with an admissible spatial size variation, let E_2 be the quadratic spatial energy, E_4 the quartic spatial energy, and E_V the potential energy. The two inertia contributions scale as $s_{\mathrm{dil}}^3 I_2$ and $s_{\mathrm{dil}} I_4$. At fixed angular momentum J, the energy along this variation is

$$E_J(s_{\mathrm{dil}}) = s_{\mathrm{dil}} E_2 + s_{\mathrm{dil}}^{-1} E_4 + s_{\mathrm{dil}}^3 E_V + \frac{J^2}{2(s_{\mathrm{dil}}^3 I_2 + s_{\mathrm{dil}} I_4)}. \tag{314}$$

With $T_a = J^2 I_a / [2(I_2 + I_4)^2]$,

$$\left.\frac{dE_J}{ds_{\mathrm{dil}}}\right|_1 = \mathcal{D} := E_2 - E_4 + 3E_V - 3T_2 - T_4 = 0. \tag{315}$$

With an active quasistatic regime channel, E_V contains the already adjusted potential. A fixed rotation rate and a fixed angular momentum define different variational problems. The finite core calculation has fixed Dirichlet data; the isolated size variation is not an additional boundary condition in that calculation.

21.3. Periodic Closure and Restoring Response of an Orbital Organization

In ansatz (211), the internal phase is $\psi = \varphi + \Omega c t$. At a fixed position, the field returns to the same orientation after

$$T_{\mathrm{orbit}} = \frac{2\pi}{c|\Omega|} \tag{316}$$

Spatial single-valuedness of a more general ansatz $\psi = m\varphi + \Omega c t$ requires $m \in \mathbb{Z}$ and $\oint d\psi = 2\pi m$. The calculated family uses $m = 1$. This winding condition describes the phase of the classical orientation field; it does not insert electron spin.

For the dimensionless rate used, $\widetilde{\Omega}^2 = 0.5767110308943739$,

$$\frac{cT_{\mathrm{orbit}}}{\ell_*} = \frac{2\pi}{\sqrt{0.5767110308943739}} \simeq 8.27372. \tag{317}$$

This is the periodicity of the stationary field ansatz. It does not prescribe the circular motion of a marked material constituent.

Beyond (315), size variation at fixed J gives an explicit restoring curvature. With $I = I_2 + I_4 > 0$, differentiating twice gives

$$\left.\frac{d^2E_J}{ds_{\mathrm{dil}}^2}\right|_1 = 2E_4 + 6E_V + \frac{J^2(6I_2^2 + 3I_2 I_4 + I_4^2)}{(I_2 + I_4)^3}. \tag{318}$$

For nonnegative energy and inertia contributions, this curvature is positive provided that at least one restoring contribution is present. At a point with $\mathcal{D} = 0$, there is therefore a local minimum along this size variation. This assumes a localized organization with admissible scaling.

The finite core calculation tests a different variational space: it holds Ω and the boundary

data fixed. Its smallest Hessian eigenvalues are 0.35304236 without and 0.346256 with an additional regime channel. They measure the local restoring response within the profile basis used. Periodic closure, size restoration, and this stationary profile test are thus calculated separately.

For temporally persistent binding, furthermore,

$$\left\langle \frac{dE_D}{dt} \right\rangle = -\left\langle \int_{\partial D} \mathbf{S}_{\mathrm{tot}} \cdot \mathbf{n}\, dA \right\rangle = 0 \tag{319}$$

must hold when all coupling contributions are included in the system whose balance is evaluated. Incoming power can replace outgoing power. The stationary envelope ansatz satisfies the balance through equal inward and outward components. The dynamical transition from free fronts and stability against general time-dependent perturbations belong to the further matter-formation calculation.

21.4. Integrated Stress and a Shared Metric Response

For the rigid-translation ansatz of a static profile in the present field model, the coefficient of the quadratic term in the translation velocity is

$$c^2 M_{ij} = E_0 \delta_{ij} + \int \Pi_{ij}\, d^3x. \tag{320}$$

For a complete localized, stationary, conserved tensor, the integrated spatial stresses vanish when the boundary contributions decay sufficiently. This follows by integrating $\partial_k(x_i \Pi_{kj}) = \Pi_{ij}$ for $\partial_k \Pi_{kj} = 0$. One then obtains the relativistic connection $M_{ij} = E_0 \delta_{ij}/c^2$. For bounded domains, the external confining and environmental contributions are included in the balance.

The retained GR description provides this universal metric response. An additional scalar environmental channel ψ would be universal only if $(dE_a^*/d\psi)/M_a$ were the same for the bodies considered. This property is not assigned here to the loading field ζ as a second gravitational interaction. Instead, the entire tensor couples to the same metric.

The role of shared clock and spatial factors is made clear by $d\tau = N(\psi)dt$ and $dl = L(\psi)|d\mathbf{x}|$. For the same size ansatz at fixed J, the conditional relation is

$$E_J = N\left[LE_2 + \frac{E_4}{L} + L^3 E_V + \frac{J^2}{2(L^3 I_2 + L I_4)} \right]. \tag{321}$$

At the reference state $N = L = 1$,

$$\frac{dE_J}{d\psi} = aE_J + b\mathcal{D}, \qquad a = \frac{d\ln N}{d\psi}, \quad b = \frac{d\ln L}{d\psi}. \tag{322}$$

For $\mathcal{D} = 0$, the response is proportional to the total energy. Metric coupling of all fields is the prerequisite for this relation and is used as a working basis in the manuscript.

For directed stresses, the trace-free spatial response also contributes to the linear energy

coupling. For $h_{ij} = \delta_{ij} + 2\sigma\delta_{ij} + 2s_{ij}$, $\operatorname{tr} s = 0$, and $\nu = \ln N$,

$$\delta E = \int [\nu\varepsilon - \sigma \operatorname{tr}\Pi - s_{ij}\Pi_{ij}^{\mathrm{TF}}]\, d^3x. \tag{323}$$

The twist contribution to this is given in (36). The metric response of these energy and stress contributions follows GR.

22. The Golden Ratio as a Geometric and Recursive Comparison

The golden ratio

$$\varphi_{\mathrm{g}} = \frac{1+\sqrt{5}}{2} = 1.6180339887\ldots \tag{324}$$

appears in different mathematical roles. For each role, the following derivations specify a concrete quantity and a selection condition: equal edges of a spatial arrangement, decay of a linear channel response, and growth of a spiral.

22.1. Equal Edges of an Icosahedral Arrangement

A family of twelve points on a shared sphere is obtained by normalizing the coordinates

$$(0, \pm 1, \pm\rho_{\mathrm{ico}}), \qquad (\pm 1, \pm\rho_{\mathrm{ico}}, 0), \qquad (\pm\rho_{\mathrm{ico}}, 0, \pm 1), \qquad \rho_{\mathrm{ico}} > 1. \tag{325}$$

For the points $u = (0, 1, \rho_{\mathrm{ico}})$, $v = (1, \rho_{\mathrm{ico}}, 0)$, and $w = (0, -1, \rho_{\mathrm{ico}})$, two neighboring edges on the unit sphere are

$$d_{uv}^2 = \frac{2\rho_{\mathrm{ico}}^2 - 2\rho_{\mathrm{ico}} + 2}{1 + \rho_{\mathrm{ico}}^2}, \qquad d_{uw}^2 = \frac{4}{1 + \rho_{\mathrm{ico}}^2}. \tag{326}$$

Their equality gives

$$\rho_{\mathrm{ico}}^2 - \rho_{\mathrm{ico}} - 1 = 0 \quad \Longrightarrow \quad \rho_{\mathrm{ico}} = \varphi_{\mathrm{g}}. \tag{327}$$

The golden ratio follows here from the requirement of equal edges within this point family. The corresponding angular relations $\cos 36^\circ = \varphi_{\mathrm{g}}/2$ and $\cos 72^\circ = 1/(2\varphi_{\mathrm{g}})$ allow an analytical description of symmetrically coupled orientations. For example, $(1/2, -1/(2\varphi_{\mathrm{g}}), -\varphi_{\mathrm{g}}/2)$ is a unit vector because $(1 + \varphi_{\mathrm{g}}^{-2} + \varphi_{\mathrm{g}}^2)/4 = 1$.

For bound field organization, this geometry is a defined symmetry reference. Its energetic selection by the field action is a variational question in which geometries can be freely compared. Edge equality alone does not replace this energetic selection.

22.2. Golden-Ratio Decay in the Weak Regime Channel

For small auxiliary responses, the nearest-neighbor chain from Section 8 becomes linear away from the excitation site:

$$\Sigma_n - \kappa(\Sigma_{n-1} + \Sigma_{n+1}) = 0, \qquad 0 \le \kappa < \frac{1}{2}. \tag{328}$$

With $\Sigma_n \propto r_\kappa^{|n|}$,

$$1 = \kappa(r_\kappa + r_\kappa^{-1}), \qquad r_\kappa = \frac{2\kappa}{1+\sqrt{1-4\kappa^2}}. \tag{329}$$

The continuous extension at $\kappa = 0$ is $r_\kappa = 0$. For golden-ratio decay, exactly

$$\boxed{r_\kappa = \varphi_g^{-1} \quad \Longleftrightarrow \quad \kappa = \frac{1}{\sqrt{5}}.} \tag{330}$$

This parameter lies in the convex domain of the channel energy; the lower Hessian bound is $1 - 2/\sqrt{5} = 0.1055728090\ldots > 0$. In the numerical reference case $\kappa = 0.2$, by contrast, $r_\kappa = 0.2087121525\ldots$.

The relation concerns the weak auxiliary response of the chain. A physical length or energy ratio between regimes additionally requires their scale mapping. An alternative mathematical selection would be the balance $w_n = w_{n+1} + w_{n+2}$ at a constant ratio $w_{n+1}/w_n = r$: it gives $1 = r + r^2$ and hence $r = \varphi_g^{-1}$. This specific balance is a separate model condition; it does not follow from recursive nesting alone.

22.3. Spiral Parameter and Distinction from the Orbital Period

For a logarithmic spiral $r(\theta) = r_0 \exp(b_{sp}\theta)$, growth by a factor of φ_g per quarter-turn selects the value

$$\frac{r(\theta + \pi/2)}{r(\theta)} = \varphi_g \quad \Longrightarrow \quad b_{sp,g} = \frac{2\log\varphi_g}{\pi}. \tag{331}$$

The pitch angle relative to the local azimuthal direction is

$$i_g = \arctan b_{sp,g} = 17.03239113^\circ. \tag{332}$$

It provides a geometric comparison value for independently determined spiral pitches. A logarithmic spiral is open; the periodic closure of a bound field organization is determined by its internal phase and dynamical conditions.

The three generic front weights $1 : 0.98 : 0.44$ parameterize initial amplitudes. The static triad uses three equally scaled knots and determines their internal phases. The stationary core family has the single phase period $cT_{orbit}/\ell_* \simeq 8.27372$. These quantities do not constitute three measured orbital periods from which the front weights or a golden ratio have been derived. The golden angle $360^\circ/\varphi_g^2 = 137.50776^\circ$ must likewise be distinguished from the spatial and internal triad angles. Each application of the golden ratio is thereby assigned a named variable and its own geometric or dynamical test criterion.

23. Reference Values, Physical Scales, and Scope

23.1. Dimensionless Specification and Units

The documented reference case uses

$$\widetilde{A} = 1.140673773053, \quad \widetilde{\beta}_4 = 0.1248600866454894, \quad \widetilde{C}_s = 0.01480496231993644, \quad \widetilde{Z}_0 = \widetilde{b}_* = 1. \tag{333}$$

These are the parameters of the model case investigated. The numerical calculations use these dimensionless quantities; the tildes are omitted there for readability. With length scale ℓ_* and energy scale E_*,

$$x = \ell_* \widetilde{x}, \qquad t = (\ell_*/c)\widetilde{t}, \qquad \varepsilon = (E_*/\ell_*^3)\widetilde{\varepsilon}, \tag{334}$$

$$A = (E_*/\ell_*)\widetilde{A}, \qquad Z_0 = (E_*/\ell_*)\widetilde{Z}_0, \qquad \beta_4 = E_*\ell_*\widetilde{\beta}_4, \tag{335}$$

$$C_s = (E_*/\ell_*^3)\widetilde{C}_s, \qquad b_* = (E_*/\ell_*^3)\widetilde{b}_*, \qquad \Omega = \widetilde{\Omega}/\ell_*. \tag{336}$$

The dimensionless gravitational strength is

$$\gamma_g = \frac{8\pi G E_*}{c^4 \ell_*}. \tag{337}$$

A shared calibration must consistently assign lengths, energies, times, and gravity. This remains open.

23.2. Physical Context

In the UQSH picture, bodies, radiation, and the environment are different organizational forms of the same nested field. A bound configuration holds internal motion together and simultaneously shapes its environment. Gravity describes the spatial response of this order. The Anun and regime equations in Section 2 specify how saturated binding and a small additional carrier response are related. The environment in motion remains effective in its own right.

The effective calculation describes how orientation, spatial tension, and loading exchange energy and can support a stationary core. The gravitational source is calculated from their total energy and stresses. The metric closure translates it into motion and clock comparison and feeds back on the field profiles.

Section 7 calculates the additional source of a reorganization and connects it to the additional acceleration. The eliminated carrier response also provides a condition for source enhancement in a single response channel. The coupled comparison in Section 17 shows how spatial adjustment changes the immediate channel contribution: inner enhancement and an outer counteracting change together determine the remaining additional source.

The weak limit explains the law of free fall and, for a compact spherically symmetric source, the $1/r^2$ exterior response. A distributed envelope source can provide an additional, approximately constant contribution to the squared orbital speed. UQSH qualitatively interprets the weak coupling strength through partial spatial and regime-dependent effectiveness. For

the present calculations, the observed gravitational connection is implemented through G; explaining its numerical value is a further question of scale organization.

23.3. Results and Physical Connection

Subproblem	Status of the present description
Shared dynamics	The action, field equations, tensor source, and mutual energy balance are given explicitly.
Saturation	The local restoring response is derived; the spatial threshold and nonlinear profile balance are determined.
Core	A stationary coupled profile is available for fixed environmental values, including regularity and numerical equation residuals.
Envelope	Transition amplitudes, matching conditions, and the weak gravitational monopole contribution are connected analytically.
Backreaction	The curved-space field equations, the clock-rate equation, and the first field adjustment are formulated.
Regime channel	The input, bounded response, backreaction, and tensor are determined from one action; a coupled stationary comparison separates the channel contribution from spatial adjustment.
Directional structure	Twist moments and the unaveraged gravitational quadrupole response are given explicitly.
Front superposition	Angular and phase contributions, a spatial amplification criterion, internal directional order, and the combined energy balance are connected analytically.
Arrival-time offset	Source position and emission time determine the arrival-time mapping; a regular selection satisfies the phase criterion in Section 10.
Matter formation	Front energy, static triad geometry, and the topological matching condition are determined separately (Sections 12 and 13).
Golden ratio	The icosahedral condition, weak-channel ratio, and spiral parameter are derived analytically (Section 22).

The channel is identified as a quasistatic constitutive extension. In addition to the base-model reference, the spatially adjusted solution with an active channel is available. For this parameter case, the integrated active source increases by 0.000262345 % while the field energy is slightly lower. The stress balance explains these signs.

The following tasks connect the model derivation to physical predictions:

1. **A coupled solution with a physical environment.** The core, nonlinear transition, exterior field, and metric must be solved with the same environmental data. The

stationary sphere calculation and the linear exterior envelope are currently different, analytically connected subproblems.

2. **Formation and general stability.** The temporal path from suitable initial states, energy release, and the response to further spatial and phase perturbations must be determined. This includes the regular domain of the full nonlinear initial-value problem. Section 11 determines the analytical loading connection and an overlap comparison with four defined initial states for this purpose. The triad angles from Section 13 add the static phase binding; the topological matching condition stated there determines the admissible transition from the initial front data. Identification with an electron requires a dynamically maintained solution, its electromagnetic charge and coupling, angular momentum, and energy and length scales to be determined together. The periodic closure and restoring curvature in Section 21.3 provide specific test conditions for this purpose.

3. **Scales and observations.** A justified assignment of E_*, ℓ_* and field organization to observable matter must yield quantitatively consistent predictions for motion, light deflection, and time measurement. A rotation-curve fit does not yet follow from the plateau expression alone.

4. **Derivation of the effective model choice.** The explicit regime responses from Section 2 are connected to q, ζ through the channel defined here and then tested at the relevant scales. The coupled Anun–environment solutions must quantitatively determine the decreasing contributions from deeper levels. The specific potential channel from Section 8 defines a first shared action. Its scale parameters and the assignment of additional regimes are determined quantitatively. GR and G remain the geometric working basis; their own microscopic derivation is not a prerequisite for this task.

5. **Radiation and additional regime statements.** An electromagnetic detector identification, photon energy, a general additional redshift law, and a complete Anun description require further derivations. Reversible channel exchange alone does not determine a universal direction of irreversible relaxation.

24. Methods and Data Basis

The continuum derivation uses the action (13) and the stress-energy tensor with the same normalization. After elimination, the regime channel changes only the local loading potential. The algebraic saturation approximation, stationary core problem, linear exterior envelope, and geometric backreaction are treated with the boundary and approximation conditions specified for each.

The stationary core results in Sections 16 and 17 are based on a Galerkin representation of the two profiles. Gradient residuals, strong-form equation residuals, and Hessian eigenvalues test stationarity and local curvature in the resolved variational space. Two quadrature resolutions check the energy and source integrals at the same profile coefficients. A quadrature check must be distinguished from convergence under refinement of the profile basis.

The static three-knot data in Section 13 belong to a separate dimensionless preliminary investigation. They determine phase differences and relative binding energies at fixed geometry. Their energy unit and texture diagnostic are not transferred to the later core family. The reported phase paths check the state dependence of the energy change, not a temporal relaxation rate.

The connection from fronts follows from the first loading variation, the angular cross term, and the spherical-chart metric. Norm-preserving rotations provide admissible initial data; the canonical momentum map and energy cross balances determine their comparison. The overlap geometry is defined analytically. A dynamic binding contribution after front passage is not reported as a numerical result in this work.

24.1. Data and Code Availability

The core calculation is documented through profile coefficients, initial and boundary data, analysis programs, and tables of radial energy and source profiles. The discrete knot comparisons have separate normalizations and result records. A public archive address for the coefficients, programs, and figure data associated with this manuscript version has not yet been assigned. An archive DOI will be added after these materials are published.

The algebraic checks concern the front energy, the arrival-time mapping, the discrete selection of arrival phases, and the golden-ratio comparison geometries. They are reported separately from field evolution and stationary solution methods.

A. Spatial Hamiltonian Method

A.1. Verification Equation and Discrete Energy

The meridional front sector can be evolved from a discrete energy. The following specification gives the operators that can be used for this purpose directly. The vacuum tangent with $\ell_{\mathrm{ang}} = 2$, $\Lambda_{\mathrm{ang}} = \ell_{\mathrm{ang}}(\ell_{\mathrm{ang}} + 1) = 6$ satisfies

$$\eta_{\tau\tau} = \eta_{rr} + 2\eta_r/r - \Lambda_{\mathrm{ang}}\eta/r^2, \qquad \pi = A\eta_\tau. \tag{338}$$

For any sufficiently smooth compact generating function g, with $x = r - \tau$,

$$\eta_*(\tau, r) = \frac{g''(x)}{r} - \frac{3g'(x)}{r^2} + \frac{3g(x)}{r^3}, \qquad \eta_{*,\tau} = -\frac{g'''(x)}{r} + \frac{3g''(x)}{r^2} - \frac{3g'(x)}{r^3} \tag{339}$$

is an exactly outgoing verification solution in the source-free region. Substitution into the operator checks the identity directly. With $\Gamma_2 = \int Y^2 d\Omega_s = 4\pi/5$, the positive energy is

$$H = \frac{\Gamma_2}{2} \int \left[\frac{r^2\pi^2}{A} + Ar^2\eta_r^2 + A\Lambda_{\mathrm{ang}}\eta^2 \right] dr. \tag{340}$$

At cell centers r_i and interior edges $r_e = r_{i+1/2}$ of a uniform grid with spacing h, we define

$$M_i = hr_i^2, \quad W_e = hr_e^2, \qquad (D_4\eta)_{i+1/2} = \frac{\eta_{i-1} - 27\eta_i + 27\eta_{i+1} - \eta_{i+2}}{24h}. \tag{341}$$

Taylor expansion about the edge gives the leading error $-3h^4\eta^{(5)}/640$. At the first and last interior edges, the two-point gradient is used. Fourth order refers to the interior gradient, not automatically to the full boundary-value and time-integration method.

With diagonal matrices M, W, the discrete energy is

$$H_h = \Gamma_2 \left[\sum_i \frac{M_i \pi_i^2}{2A} + \frac{A}{2}(D_4\eta)^{\mathsf{T}} W (D_4\eta) + \frac{A\Lambda_{\mathrm{ang}} h}{2} \sum_i \eta_i^2 \right]. \tag{342}$$

The canonical momenta are $\Gamma_2 M_i \pi_i$. Hamilton's equations are therefore

$$\dot\eta = \pi/A, \qquad \dot\pi = -AM^{-1}(D_4^{\mathsf{T}} W D_4 \eta + \Lambda_{\mathrm{ang}} h \eta). \tag{343}$$

Substitution into $\dot H_h$ cancels the kinetic and spatial contributions exactly. The dot here denotes ∂_τ. Time discretization and detector readout have their own errors. A quantitatively accurate energy flux additionally requires comparison of its time series with (339).

A.2. Positive Coupling Coefficients and the Nonlinear Operator

For interior edge interpolation,

$$(I_4 a)_{i+1/2} = \frac{-a_{i-1} + 9a_i + 9a_{i+1} - a_{i+2}}{16} \tag{344}$$

is used. Its leading error is $-3h^4 a^{(4)}/128$. Because $C, Z > 0$ for finite real loading, we set

$$\widehat C = \exp(I_4 \log C), \qquad \widehat Z = \exp(I_4 \log Z). \tag{345}$$

At the two outermost interior edges, the logarithmic values are interpolated with the symmetric two-point mean. The edge coefficients thus remain positive. The interpolation weights must be distinguished from a distribution of edge energy among cells.

Write $D = D_4$, $I = I_4$, $g_\vartheta = D\vartheta$, $g_\zeta = D\zeta$, $v = p_\vartheta/(AC)$, $u = p_\zeta/Z$, $\beta_C = C'/C$, and $\beta_Z = Z'/Z$. The radial sector energies are

$$E^h_{\vartheta,r} = 4\pi \left[\sum_i M_i \left(\frac{p_{\vartheta i}^2}{2AC_i} + V_i \right) + \frac{A}{2} \sum_e W_e \widehat C_e g_{\vartheta e}^2 \right], \tag{346}$$

$$E^h_{\zeta,r} = 4\pi \left[\sum_i M_i \left(\frac{p_{\zeta i}^2}{2Z_i} + U_i \right) + \frac{1}{2} \sum_e W_e \widehat Z_e g_{\zeta e}^2 \right]. \tag{347}$$

Their variation with the canonical weights $4\pi M_i$ determines the full radial operator $\mathcal{F}_0^h$:

$$\dot\vartheta = v, \qquad \dot\zeta = u, \tag{348}$$

$$\dot p_\vartheta = -AM^{-1} D^{\mathsf{T}} (W \widehat C g_\vartheta) - V_\vartheta, \tag{349}$$

$$\dot{p}_\zeta = -M^{-1}D^{\mathsf{T}}(W\widehat{Z}g_\zeta) + \frac{A}{2}C'v^2 + \frac{Z'}{2}u^2 - U' - \frac{A}{2}M^{-1}\{\beta_C I^{\mathsf{T}}(W\widehat{C}g_\vartheta^2)\} - \frac{1}{2}M^{-1}\{\beta_Z I^{\mathsf{T}}(W\widehat{Z}g_\zeta^2)\}. \tag{350}$$

All unmarked products and quotients are pointwise. The terms containing I^{T} arise from variation of the state-dependent edge coefficients. The sector balance follows from the same energy:

$$\dot{Q}_r^h = \frac{4\pi A}{2}\left[\sum_i M_i C_i' v_i^2 u_i - \sum_e W_e \widehat{C}_e g_{\vartheta e}^2 (I(\beta_C u))_e\right], \tag{351}$$

$$\dot{E}_{\vartheta,r}^h = -\dot{Q}_r^h, \qquad \dot{E}_{\zeta,r}^h = \dot{Q}_r^h. \tag{352}$$

The final equalities hold for the closed boundary conditions of the finite Hamiltonian energy. For open boundaries, their fluxes are additionally included in the balance.

A.3. Flux Across a Cut and Coupled Perturbation Evolution

Let $\chi_i = 1$ to the left of a detector surface and zero elsewhere. The positive edge energy is distributed evenly among its supporting cells; α_e is the fraction on the left. Define

$$B_D(w) = D(\chi w) - \alpha D w, \qquad B_I(w) = I(\chi w) - \alpha I w. \tag{353}$$

Differentiating the energy assigned to the left gives the outward-positive fluxes

$$L_\vartheta^h = 4\pi A \sum_e W_e \widehat{C}_e [g_{\vartheta e} B_D(v)_e + \tfrac{1}{2} g_{\vartheta e}^2 B_I(\beta_C u)_e], \tag{354}$$

$$L_\zeta^h = 4\pi \sum_e W_e \widehat{Z}_e [g_{\zeta e} B_D(u)_e + \tfrac{1}{2} g_{\zeta e}^2 B_I(\beta_Z u)_e]. \tag{355}$$

The cellwise exchange assignment is

$$q_i^h = \frac{4\pi A}{2} u_i [M_i C_i' v_i^2 - \beta_{Ci}(I^{\mathsf{T}}(W\widehat{C}g_\vartheta^2))_i]. \tag{356}$$

With $\dot{Q}_{\text{left}}^h = \sum_i \chi_i q_i^h$,

$$\dot{E}_{\vartheta,\text{left}}^h + L_\vartheta^h + \dot{Q}_{\text{left}}^h = 0, \qquad \dot{E}_{\zeta,\text{left}}^h + L_\zeta^h - \dot{Q}_{\text{left}}^h = 0. \tag{357}$$

The additional terms containing B_I capture the time dependence of the distributed coupling coefficients. They vanish in the vacuum.

The angular energy supplements (346)–(347) with

$$H_{S^2}^h = \frac{h}{2}\int d\Omega_s \sum_i [A C_i |\nabla_{S^2}\vartheta_i|^2 + Z_i |\nabla_{S^2}\zeta_i|^2]. \tag{358}$$

This gives the discrete version of (111): $\mathcal{F}_0$ is replaced by (348)–(350), and the factor r^{-2} in

the explicit angular terms is replaced by hM^{-1}. The mean-energy coefficient is

$$E_a^{h,[2]} = DE_{a,r}^h[y_0]m + \frac{1}{10}D^2E_{a,r}^h[y_0][s,s] + \frac{4\pi\Lambda_{\mathrm{ang}}h}{10}\sum_i K_{a,i}s_{a,i}^2. \tag{359}$$

Similarly, the exchange coefficient arises from the first two variations of $\dot{Q}_r^h$ and the additional term $-4\pi A\Lambda_{\mathrm{ang}}h\sum_i C_i'u_i\eta_i^2/10$.

For the operator derivatives, the chain rule is explicit: at $\widehat{K} = \exp(I\mathcal{L}_K)$, $\mathcal{L}_K = \log K$, and in direction ξ, let $J_1 = I(\mathcal{L}_K'\xi)$, $J_2 = I(\mathcal{L}_K''\xi^2)$, $J_3 = I(\mathcal{L}_K'''\xi^3)$. Then

$$D\widehat{K} = \widehat{K}J_1, \quad D^2\widehat{K} = \widehat{K}(J_1^2 + J_2), \quad D^3\widehat{K} = \widehat{K}(J_1^3 + 3J_1J_2 + J_3). \tag{360}$$

The background, tangent, and mean response follow from the same discrete energy. This appendix defines the spatial method; time integration and its error control are chosen for the respective initial-value problem.

B. Glossary of Terms

This glossary explains the UQSH terminology used in this work and the concepts of the effective field model. The definitions refer to the theoretical framework chosen here [1, 2]; the section references point to their mathematical use. The notation overview is provided in Section 1.4.

Term	Meaning in this work
Active gravitational source	The combination $Q = \varepsilon + \sum_i p_i$ of energy density and principal spatial stresses in the orthonormal system used. It determines the quasistatic weak gravitational response; the full metric requires the entire stress-energy tensor (Sections 5 and 6).
Arrival-time offset	The temporal offset of matched front features at the detector due to their emission times, emission positions, and propagation paths. For straight propagation at c, $t_i^{\mathrm{arr}} = t_i^{\mathrm{em}} + |\mathbf{x}_D - \mathbf{x}_i|/c$. Such an offset can occur even when the source cadence is unchanged (Section 10).
Anun	In the conventional picture, an Anun is a black hole. The component *An* refers, on the one hand, to Albert Einstein, formed from the first letter of *Albert* and the last letter of *Einstein*. On the other hand, *An* stands not only for the beginning of the first name and the end of the last name, but symbolically for both beginning and end. This double meaning fits UQSH directly because a regime transition has precisely this structure: what appears to one regime as the end of its maximal saturation may become the beginning of a new coupling capacity for the next higher superregime. It is complemented by the *u*, representing the physical universality of the universe in every regime and of the Universal Quantum Foam Hypothesis. The final *n* can also be read as a mathematical regime or dimensional index whose nested levels carry physics that is relative yet the same. In the Phraqtal Field, it denotes a maximally saturated, continuously moving, spatially extended bound configuration in its own regime. Its supporting subregimes may remain close to their ordinary environmental state in motion. Further absorption changes the extent, boundary organization, and exchange of the saturated region (Section 2).
Loading	The dimensionless local field quantity ζ of the effective model. It changes the orientation stiffness $C(\zeta)$ and has its own kinetic energy, spatial gradient energy, and potential energy. Its assignment to a regime-dependent saturation coordinate is specified separately (Section 3).
Core	The central, nonlinearly organized region of a calculated field profile. The stationary core solution is determined on a finite domain with prescribed boundary data. Classifying such a core as an Anun additionally requires the regime-dependent saturation criterion (Sections 3 and 16).
Dedensification	During dedensification, the tension and modulation components of a front that are effective in the observational regime decrease; energy is redistributed into finer carrier organizations. Exchange across regimes and geometric distribution over a larger front surface are accounted for separately. Cadence is captured through the arrival sequence (Section 10).

Continued on the next page

Term	Meaning in this work (continued)
Temporal thinning	An increase in the time intervals between the resolved arrival events that remain effective. With an unchanged source cadence, it can arise from a changed selection of geometrically offset front contributions. The mean interval and the uniformity of the remaining sequence are distinct observables (Section 10.3).
Field envelope	The exterior field organization matched to the nonlinear core. The envelope treated analytically here consists of inward and outward components of equal strength and has no net energy flux on average. Its spatial matching is described on a finite domain (Section 18).
Field knot	A spatially concentrated, structured field configuration with internal orientation and twist. Binding energy, dynamical stability, and topological class characterize different properties of a knot. The three-knot data used describe static binding comparisons (Section 13).
Free field organization	An environment that is unsaturated in the regime considered, with its own motion, tension structure, and radiation. It is supported by finer regimes and forms the dynamic reference state for additional binding or saturation (Section 2).
Geometric dilution	The distribution of an outgoing excitation over a growing spherical surface. In the free radial limit, the field amplitude decreases as $1/r$. The available energy integrated over a complete spherical surface and a complete pulse, as well as the intervals between matched profile features, are preserved in this limit (Section 9).
Gravity-Light	The UQSH term for the gravitational effectiveness of the same field organization in motion that manifests as radiation or bound matter. In this work, its geometric response is described through the stress-energy tensor and the Einstein equation (Sections 4 and 2.3).
Hopf class	The topological class of a smooth spatial orientation structure $q \in S^2$ with a fixed vacuum boundary. It captures a global linking of the field structure and is preserved under smooth norm-preserving evolution with this boundary. Numerical Hopf integrals are evaluated as separate discretization diagnostics (Section 13.1).
Coupling filter and effectiveness	The conditions under which an existing field organization couples to the regime considered and the chosen readout channel. In the arrival-time calculation, w_a describes the adopted effectiveness of a front contribution; a detection criterion determines the recorded events (Section 10.3).
Light Bubbles	A descriptive term for spherically outgoing tension fronts. A Light Bubble denotes the advancing field excitation with its modulatable tension skin. Its distribution may depend on angle; the front is therefore not a spherical shell excited with equal strength everywhere (Section 9).
Light in Light	A term for modulation within an already dynamic field organization that is itself supported by finer regimes. In the title, the expression connects the spectral modulation content to its support by the Light Bubbles and the underlying regimes.
Matter binding and matter formation	Matter binding denotes the spatial persistence of organized field motion through its internal coupling and restoring response. Matter formation denotes the dynamical transition into such a bound organization. Front excitation, static binding, and the topological transition are each investigated separately (Sections 11 and 13).

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Term	Meaning in this work (continued)
Modulation and modulation frequency	The spatial or temporal structure carried by a tension front. Modulation includes changes in field strength, orientation, phase, and twist. Its modulation frequency describes the temporal repetition of such a pattern; the source cadence, by contrast, counts the emission of matched fronts (Sections 9 and 10).
Orientation field	The unit field $q \in S^2$ describes the local internal alignment of the effective field organization. Its spatial and temporal changes enter the action, energy, and stresses. It must be distinguished from the phenomenological classification quantities of a galactic Q-switch (Sections 1 and 3).
Periodic closure	The return of the internal field organization to the same state after one orbital period. For the rotating ansatz, this condition is formulated through the temporal phase. Persistent binding additionally requires a balanced power budget and an appropriate restoring response to perturbations (Section 21.3).
Phraqtal Field	The *Phraqtal Field* denotes the physically real organizational form that is readable in a regime-dependent way. It is the carrier of physics. Clear distinctions from the ether and from Wheeler's quantum foam are described in previous works [2]. The earlier terms *Qu-field* and *Qu-foam* remain historical names within the UQSH research context. The new name specifies the physically effective field organization: *Ph* refers to its physical and holographically readable manifestation, and the lowercase q to the Qu-ground. The Qu-ground is not developed further here. It nevertheless remains part of UQSH but is not needed for this work [2]. The separately designated quantities q, ζ, and the metric constitute the effective portion of this framework investigated here (Section 3).
Source cadence	The time interval T_Q between recurring emission events of an identified source sequence. Several spatially offset component sources can jointly provide additional arrival phases. Their event rate recorded at the detector and their spectral modulation frequency are determined separately (Section 10).
Regime; subregime and superregime	An organizational and coupling level with its own resolution and reorganization order. R_n denotes the chosen observational regime, R_{n-k} finer subregimes, and R_{n+k} coarser superregimes. The index separation counts regime levels, not spatial distance. The ansatz assumes no ultimate smallest or largest level (Section 2).
Regime channel	The specific chain of scalar auxiliary responses with coupling between neighboring levels used in this work. Its quasistatic adjustment gives the effective potential U_{eff}. It describes a particular contribution to the regime response within the chosen action (Section 8).
Regime coupling	Interaction between neighboring organizational levels through which energy and changes in tension are transmitted and returned. The description requires a mapping of resolution, quantities, and response times between the regimes involved (Sections 2 and 21).
Relaxation	Reorganization of existing tension differences through motion, scattering, and energy exchange. Released energy remains active in the receiving field components and can trigger further reorganization or feedback there. The direction of exchange in each case follows from the state (Sections 2 and 5).

Continued on the next page

Term	Meaning in this work (continued)
Saturation	A regime-dependent state in which additional local loading increasingly leads to spatial redistribution, scattering, and exchange with neighboring regimes. The upper local organizational form depends on this coupling. A bounded saturation coordinate is not an absolute upper bound on the energy of the entire nested field (Sections 2, 13, and 14).
Spectral carrier	The role of a tension front whose spatial and temporal modulation carries the content of a measurable spectrum. Which components become visible depends on coupling to the detector. Spectral analysis and the counting of individual arrival events are separate readout quantities (Sections 9 and 10).
Tension front and tension skin	The tension front is the advancing transition region of a field excitation. Its spatially extended layer that is effective in the regime considered is called the tension skin. It carries tension, orientation, and modulation and may be distributed nonuniformly across its surface. Free reference propagation occurs at c (Sections 9 and 10).
Carrier regime and nesting	Finer regimes provide the organization on which the described front, bound configuration, and environment depend. Because these carriers themselves have further underlying support, a recursive sequence of scales arises. The participation of a level and the spatial reach of its effect are distinct quantities (Section 2).
Triadic coupling	The interaction of three field organizations with different excitation, deflection, and restoring roles. Spatial encounter angles, internal directions, phases, and energy contributions determine their contribution to binding. The static three-knot configuration and the dynamic front scenario are treated separately (Section 13).
UQSH	Universal Quantum Foam Hypothesis. A theoretical framework in which radiation is described as free field excitation, matter as bound field organization, and gravity as the geometric response of that organization. The regime concepts organize its support across nested scales.
Twist	Nonparallel changes in the internal field orientation. In the effective model, it is captured by $J_{\mu\nu} = \nabla_\mu q \times \nabla_\nu q$. Its spatial and temporal components contribute to energy, momentum, and stresses; local twist alone does not determine the global Hopf class (Sections 4 and 5.1).
Action and field action	In the mathematical part, the action denotes the functional whose variation determines the field equations. The stress-energy tensor, which couples to the geometric gravitational equation, is calculated from the same field action. In physical usage, the action of the field denotes the resulting change in motion and organization (Section 4).

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